watsonx.governance Hands-On Workshop

*Introduction to AI Governance via **Building**, **Deploying**, and **Managing** AI models.*

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For more information, visit us at: https://www.ibm.com/products/watsonx-governance

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Introduction to watsonx.governance

watsonx.governance is a Modern AI Governance toolkit that provides a sleek and intuitive User Interface. Whether Users are looking for responsible AI workflows, or to reduce costs and comply with regulations, watsonx.governance provides the capabilities to empower everyone in the organization with the insights needed to positively impact governance.

This workshop is designed to give you an opportunity to discover the ease of use of watsonx.governance, IBM's AI Governance solution that enhances the efficiency and capabilities of AI developers, users, policymakers and ethicists alike.

In this workshop, you will be creating a Use Case which uses AI to summarise a Car Insurance Claim. The scenario will allow you to experience the following capabilities in watsonx.governance:

- Creation of a Use Case including automated workflows
- Performing Risk and Compliance Assessments
- Gaining approvals for the sign off to Development
- Development of a prompt template
- Test your prompt, try different parameters and models.
- Evaluate your prompt against gold standard summaries.
- Track your prompt through its lifecycle within the AI use case. This includes version control.
- Promote and evaluate your prompt template in a pre-production deployment space.
- Managing the prompt in production.

Introduction to your lab scenario

In this lab, your client is a large insurance company that is attempting to infuse AI into their business, while complying with best practices and regulations. Using the watsonx.governance platform, you will demonstrate how, with a single solution, they can govern, monitor, and document both traditional predictive models and new generative AI models.

Phase 1: Plan

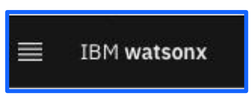
We define a Use Case as a collection of information to solve a business problem, in this case an Insurance Claim Summarisation. In practice, an organization would identify a business need for an AI model, and create a use case to track that effort.

The watsonx.governance solution allows organizations to group and track their models based on use cases, or issues that models are attempting to solve. Each use case stores and organizes data and lifecycle information for candidate models in planning, design, development, and production phases.

In the planning phase, you will use pre-configured workflows to populate the necessary information about the Use Case itself.

Create an AI Use Case

1. Please select the hamburger menu on the top left.



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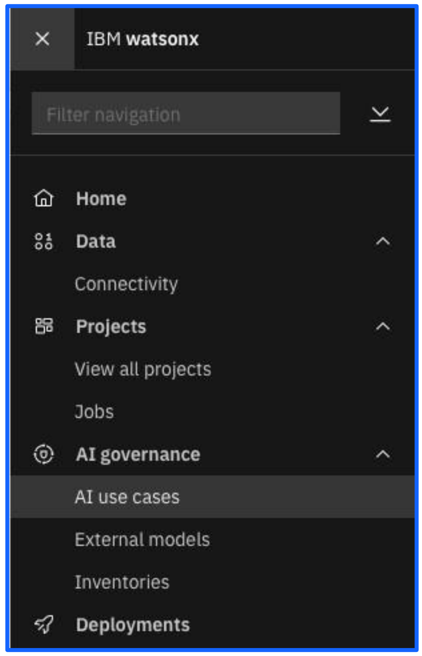
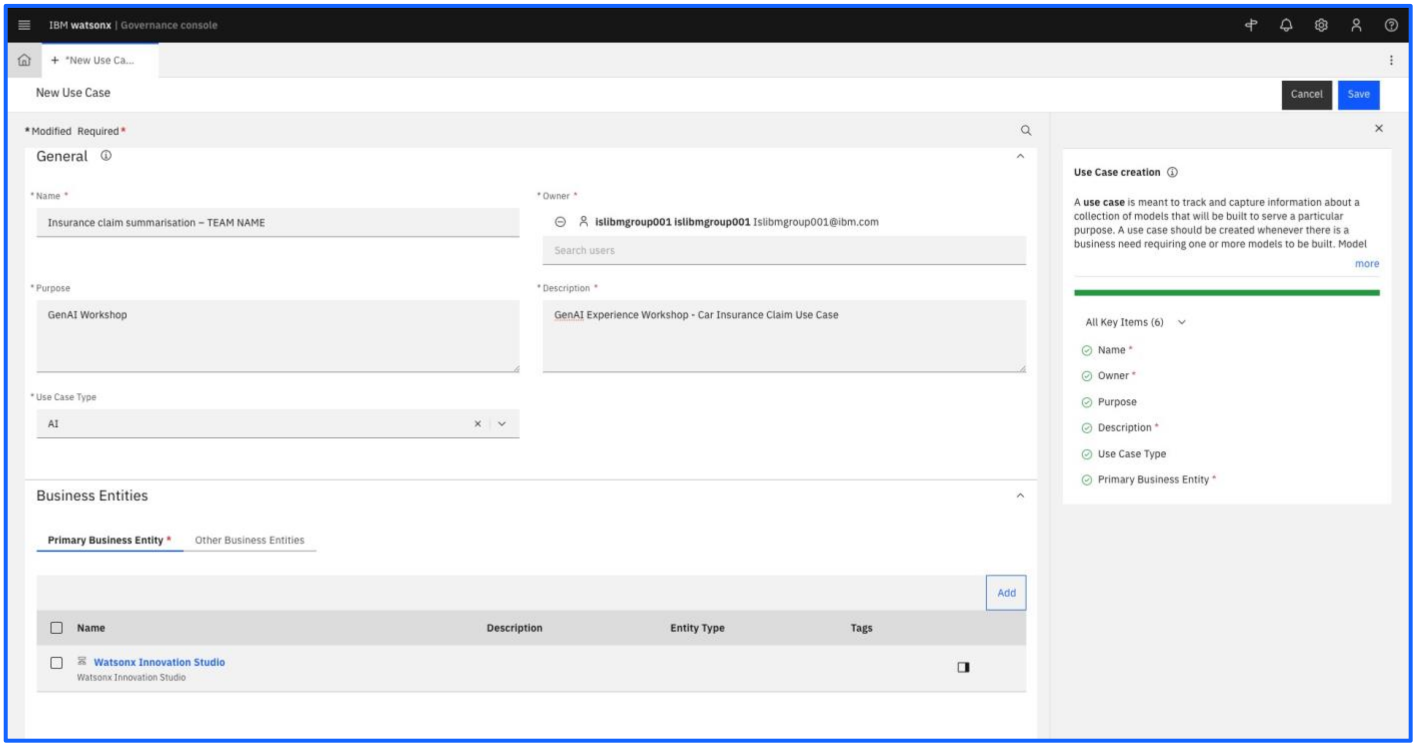
2. Select **AI use cases** from the menu.

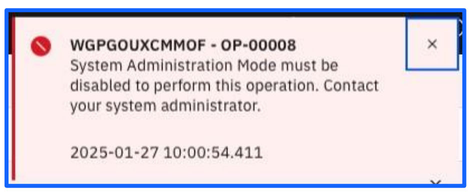


3. Create a new AI use case. At this point you will be redirected to the watsonx.governance console however once the steps are completed the Use Case will also be visible in your list of AI Use Cases screen. This is an example of the bi-directional flow between the Evaluation/Monitoring and AI Factsheet modules and the Governance Console

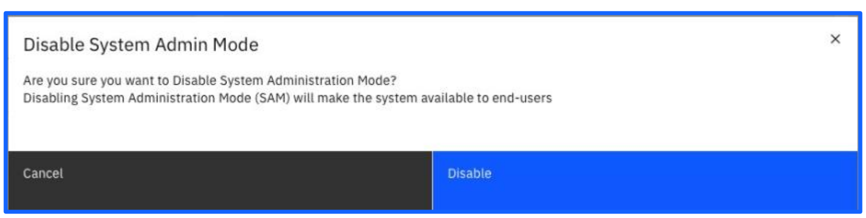
4. Fill out the Use Case using the following parameters:
   - Give it a **Name** such as “Insurance claim summarisation – (*insert your team name*)”
   - Add yourself as the **Owner**
   - **Primary Business Entity** can be set to any listed
   - Click **Save**
5. Next, fill out details which are applicable to the Use Case itself, e.g.:
   - **Description** set “GenAI Experience Workshop - Car Insurance Claim Use Case”
   - *Any others?*



NOTE: If you receive the following error:



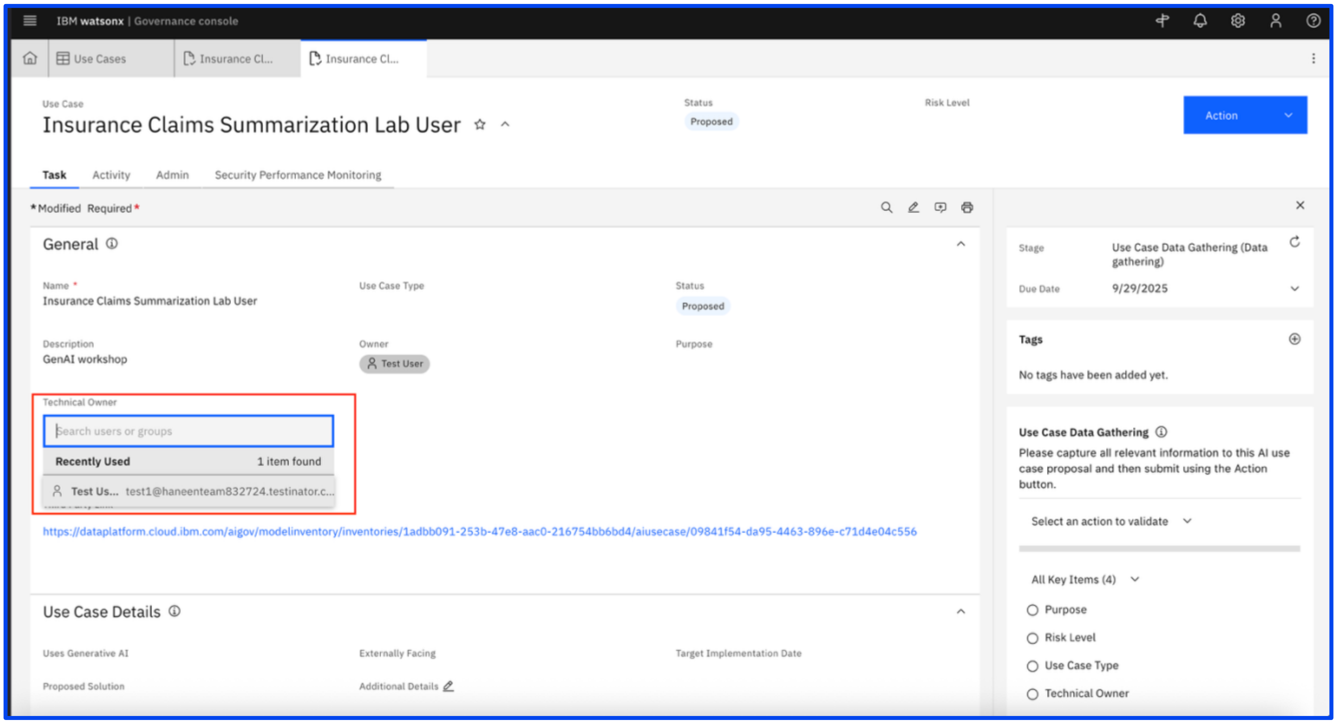
Then from Setting disable **"System Admin Mode"**



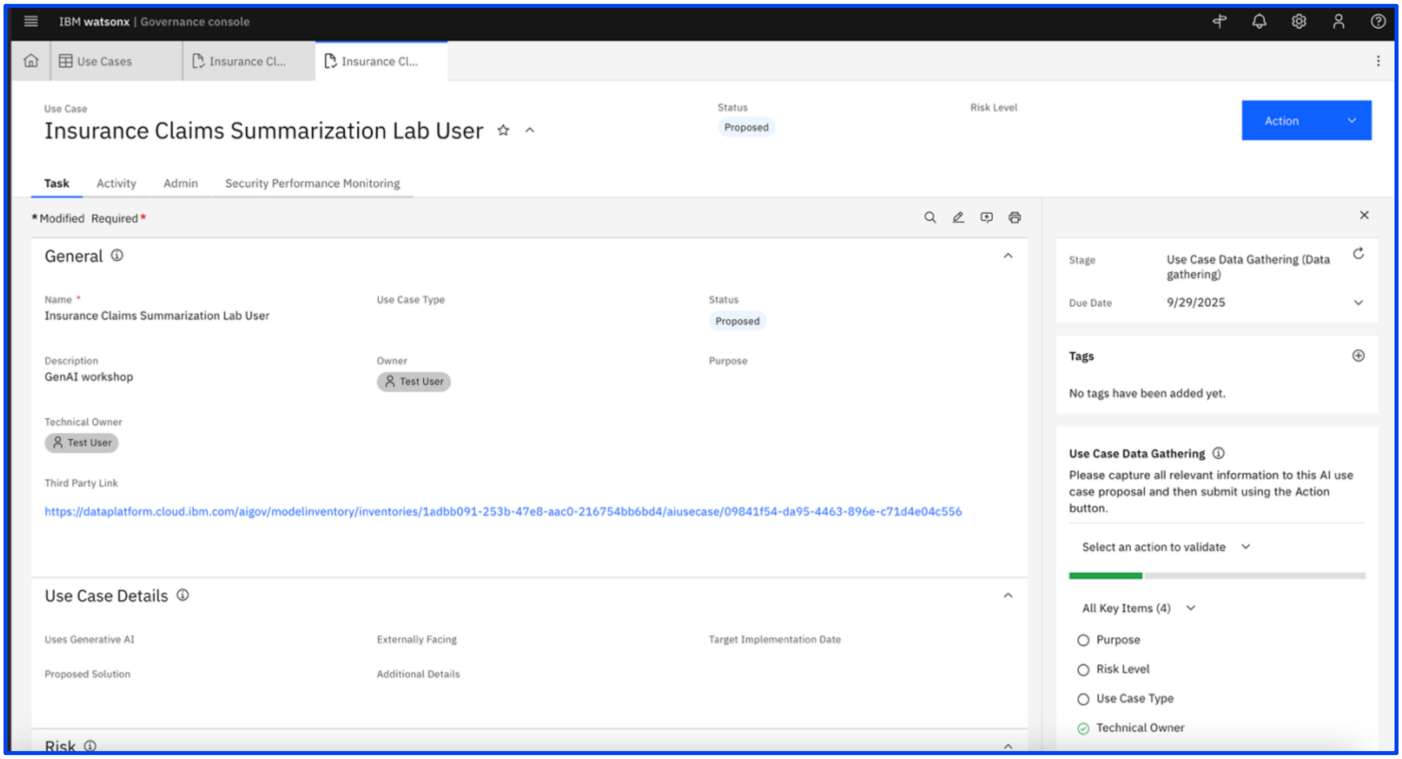
Note: you may also get an error indicating someone else has updated the Use Case/Object. If you do, please refresh the screen.

6. Whilst in Edit mode  set the **Technical Owner** to YOUR USER and hit 

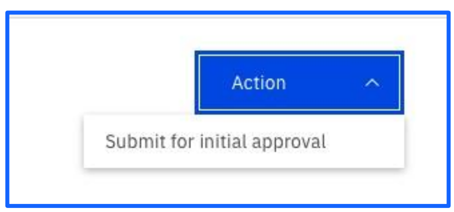
at the top right of the screen.


IBM watsonx | Governance console
Use Cases
Insurance Cl...
Insurance Cl...
Use Case
Insurance Claims Summarization Lab User
Status
Proposed
Risk Level
Action
Task
Activity
Admin
Security Performance Monitoring
*Modified Required *
General
Name *
Insurance Claims Summarization Lab User
Use Case Type
Status
Proposed
Description
GenAI workshop
Owner
Test User
Purpose
Technical Owner
Search users or groups
Recently Used
1 item found
Test Us... test1@haneenteam832724.testinator.c...
https://dataplatform.cloud.ibm.com/aigov/modelinventory/inventories/1adbb091-253b-47e8-aac0-216754bb6bd4/aiusecase/09841f54-da95-4463-896e-c71d4e04c556
Use Case Details
Uses Generative AI
Externally Facing
Target Implementation Date
Proposed Solution
Additional Details
Stage
Use Case Data Gathering (Data gathering)
Due Date
9/29/2025
Tags
No tags have been added yet.
Use Case Data Gathering
Please capture all relevant information to this AI use case proposal and then submit using the Action button.
Select an action to validate
All Key Items (4)
Purpose
Risk Level
Use Case Type
Technical Owner

7. The Use case should look like this:



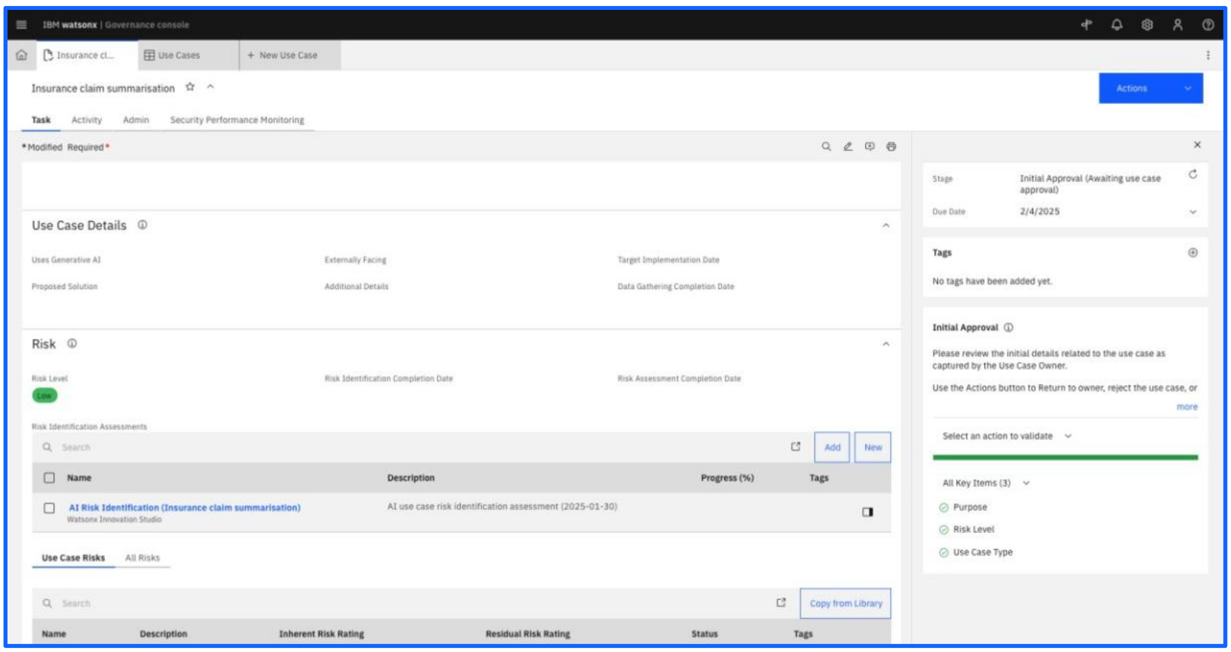
8. Now that you have successfully created the Use Case you can **Submit it for Initial Approval** by navigating to the **Actions** button at the Top right of the screen:



The Action button only works when the required information is captured. This is all driven by customisable Workflows which are set up to match an organisations business process'. This is outside the scope of the labs however if you wish to see this, please ask the table facilitator to demonstrate.

9. On the pop-up window select **Continue**:

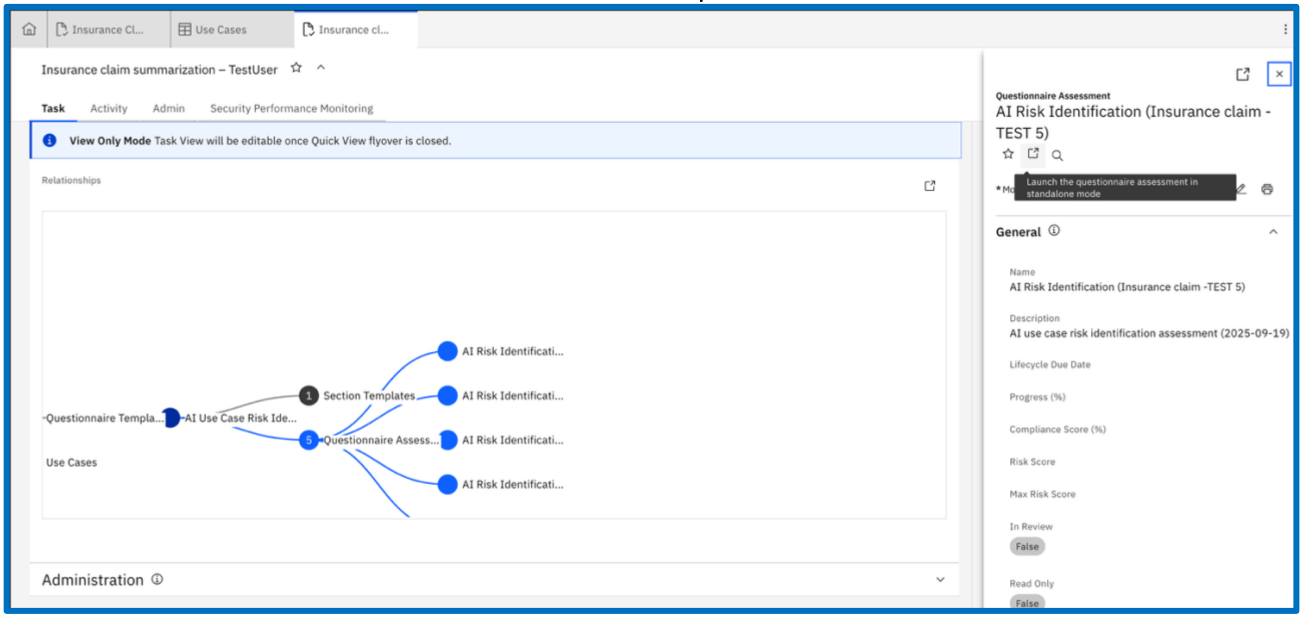
10. Once the Use case is Submitted for Approval, the Use Case gets updated and under the **Risk** section further down the screen, an **AI Risk Identification Assessment** has been automatically created:



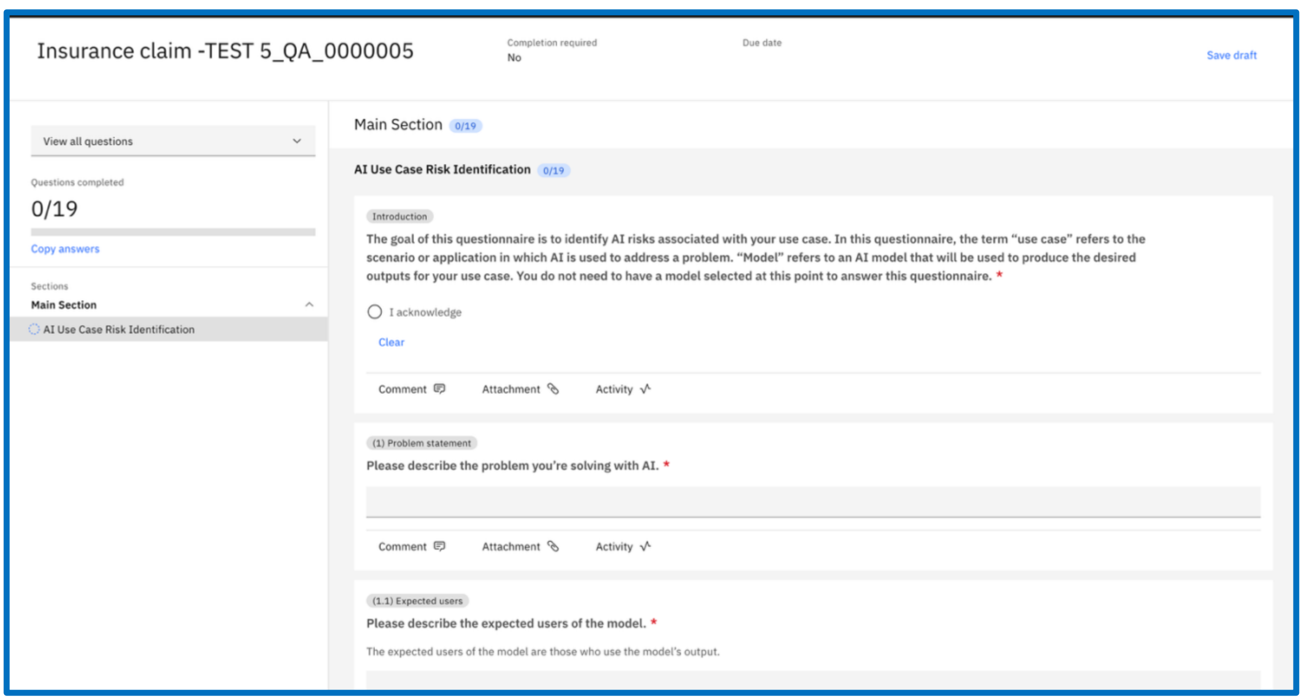
Risk Assessment

As part of the planning phase, we need to perform a risk assessment of our AI Use Case. The following steps show how an organisation can use a pre-build risk assessment to identify and manage the AI Risks associated with their Use Case.

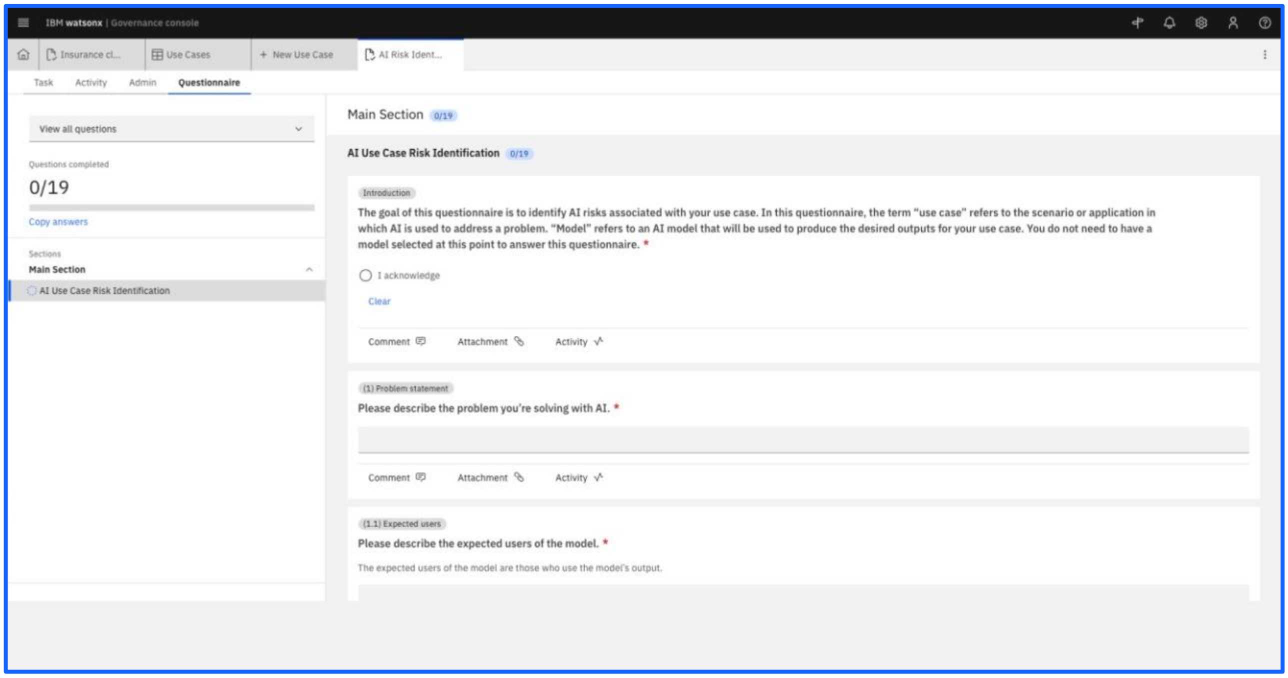
1. Select the **AI Risk Assessment** by scrolling down to **relationships** and selecting Questionnaire Assessment → AI Risk Identification and Launch the questionnaire.



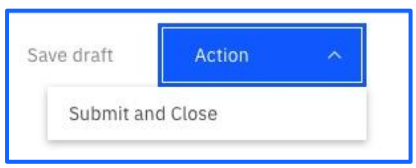
2. Go ahead by reading and answering the questions. For the purposes of our exercise, it is more important to understand the nature of the **Questions** themselves, therefore feel free to provide answers such as **“ABC”** & **“No”** :



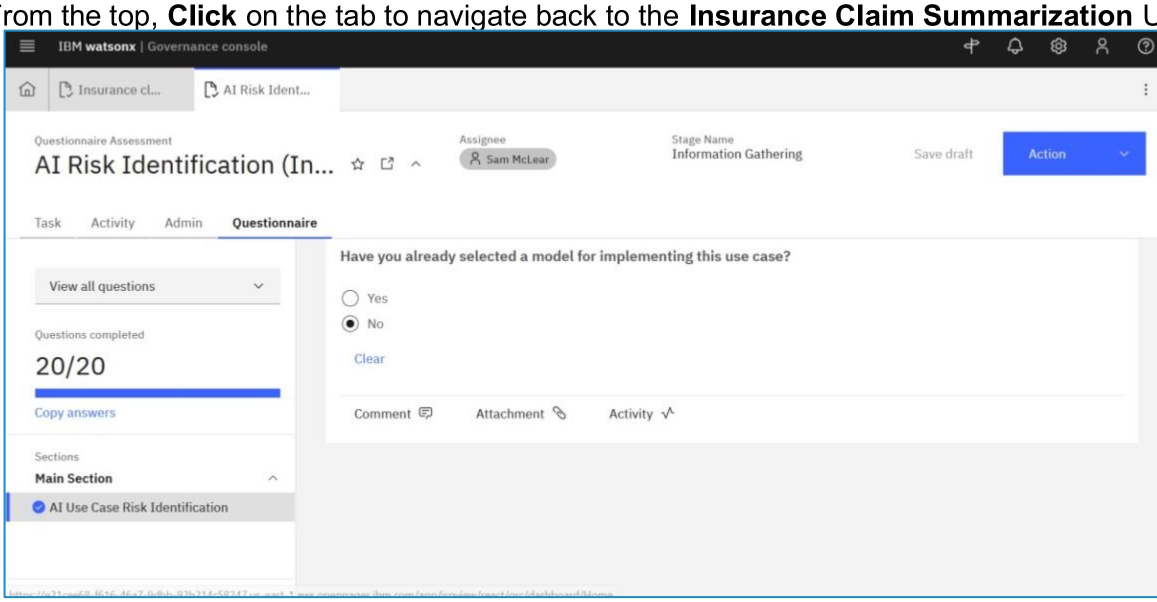
1. Select the **Assessment**:



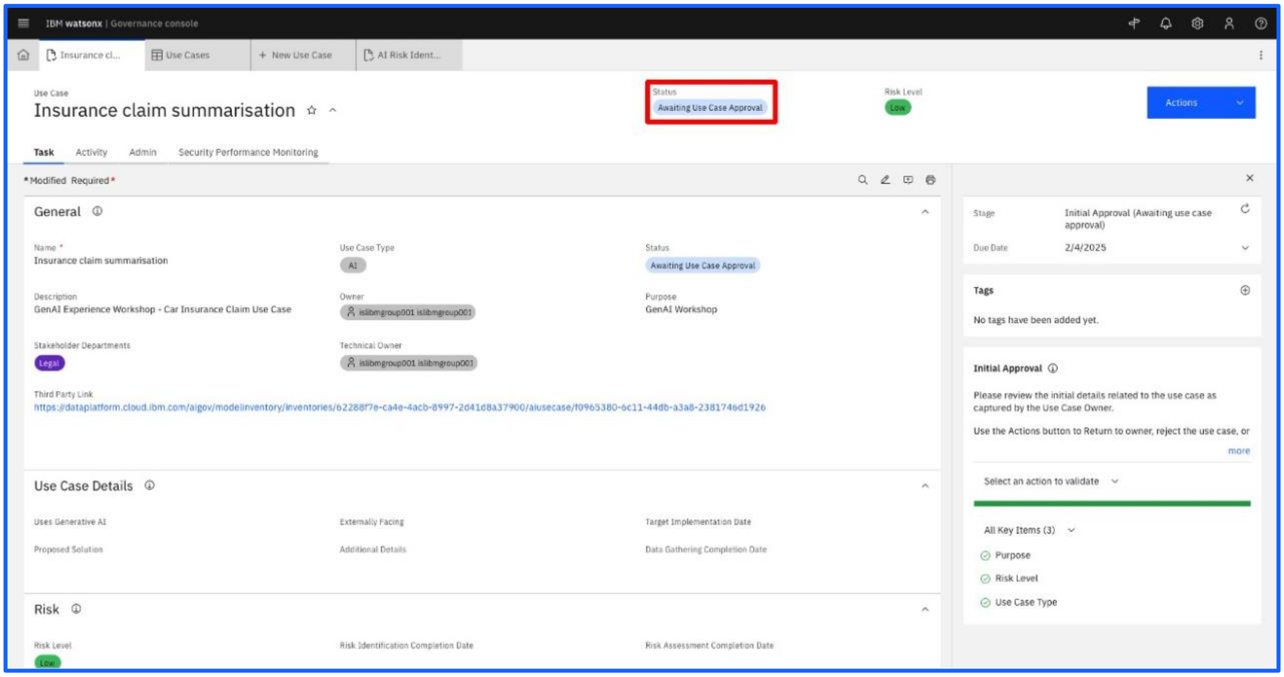
2. Once all **Questions** are answered, go ahead and select **Submit and Close** under the **Action** button:



3. From the top, **Click** on the tab to navigate back to the **Insurance Claim Summarization** Use Case:

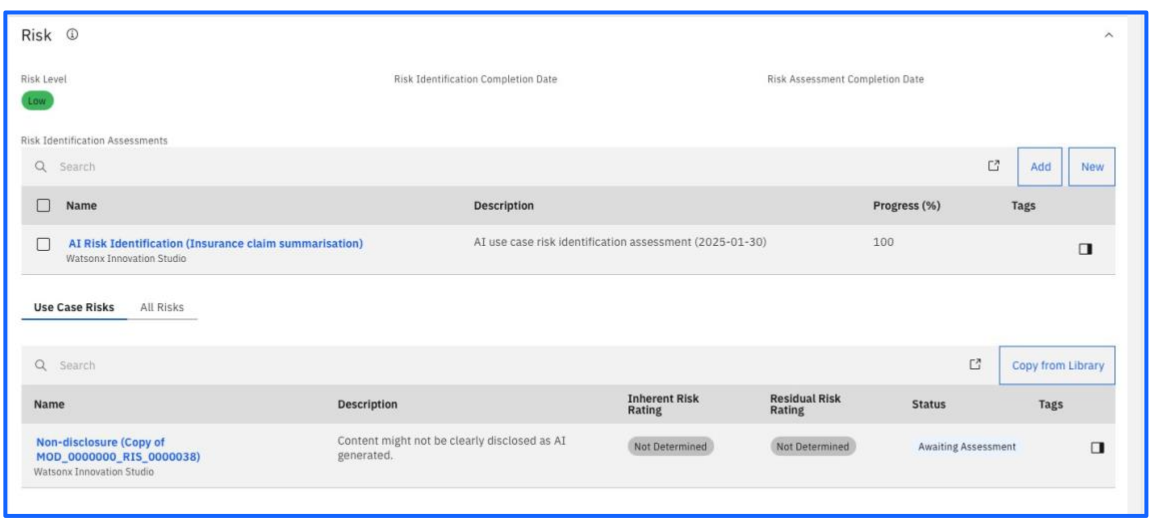


Note that the **Status** from **Proposed** has progressed to **Awaiting Use Case Approval**.

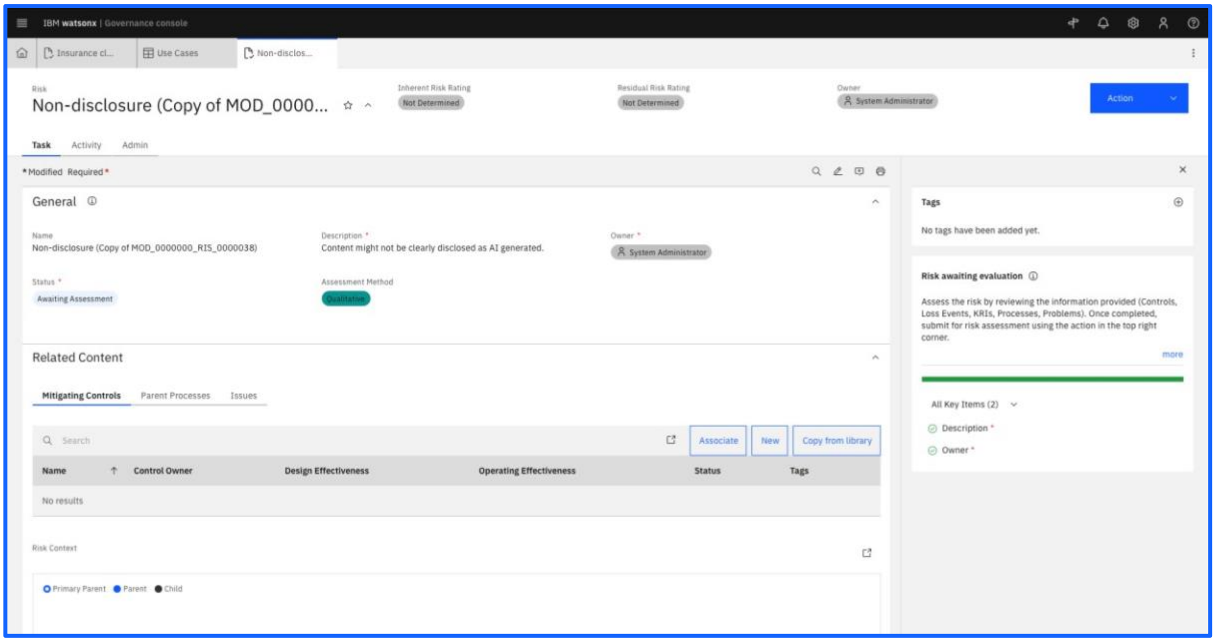


4. Additionally, further down in the **Risk** section, we can see one (or many depending on your answers) Use Case Risk identified.

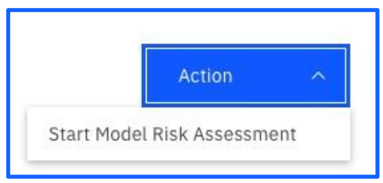
The risk(s) was populated in the Use Case based on our answers on the Questionnaire in the previous step. The risks are configured as a Response Action to the question and come from the AI Risk Atlas, which is an IBM Research produced set of risks associated with Gen AI and Traditional ML. It can be found here: [AI risk atlas - IBM Documentation]



5. Next we will assess the risk in the context of our Use Case**.** **Open** the Use Case Risk by selecting it:

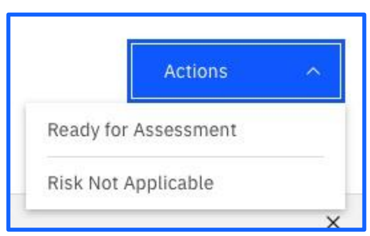


6. Go ahead and select **Start Model Risk Assessment** under the **Action** button:



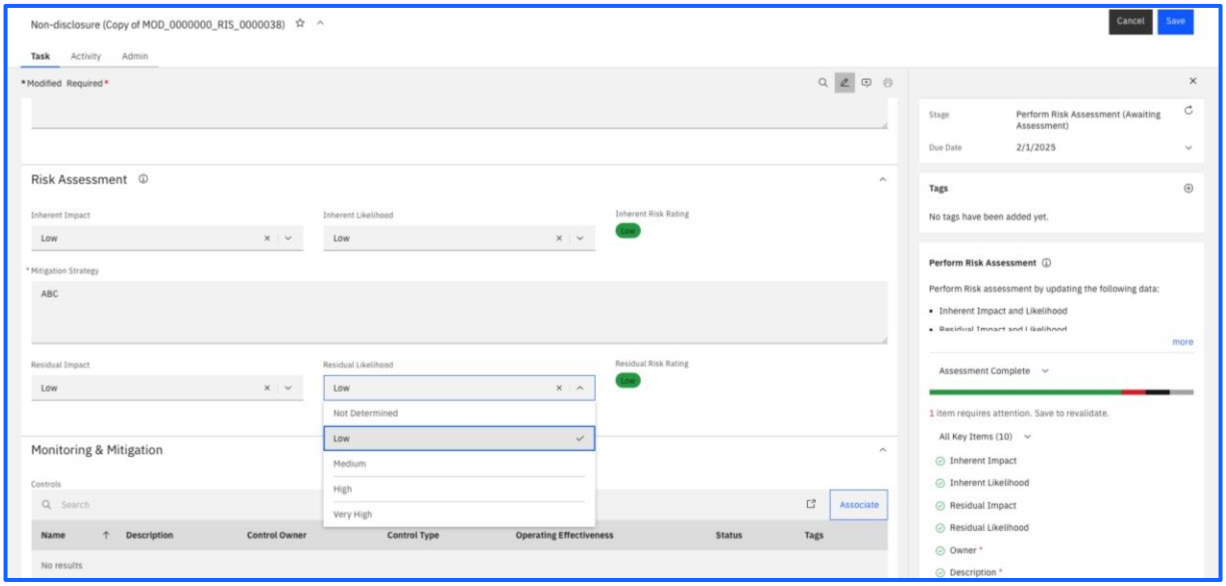
At this stage, the person responsible for the risk assessment (e.g. a Risk Owner) would assess the risk based on controls that exist in the environment. You can see there are options to associate any mitigation controls with the risk itself. This is outside the scope of this lab.

7. Once the **Risk Assessment** is initiated proceed by flagging it as **Ready for Assessment** under the **Action** button:



8. Switch to Edit mode and provide **Low** ratings for **Inherent Impact, Inherent Likelihood, Residual Impact, Residual Likelihood,** as well as a placeholder **"ABC"** for **Migration Strategy.**

The risk level is a subjective rating and calculates an automatic overall risk rating for the specific risk.



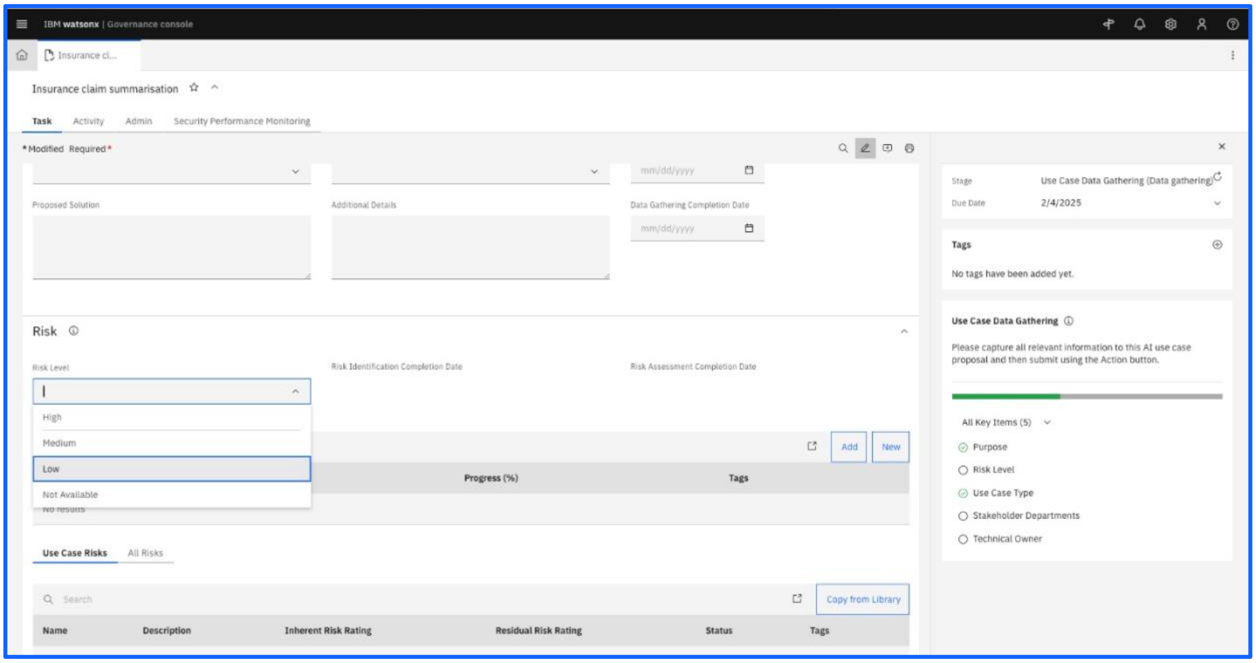
9. Finally, **Save** and proceed as **Assessment Complete.** Feel free to add a Workflow Comment with this.

10. Once the Risk Assessment is completed, navigate back to the Use Case.

You will see the Risk(s) associated with the Use Case and the fact it is “Accepted”. You will also a series of dashboards and risk ratings which can be configured to show the current risk profile of your Use Case. This is a useful visualisation.

It is now the responsibility of the Model Owner to provide a Risk Rating commensurate with the outcome of the Risk Assessment process. This is subjective based on the Risks visible in the Use Case.

11. To do this, scroll down to the **Risk** section, switch to Edit mode  and set the Risk Level to **Low**:



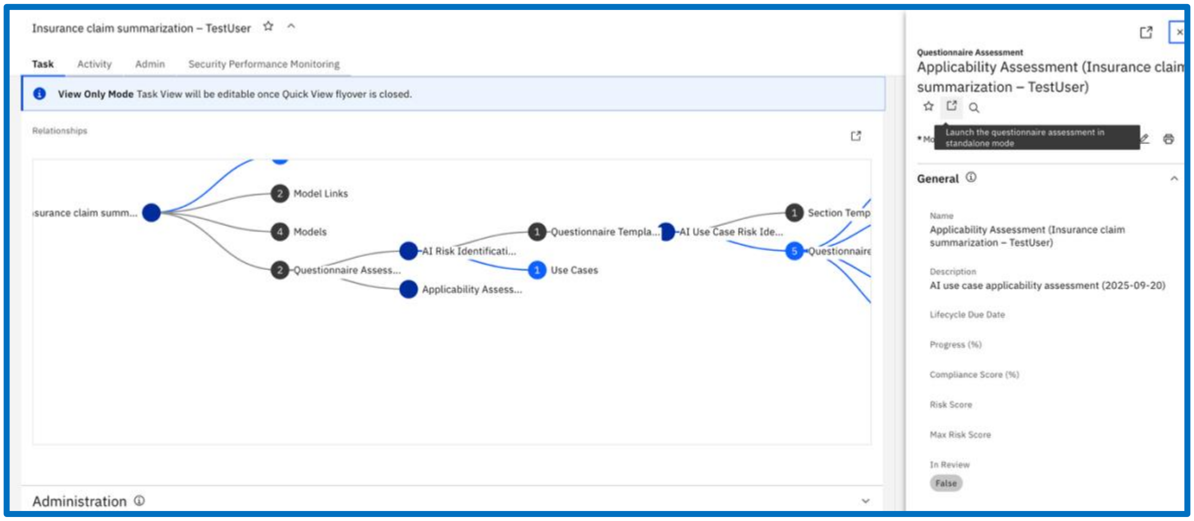
While this risk assessment comes out of the box, it is simple to create your own Risk Assessments based on your own set of organisational risks. This is outside the scope of this lab.

EU AI Act applicability Assessment

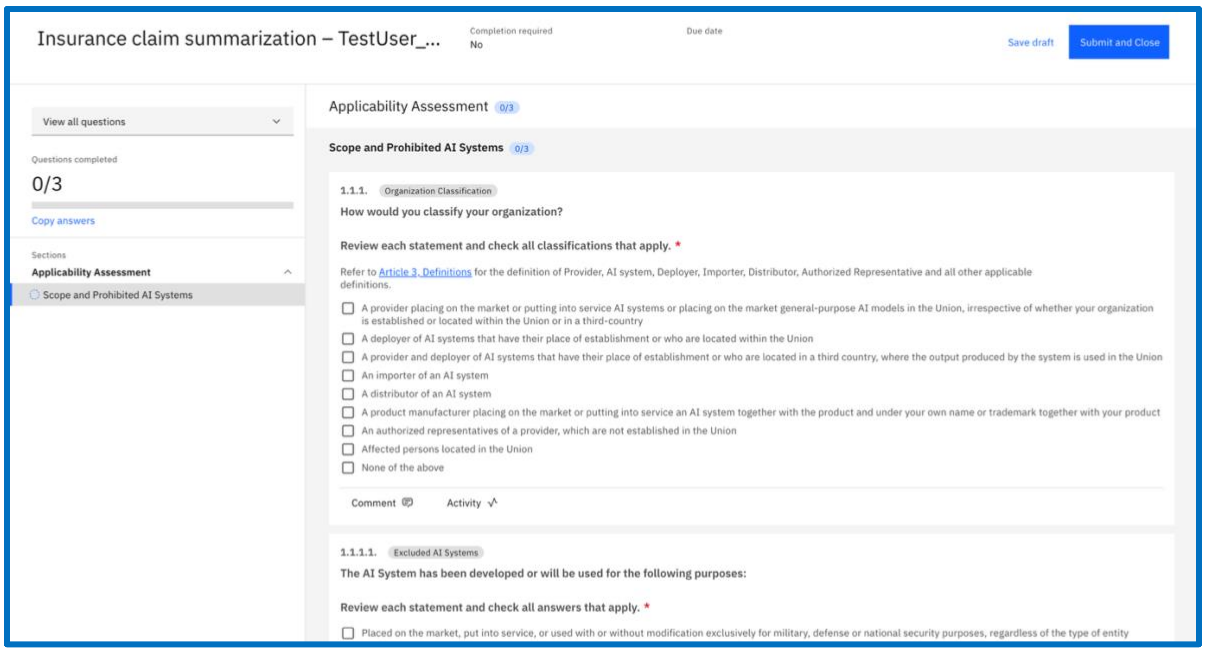
The EU AI Act is a regulation affecting AI Use Cases. It categorises Use Cases depending on the purpose of them from Prohibited, High, Medium and Out of Scope. It is important to understand the applicability of your AI Use Case to the regulation. as there are regulatory fines for non-compliance. If the result was prohibited, you would immediately know that this would never be a production use case hence saving yourself time developing this Use Case.

This pre-configured assessment helps an organisation know under which category of the act their AI Use Case falls under.

1. Complete the **Applicability Assessment** under **relationships** and selecting Questionnaire Assessment → Applicability Assessment and Launch the questionnaire.



2. Complete and submit the questionnaire assessment.

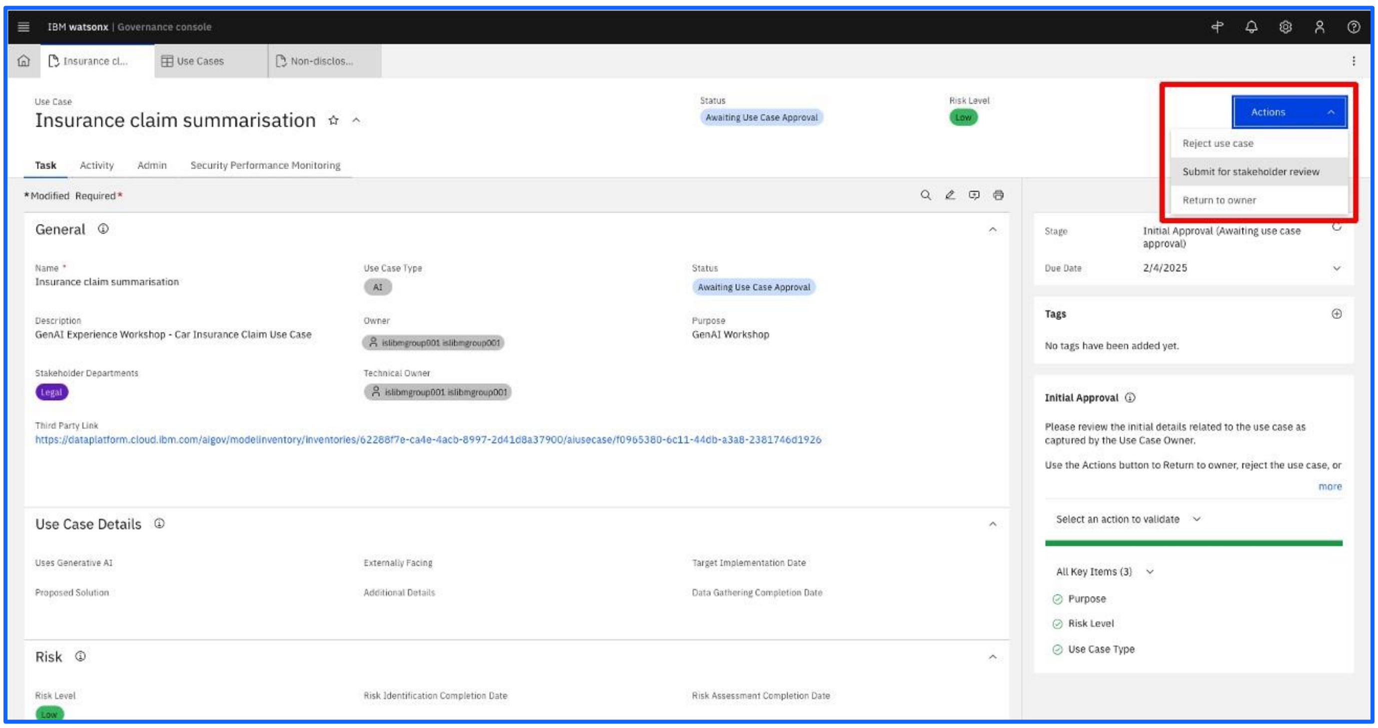


Submit for stakeholder review

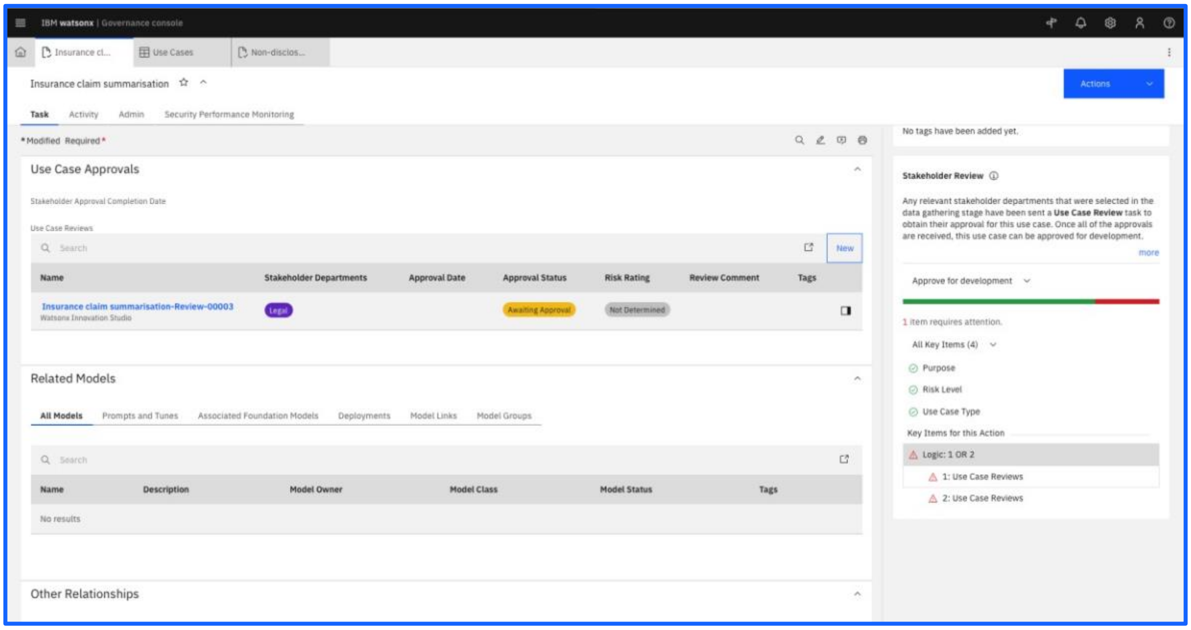
Assuming this wasn't prohibited, we are now ready to submit this for stakeholder review. The list of stakeholders indicate the different disciplines involved in the sign off of an AI Use Case. This is a final step before the AI Use Case is signed-off for development and ensures that the requisite stakeholders in an organisation have the opportunity to review all the info relating to the Use Case. If they are not happy with this they can reject it or request further information.

For this step, you will be taking on the persona of someone responsible from the legal team signing off on the Use Case.

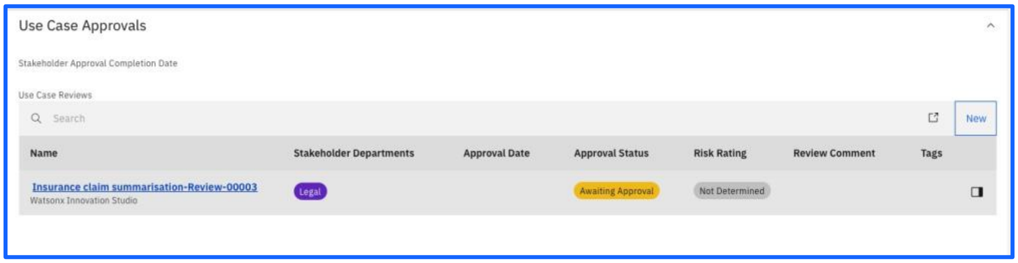
1. Switch back to the **Use Case** and proceed with **Submit for stakeholder review**:



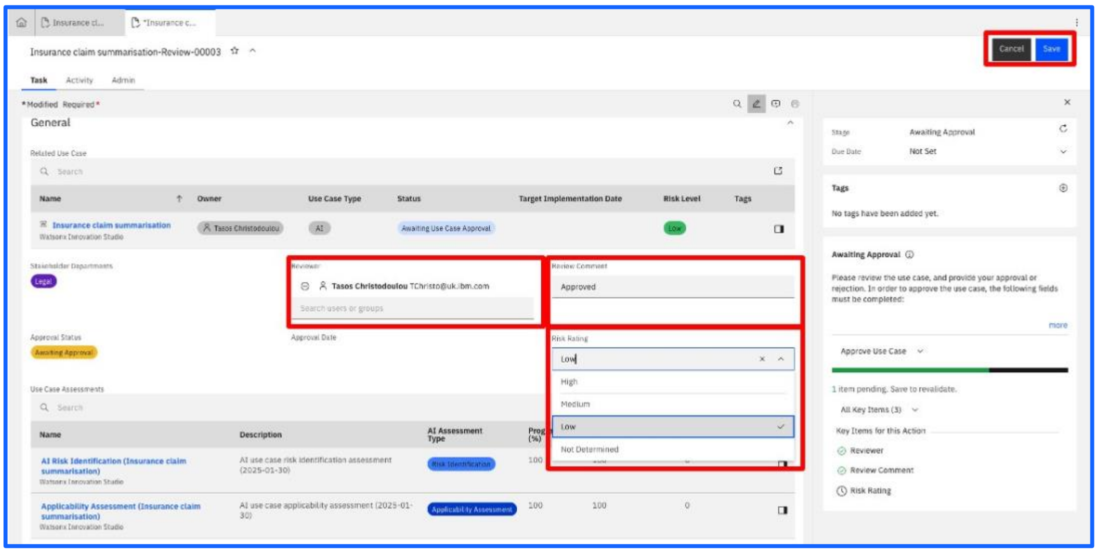
2. Once submitted, it is ready to be reviewed by one of the stakeholders, as it can be seen in the **Use Case Approvals** section:


IBM watsonx | Governance console
Insurance claim summarisation
Actions
Task
Activity
Admin
Security Performance Monitoring
Use Case Approvals
Stakeholder Approval Completion Date
Use Case Reviews
New
Name
Stakeholder Departments
Approval Date
Approval Status
Risk Rating
Review Comment
Tags
Insurance claim summarisation-Review-00003
Legal
Awaiting Approval
Not Determined
Related Models
All Models
Prompts and Tunes
Associated Foundation Models
Deployments
Model Links
Model Groups
Description
Model Owner
Model Class
Model Status
No results
Other Relationships
No tags have been added yet.
Stakeholder Review
Any relevant stakeholder departments that were selected in the data gathering stage have been sent a Use Case Review task to obtain their approval for this use case. Once all of the approvals are received, this use case can be approved for development.
more
Approve for development
1 item requires attention.
All Key Items (4)
Purpose
Risk Level
Use Case Type
Key Items for this Action
Logic: 1 OR 2
1: Use Case Reviews
2: Use Case Reviews

3. Open the **Use Case Review** by selecting it:



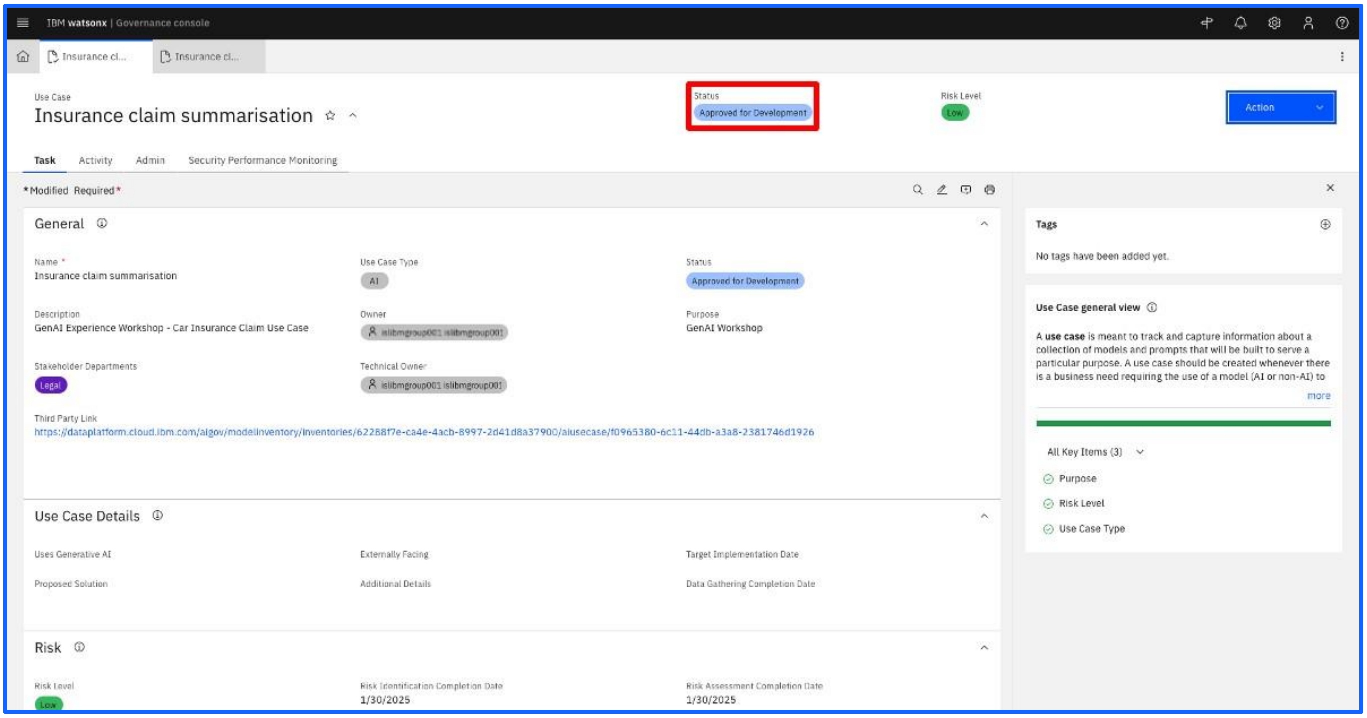
4. Set the **Reviewer** as your current username, for **Review Comment** type **"Approved",** set the **Risk Rating** as Low and press **Save**:



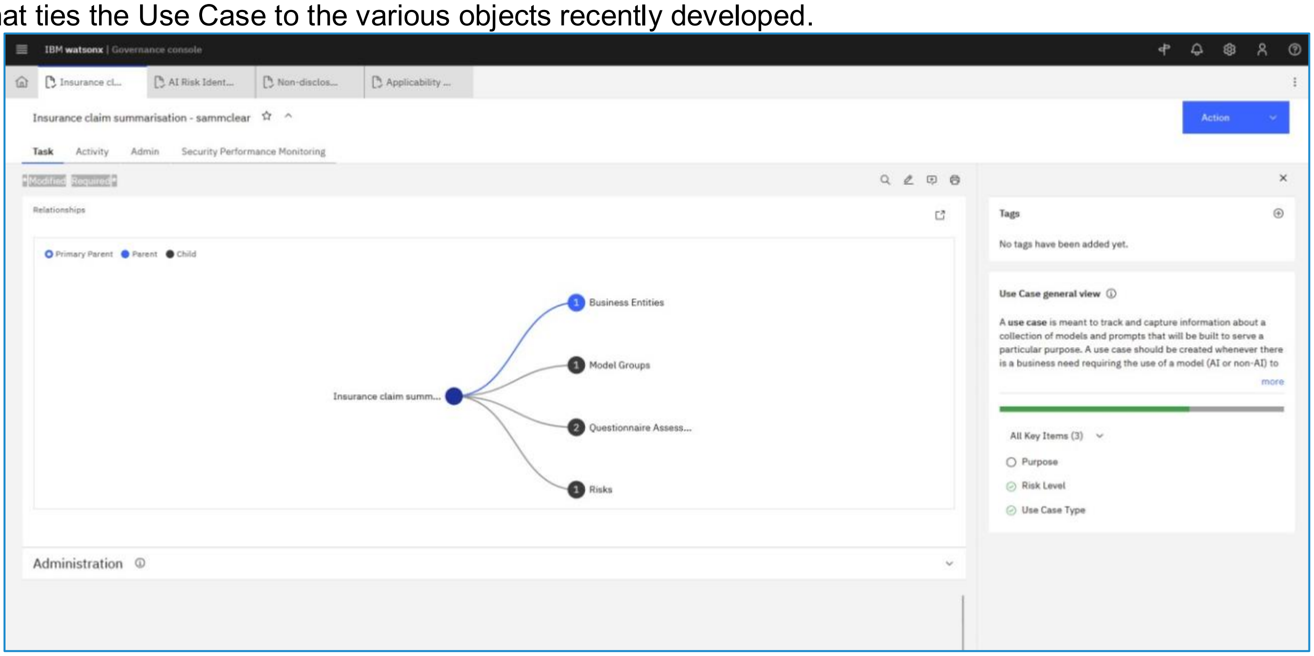
5. Once Saved, **Approve Use Case** under the Action Button:

6. Return to the Use Case. You can now **Approve for Development**:

7. The Use Case has progressed through the Plan phase and had the necessary approvals from key stakeholders including Risk Management and Legal. This Use Case is now “Approved for Development”.



8. Scroll down to the bottom of the Use Case to see that there has been a relationship diagram developed that ties the Use Case to the various objects recently developed.

9. 

Well done, you have successfully completed Phase 1 of the AI Use Case lifecycle, the Planning phase. You have identified:

- the purpose

- technical owners
- the Risk Level
- whether it is compliant to the EU AI Act - stakeholder approval.

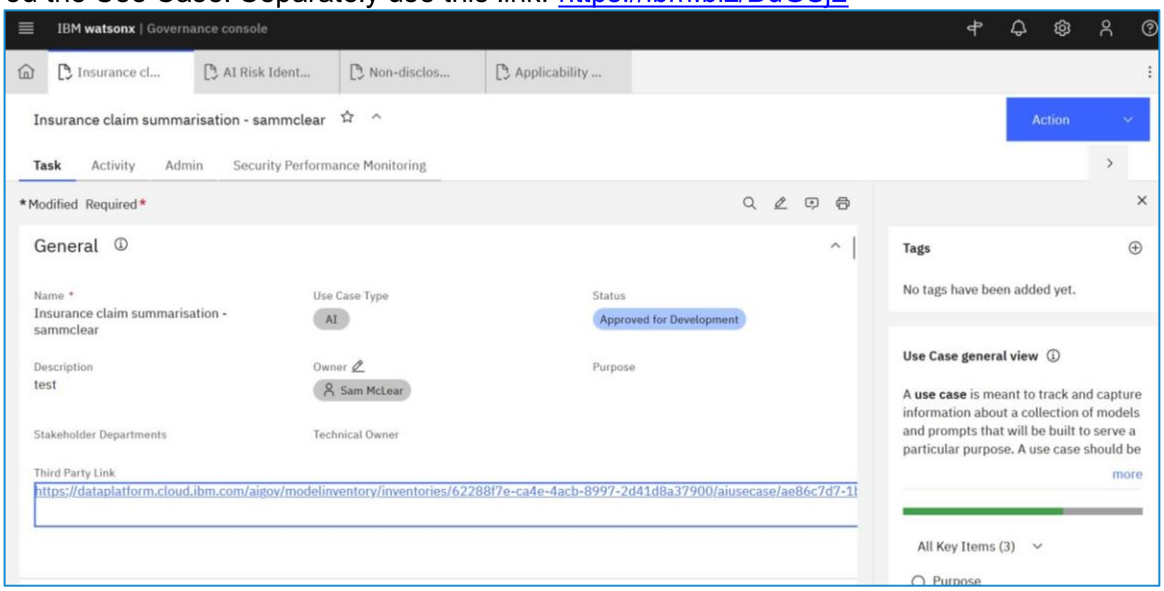
Please pause here before we explain the next Phase.

Phase 2: Design

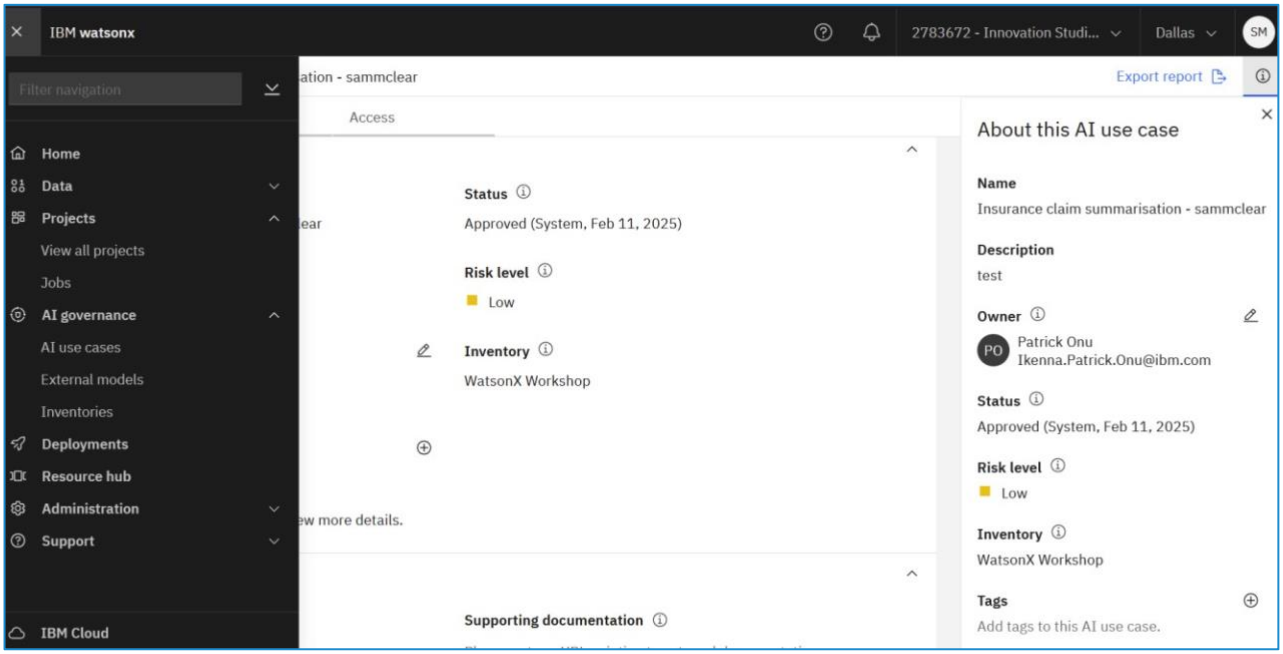
Create a Prompt Template

Let's begin creating our first prompt using Watsonx.AI. Watsonx.AI is IBM's AI studio that allows you to build, tune and deploy AI models.

You can click on the third party link within the Use Case to navigate to the Watsonx.AI platform which will show you the Use Case. Separately use this link: https://ibm.biz/BdGSj2



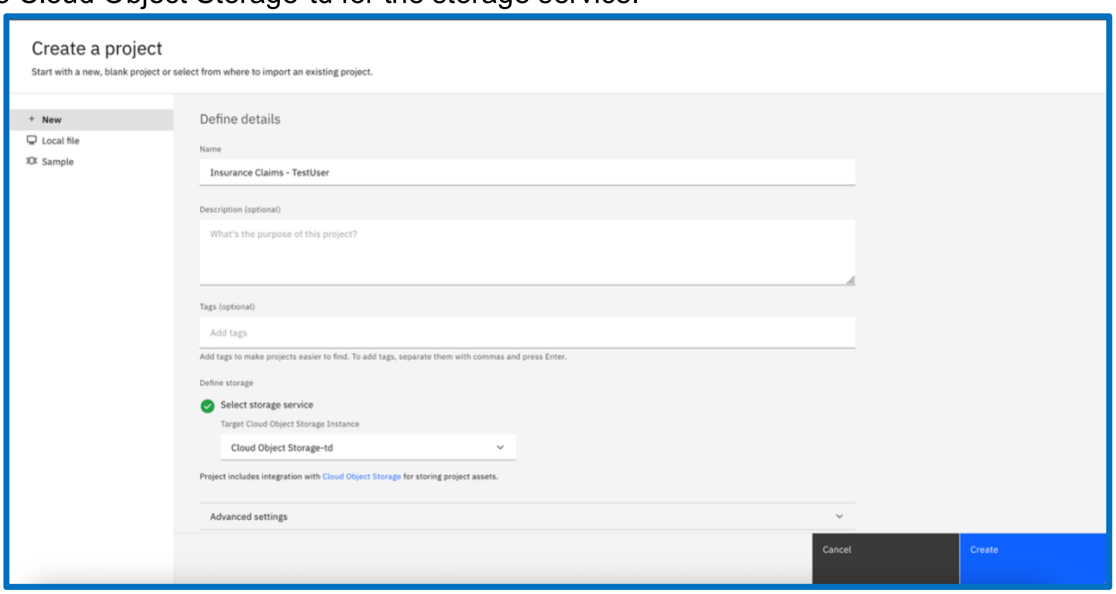
From IBM watsonx.AI, use the hamburger menu to navigate to your Team's project e.g. Team 1.



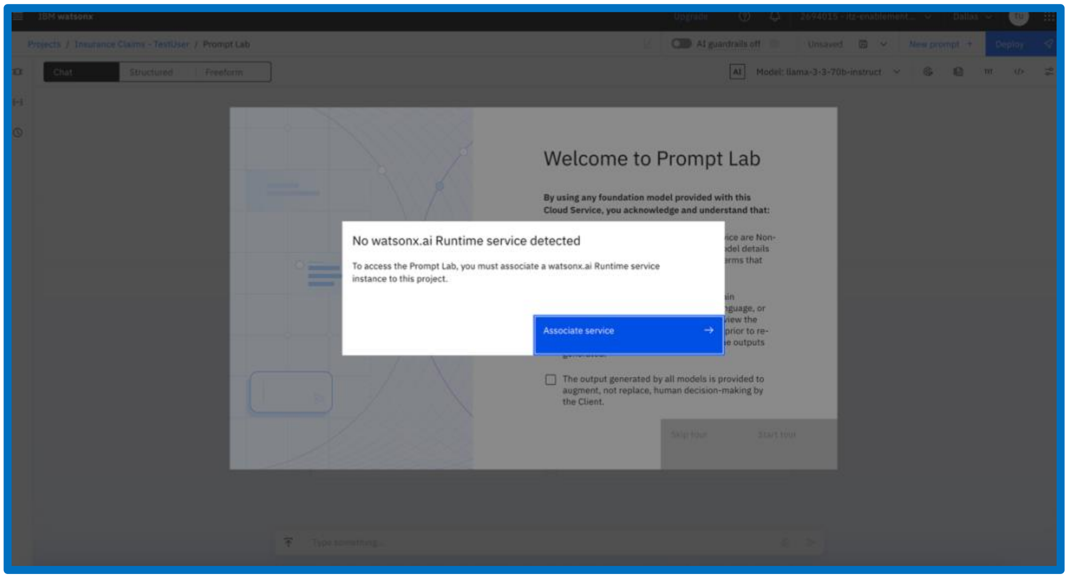
Projects are spaces to collaborate with others and build data and AI assets. You will see multiple options along the top which include information about the project, the assets associated with the project, any current jobs that are running and to allow the management of that asset.

Create a New Project, name the project “Insurance Claims – YOURNAME”

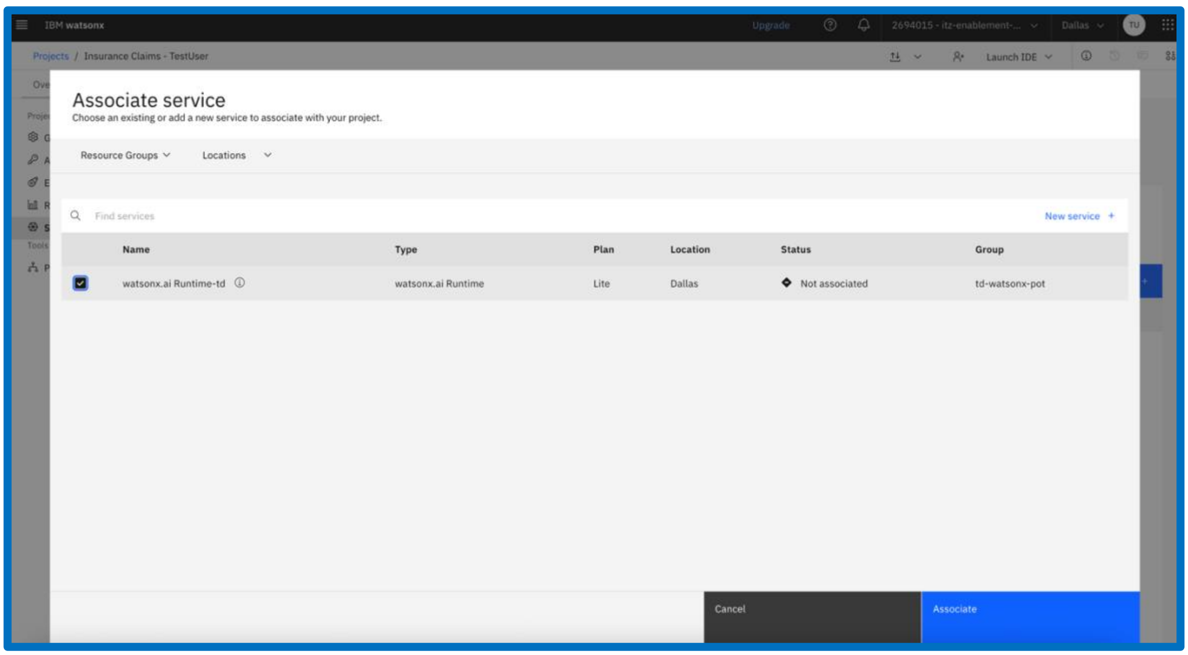
Select the Cloud Object Storage-td for the storage service.



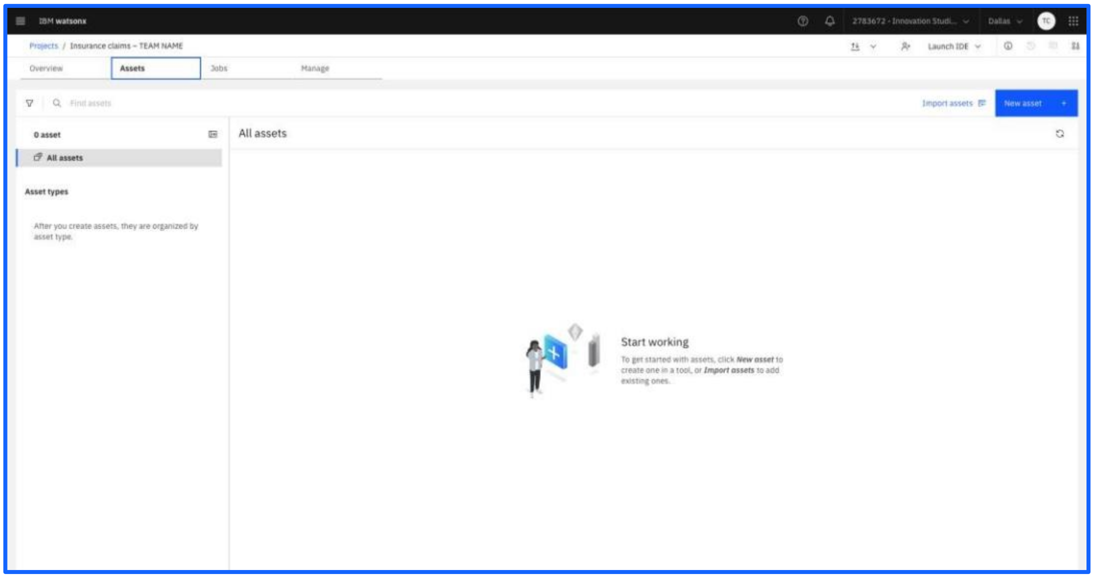
Head over to the **Assets** tab of the project home page. Once there, you will be prompted to associate a watsonx.ai Runtime service, select **Associate Service**.



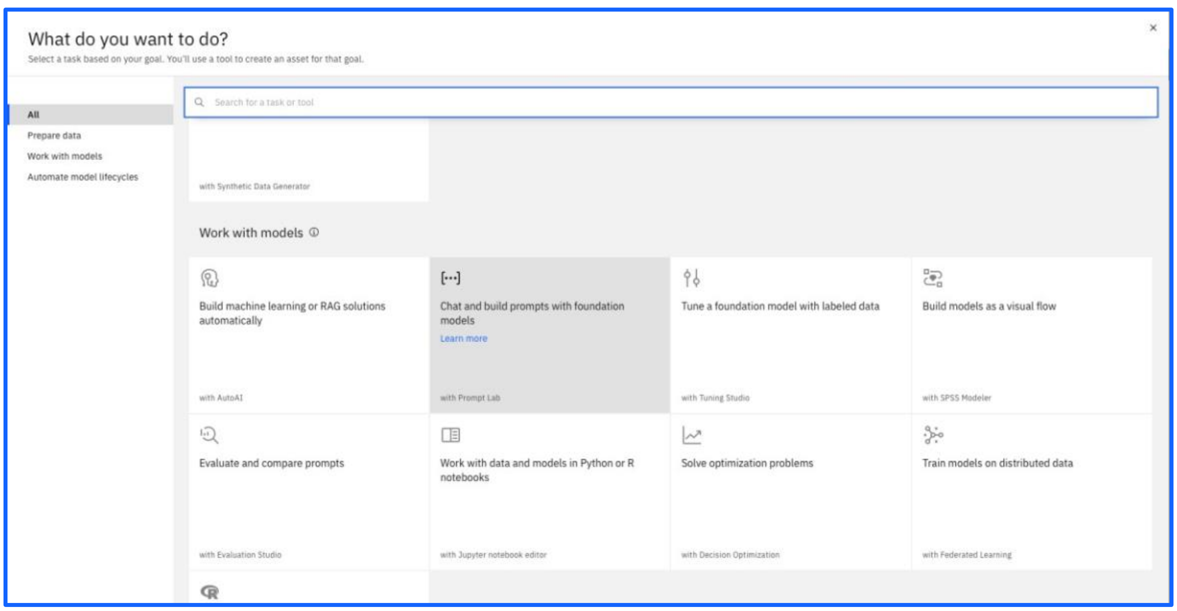
Select the “watsonx.ai Runtime-td” service and click **Associate**.



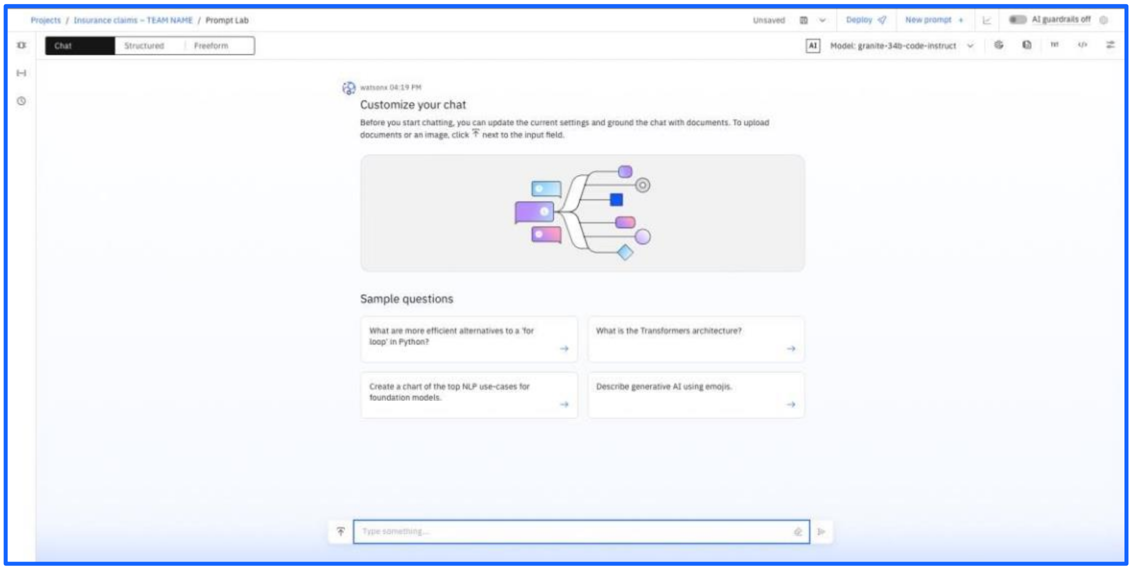
Head back over to the **Assets** tab of the project home page. Once there, click on **New Asset**. Note that your project will already include assets for a later part of the lab.



Now, if you scroll down on the menu that appears, you should see a card that says **Chat and build prompts with foundation models.**



You will be taken into the Watsonx.AI prompt lab. If there is an option to take a tour, skip it. You will be met with the prompt lab. This screen has numerous options on how you can configure your prompt to suit your needs.

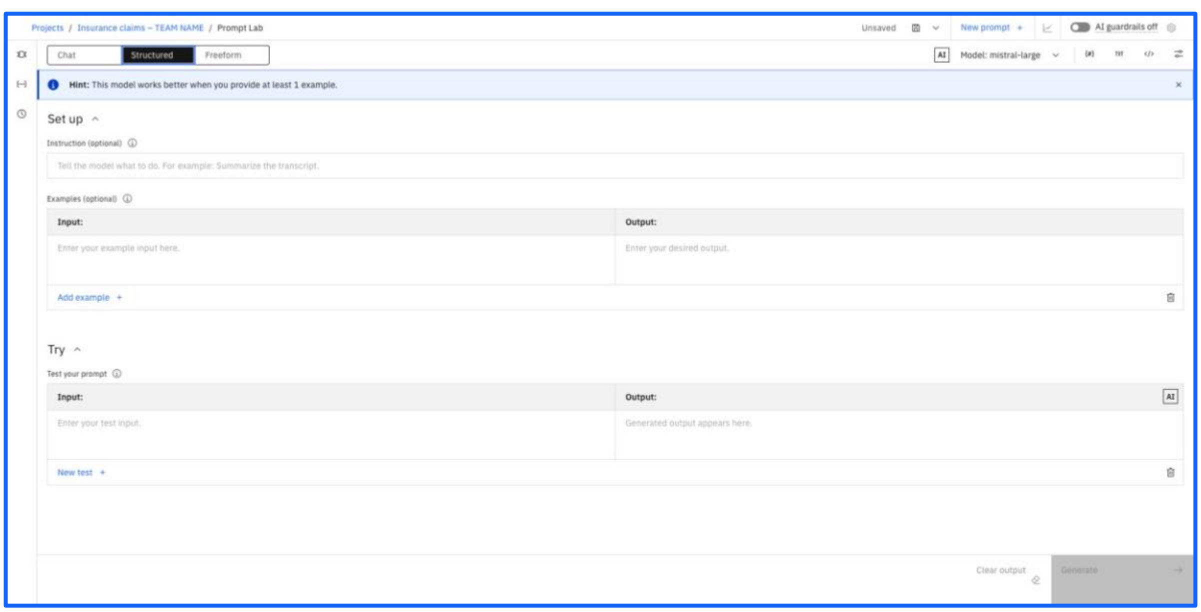


Right above that, you see the option to select “Chat”, "**Structured**" or "Freeform".

Structured prompts have easily identifiable sections to place examples, instructions, and prompt variables (which we will discuss later). Freeform prompts have a looser and more flexible structure. With Chat mode, you can chat with the foundation model to see how the model handles dialog or question-answering tasks.

This lab will walk through utilising the **Structured** prompt functionality, however this is possible to do with freeform as well, so choose whichever option is the most comfortable for you!

You will notice that there is a box labelled **Instruction** which is where we will be placing the bulk of our prompt to be evaluated later.



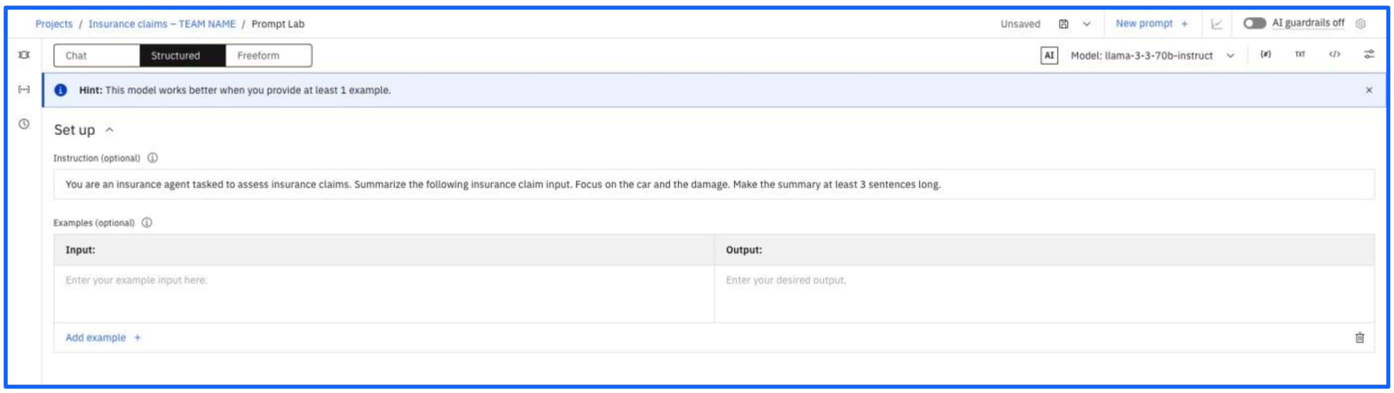
You will also see an icon with a graph at the top right , which is the button to run an evaluation that we will discuss later. You will be able to see the status of your work at the top right as well, indicating whether your prompt session work is Saved or Unsaved.

Select the “llama-3-3-70b-instruct” model from the menu on the top right.

Inside of the Instruction box, enter your first prompt that we will be evaluating throughout this lab. For now, enter the prompt shown in the screen below. This is the instruction given to the model.

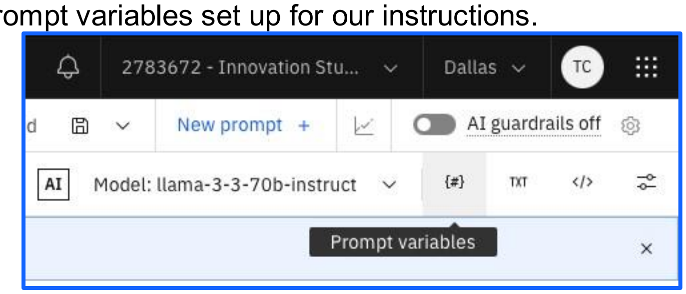
Copy & Paste from in between the quotation marks:

“You are an insurance agent tasked to assess insurance claims. Summarise the following insurance claim input. Focus on the car and the damage. Make the summary at least 3 sentences long.”

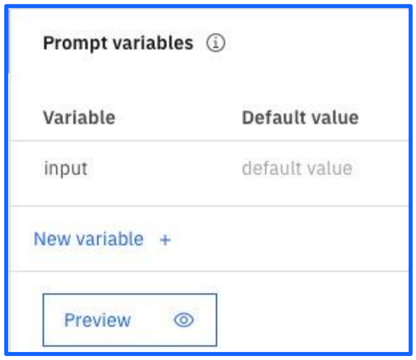


Prompt Variables

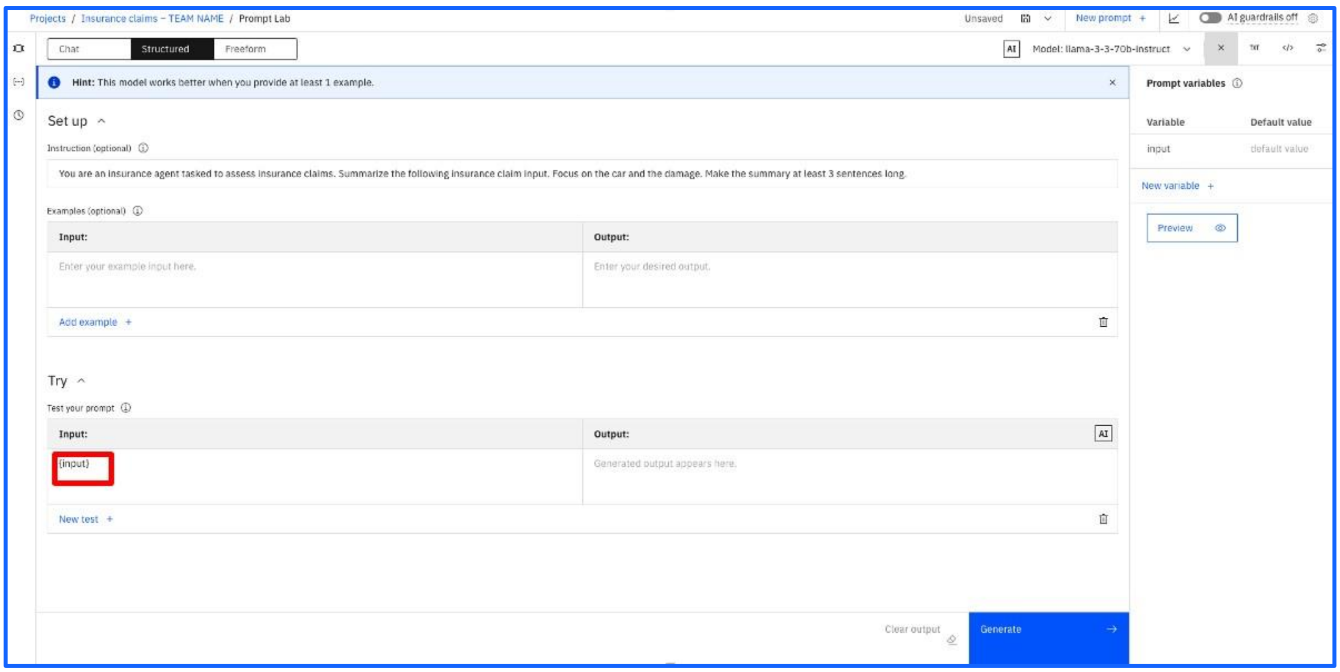
Now, we need to ensure that we are able to test our prompt against validation and test data. To do so, we need to ensure we have prompt variables set up for our instructions.



Once you have opened up this sidebar, you will see two empty boxes pop up, with **Variable** and **Default Value** able to be filled out. Use "input" as the variable name and leave the default value empty.



By scrolling down your page, you will notice an area on the screen titled **Try**. Inside of this, you want to put a variable titled "{input}". This variable will be replaced by input insurance claims.



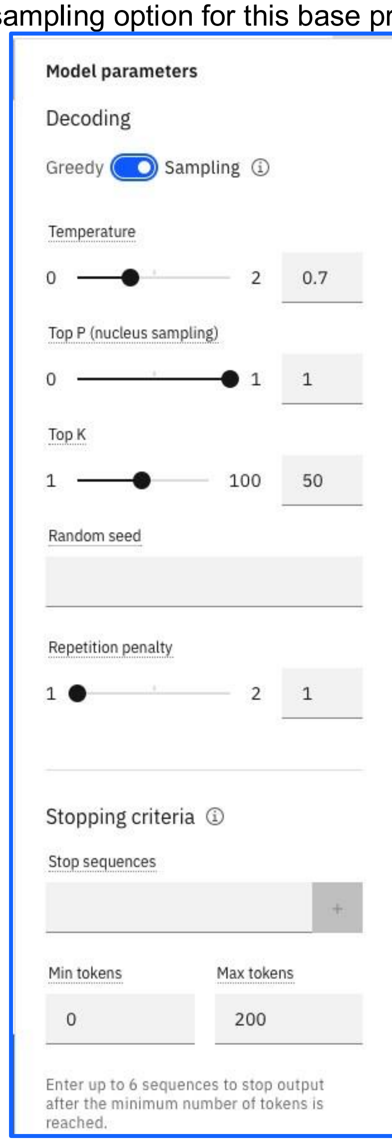
Model Parameters

Next to where you clicked prompt variables, you should see an option that says **Model parameters**. Click on that next. We will not be making any serious changes here just yet but mostly giving a tour of this section so you can make modifications later.

Most importantly, when you select the button highlighted below (top right), you will see a toggle for **Greedy** or **Sampling**.

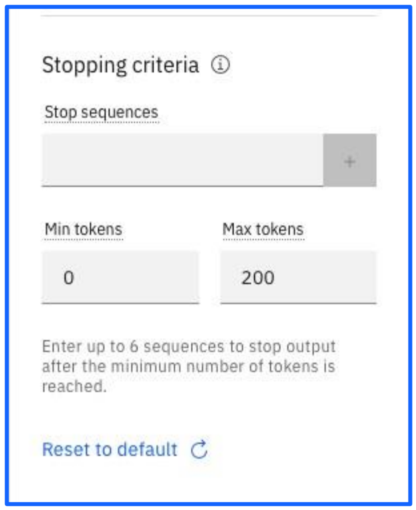
Selecting greedy will allow your model results to be reproduceable across experiments, while selection sampling will give you more variability.

For our task, we will be choosing the sampling option for this base prompt.



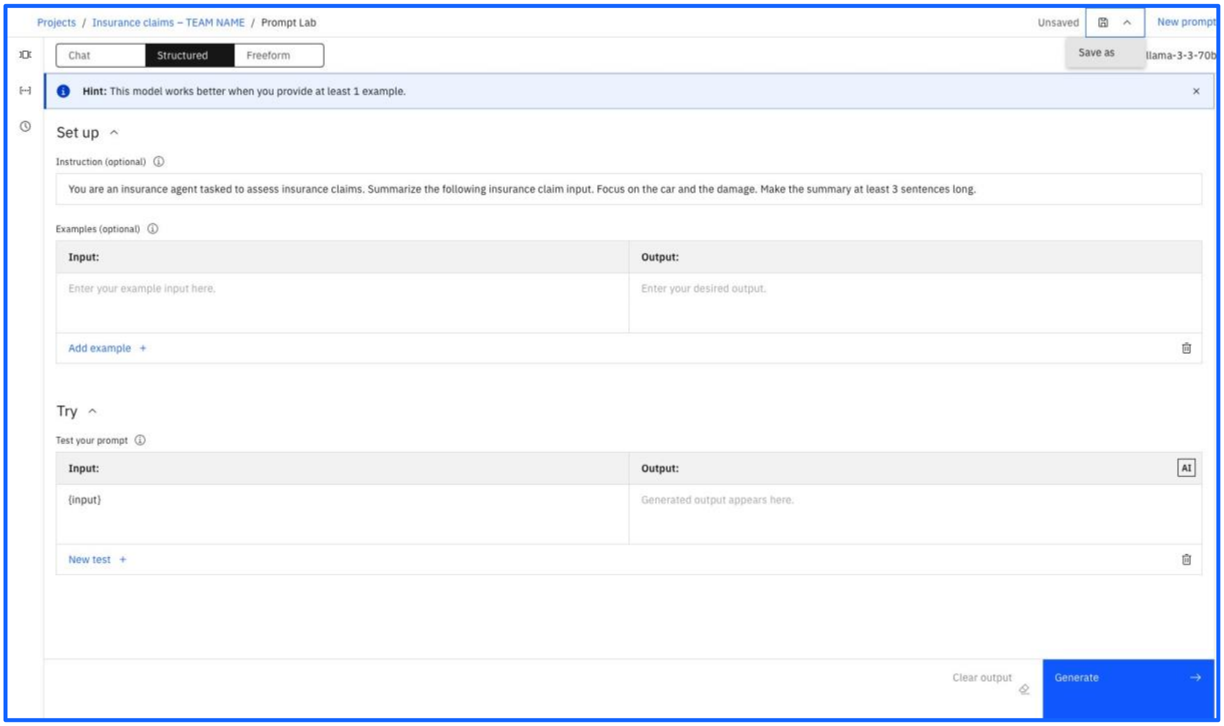
Now that you have chosen sampling, you will see several new options pop up that are specific to the sampling decoding method. **Temperature** is the most important factor here, where setting a higher temperature will give higher variation and more unique outputs. Hover over the info icons to gain more information on **Top P** and **Top K** methods as well.

Lastly, scroll all the way down the **Model parameters** sidebar to find the **Stopping criteria** as well as **Token limits**. For LLMs, tokens ~ words, so we want to set that higher (e.g. 200) to get more detail in our summaries of the insurance claims.

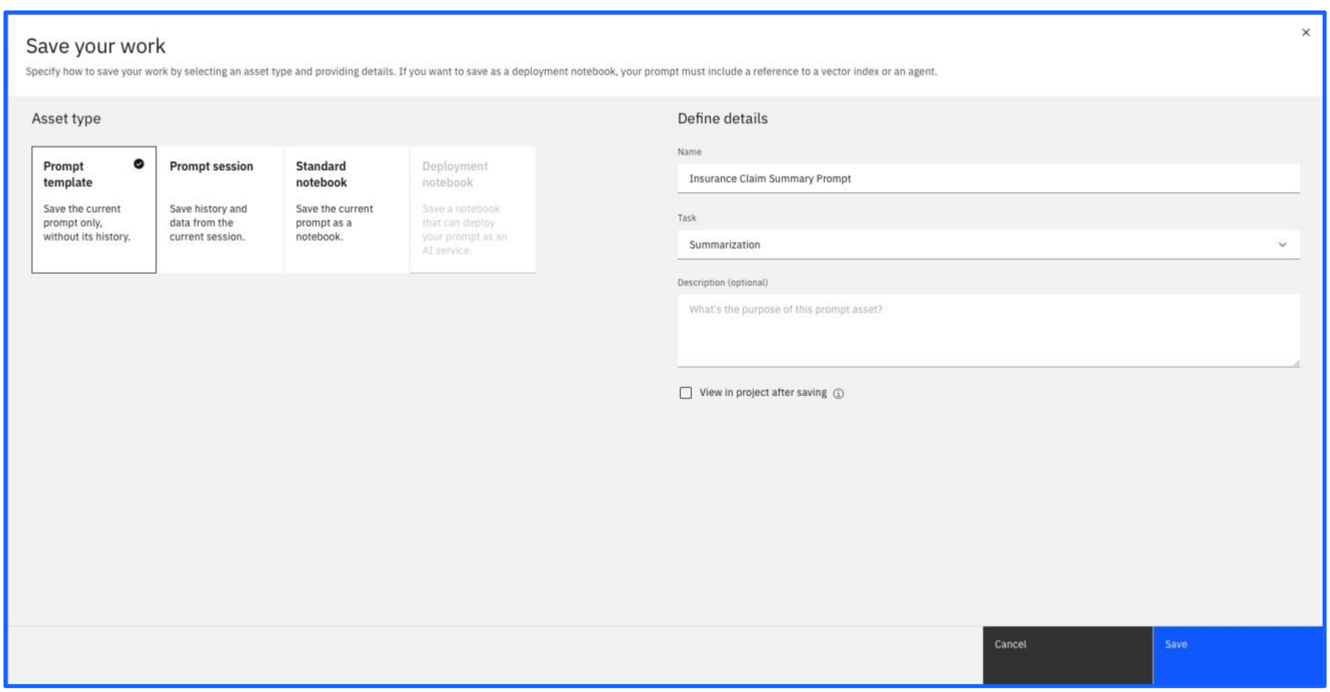


Saving Prompt

Once you are all finished, click the **Save** icon dropdown in the top right, and then click **Save as**. This will take you to the following screen, where you will select your asset type, prompt template name, and task.



Make sure to select the **Prompt template** option, then fill out a name in a similar fashion shown in the picture. From there, select **Summarization** for your task, and make sure to select **View in project after saving**.



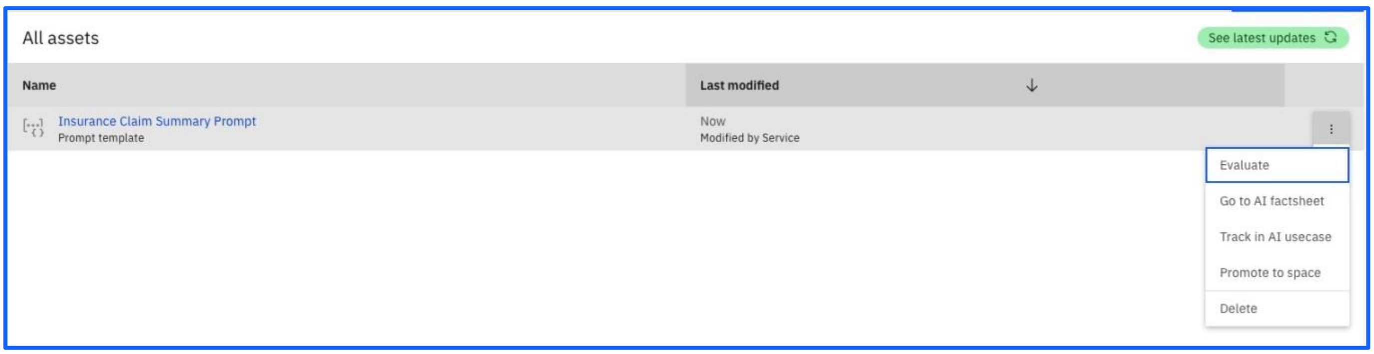
You have now successfully designed your first prompt template.

Phase 3: Development

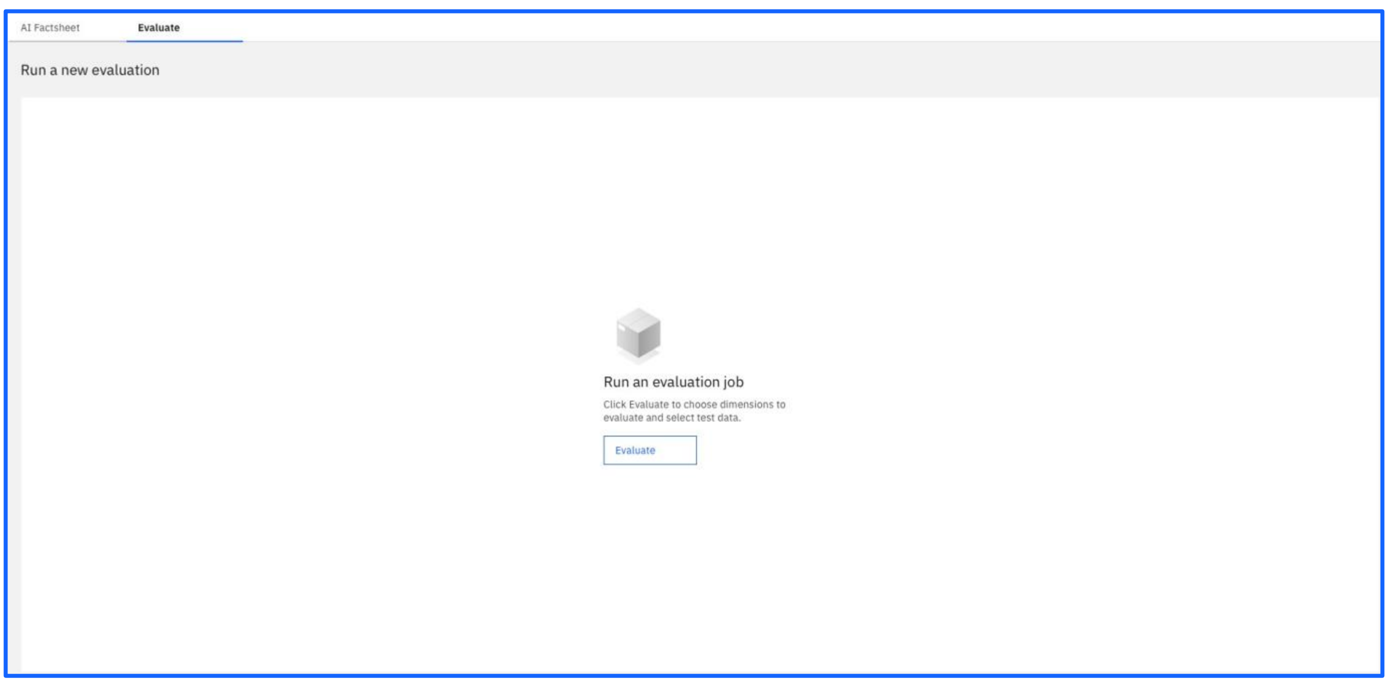
Evaluating your prompt template

The goal of this part of the lab is to iterate on our first prompt that we have created, and work to achieve better results so that we can deploy our prompt and eventually bring it to production.

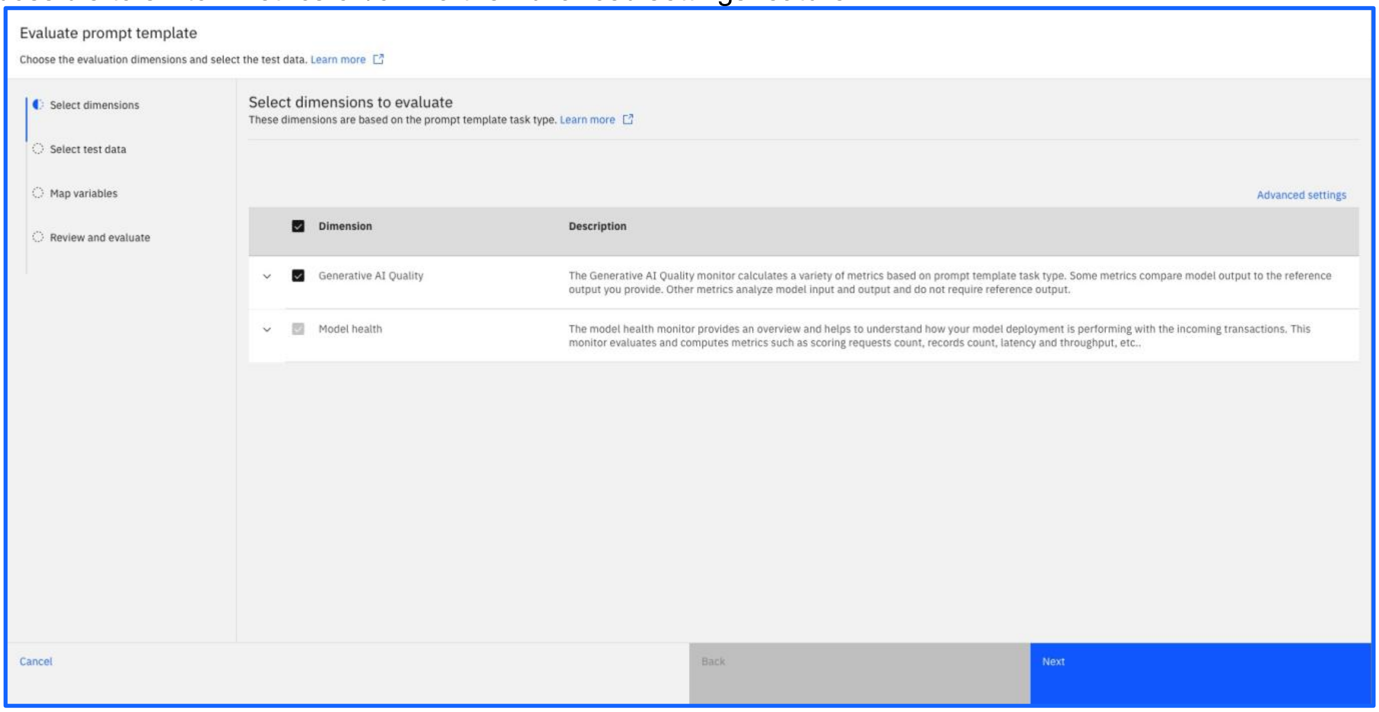
Now you should see your finished prompt template on the screen when you are taken back to the **Assets** page. Verify that you have completed the previous steps correctly by clicking the 3 dots to the right of the prompt and choose **Evaluate**. We are going to be testing out our new prompt!



You should now see a screen pop up with an evaluate button in the middle. Select that button.

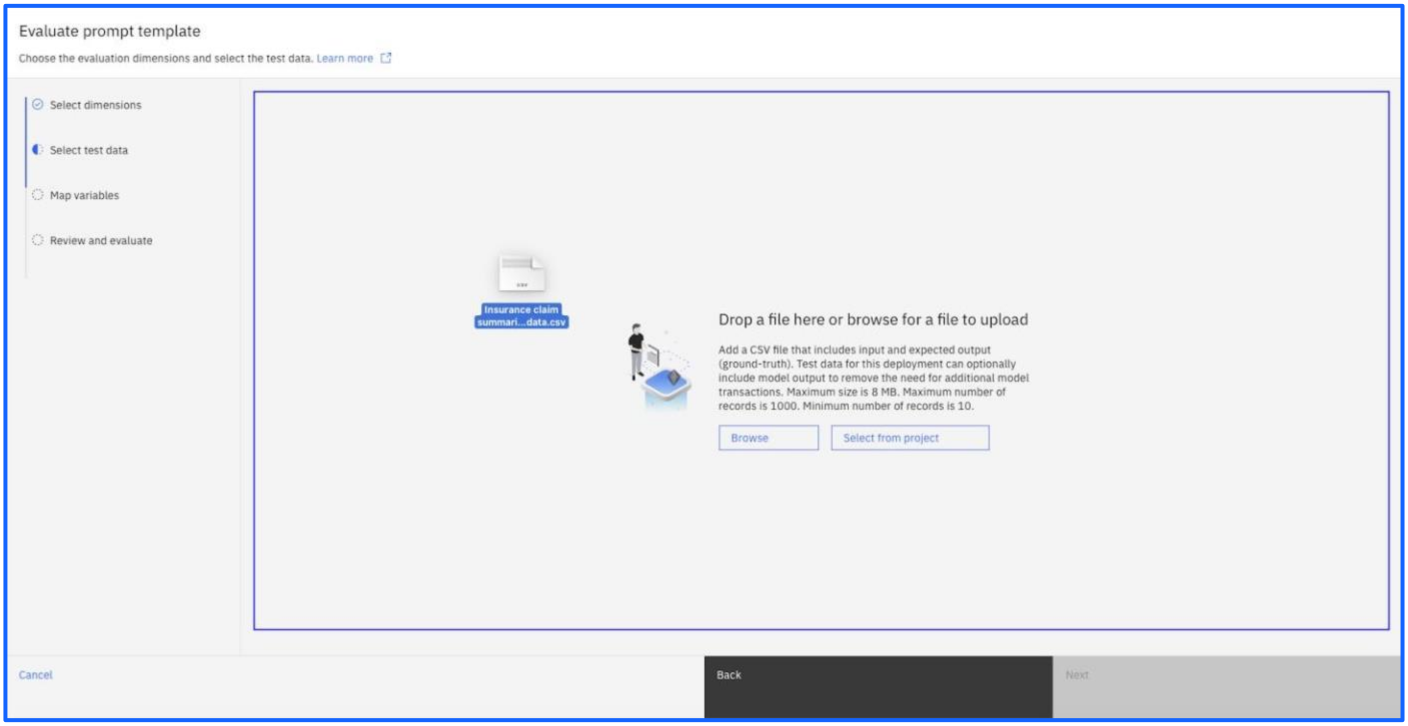


Now you will see a screen asking you which dimensions to evaluate. At this point, we are only able to evaluate GenAI quality (model health is enabled by default). By clicking on the arrow next to the Generative AI Quality, you can see which metrics are being calculated and what the definition of those are. It is also possible to switch metrics on/off via the Advanced settings feature.



Now it's time to choose our test data. For this tutorial, we will be using the **Insurance claim summarisation validation data.csv** from your lab's Data folder.

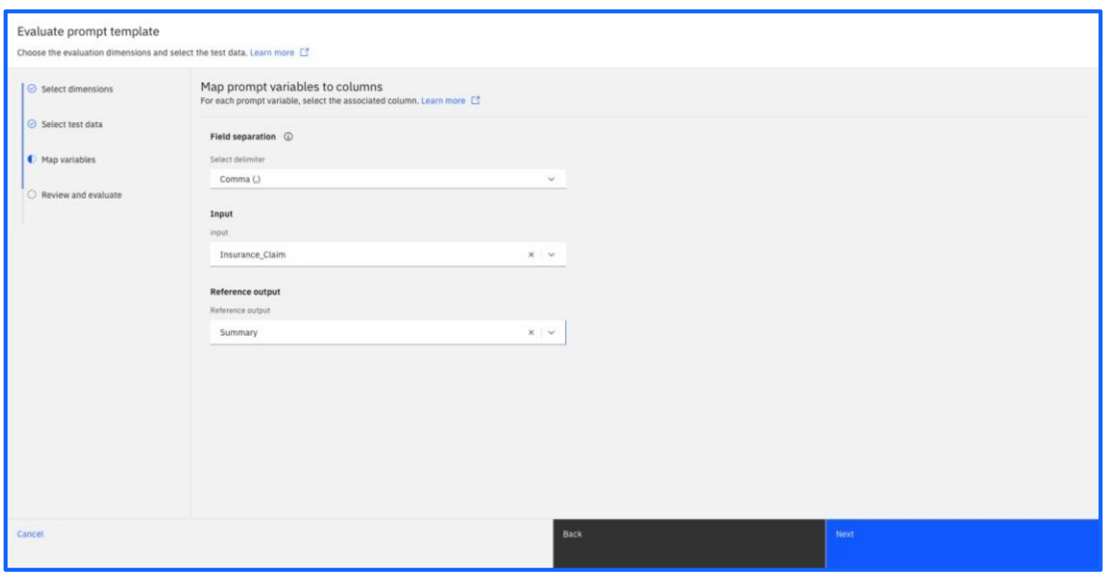
Choose Select from Project and locate the file.



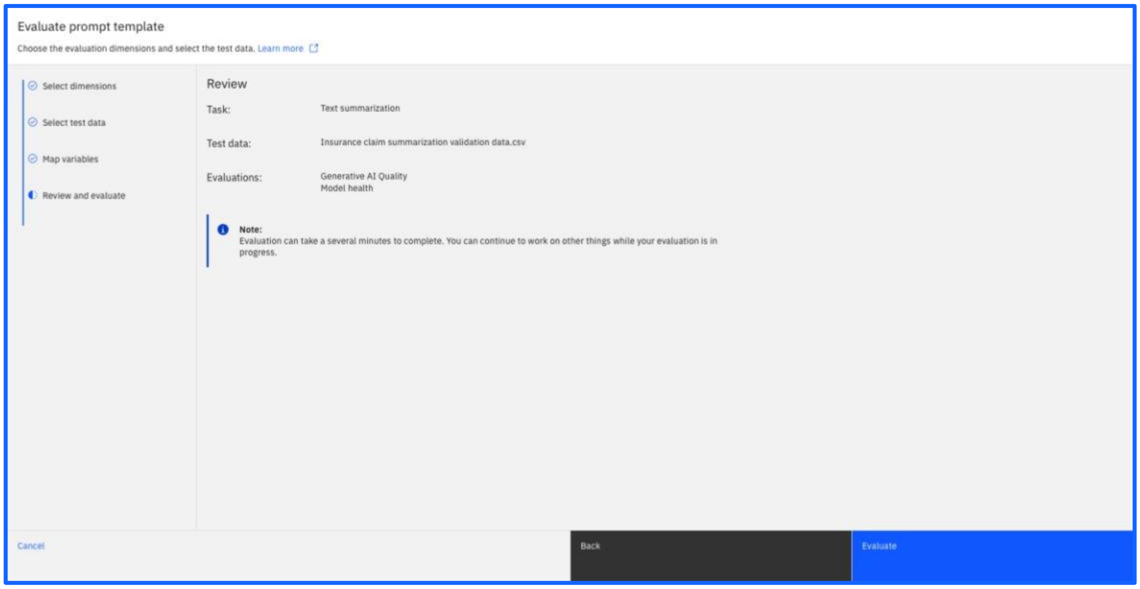
Now it's time to map the input variables to the input variable that we created earlier. Additionally, I chose **Input** to be mapped to the **"**Insurance_Claim" column in my file. This will ensure that my prompt is focused on generating the correct data.

Lastly, **Reference output** is the column of the data that gives the expected output from our LLM where metrics will be evaluated against. This is how we can measure how effectively we have structured our prompts.

I chose the reference output to be mapped to 'Summary' – this is the column of data that gives the expected output to the LLM.

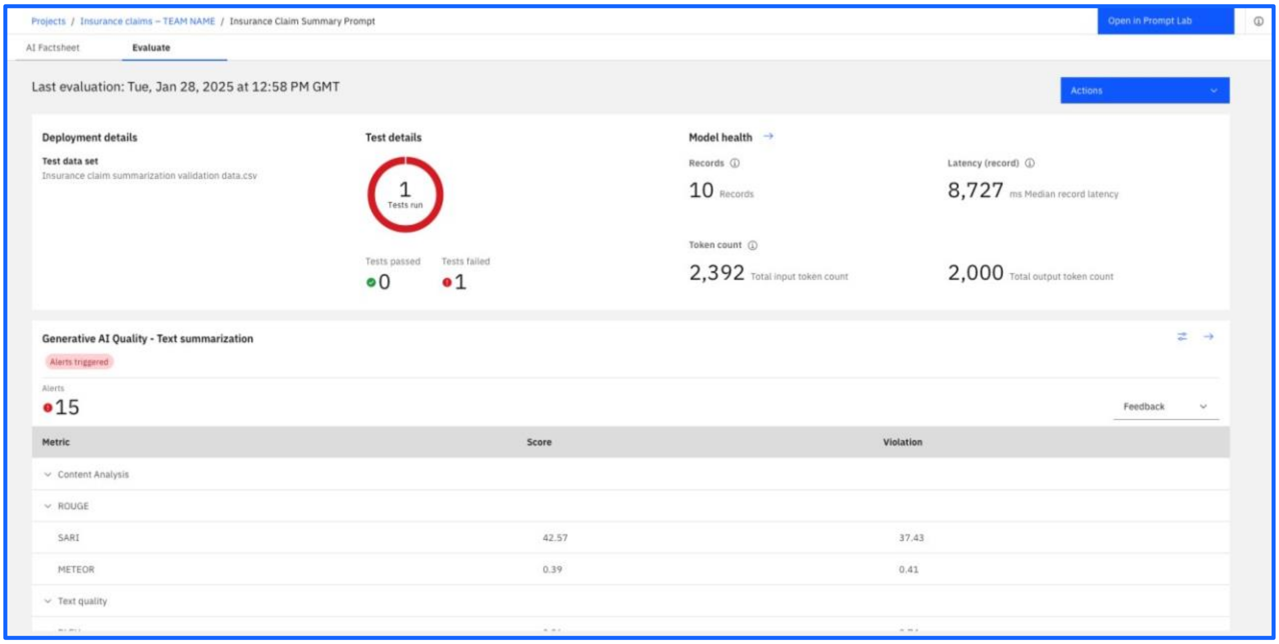


After that, you will see a screen to review the information you have entered. Once you verify that you have entered everything correctly, hit the ***Evaluate*** button. and wait until your prompt has finished being evaluated. This process can take up to a few minutes.

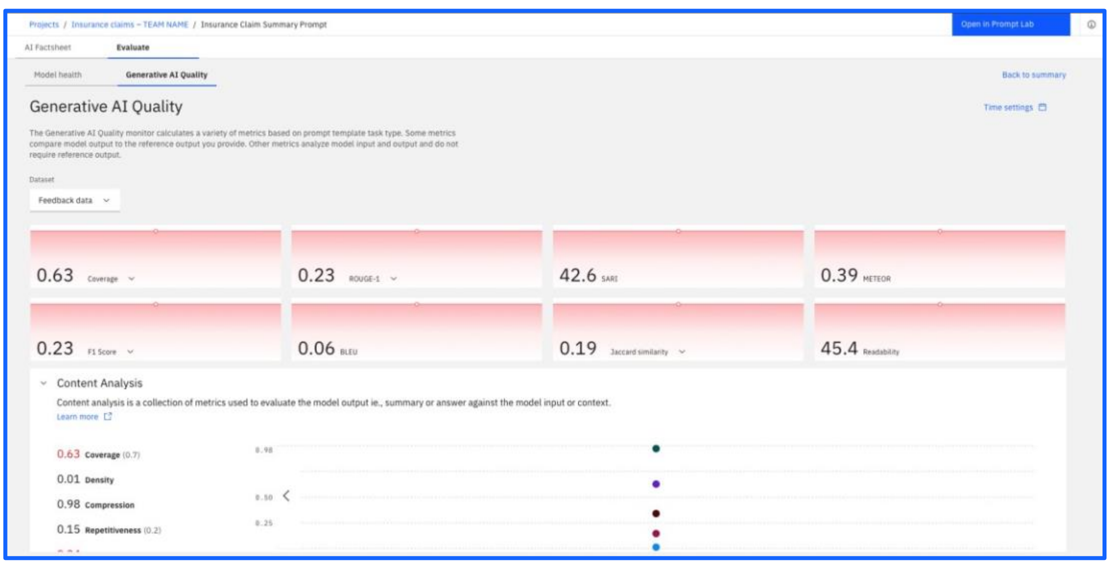


Evaluation Results

Once the evaluation has finished, you should see a screen pop up displaying the results. As you can see here, our evaluation was not very successful, triggering a number of alerts and failing its test. This is to be expected!



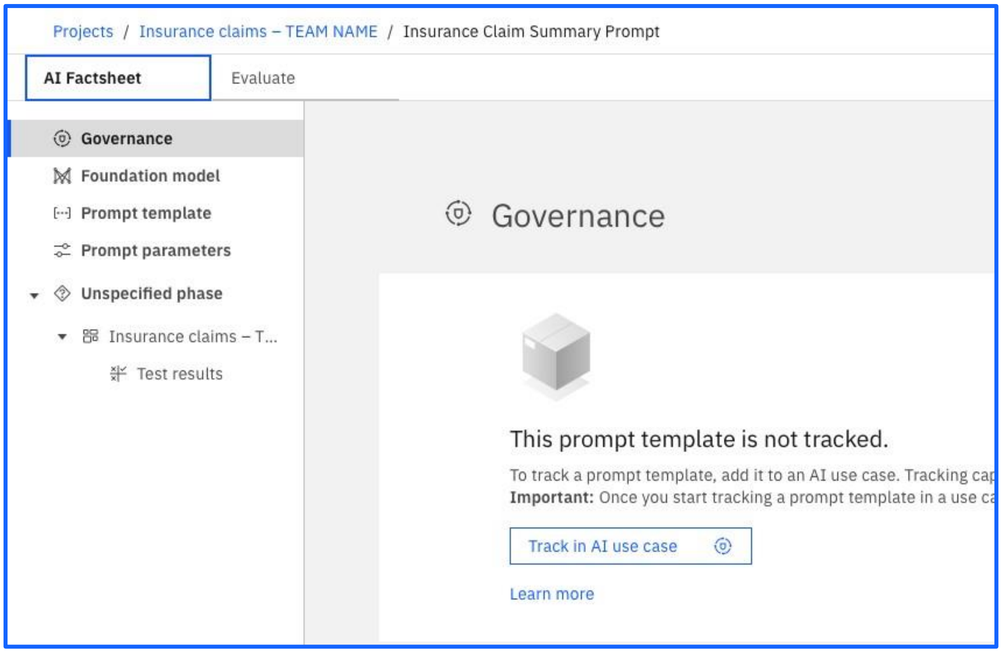
Click on the blue arrow on the right of **Generative AI Quality – Text Summarisation** to access a more detailed view of the evaluation results. Make sure to check both the **Model health** and **Generative AI Quality** tabs.



In this lab's use case, the LLM's task was "Summarisation" which can be measured using a range of quality metrics. We recommend taking time now to scan through this documentation on the generative AI quality metrics supported by watsonx.governance.

AI Factsheet

Up in the top left corner of your screen, choose the **AI Factsheet** tab, which will show us a better breakdown of the results we got. The first thing you will see on the page is an option to track your prompt in an AI use case.



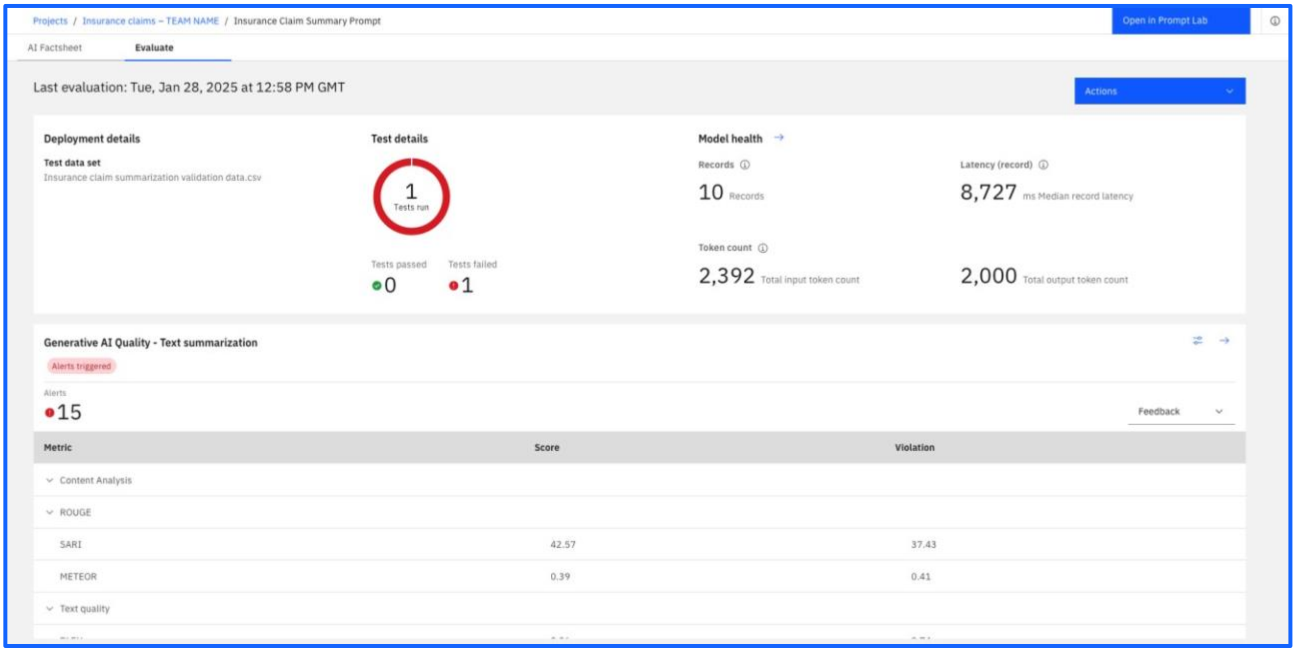
First explore the page by clicking into each of the tabs:

- The *Governance* tab collects basic information such as the name of the AI use case, the description, and the approach name and version data.
- The *Foundation model* tab displays the name of the foundation model, the license ID, and the model publisher.
- The *Prompt template* shows the prompt name, ID, prompt input, and variables, whilst the Prompt parameters collect the configuration options for the prompt template, including the decoding method and stopping criteria.
- Finally, the *Evaluation / Unspecified phase* tab displays the data from evaluation, including alerts and metric data from the evaluation. Within the *Test* sub-tab you'll see a screen similar to the one shown below:

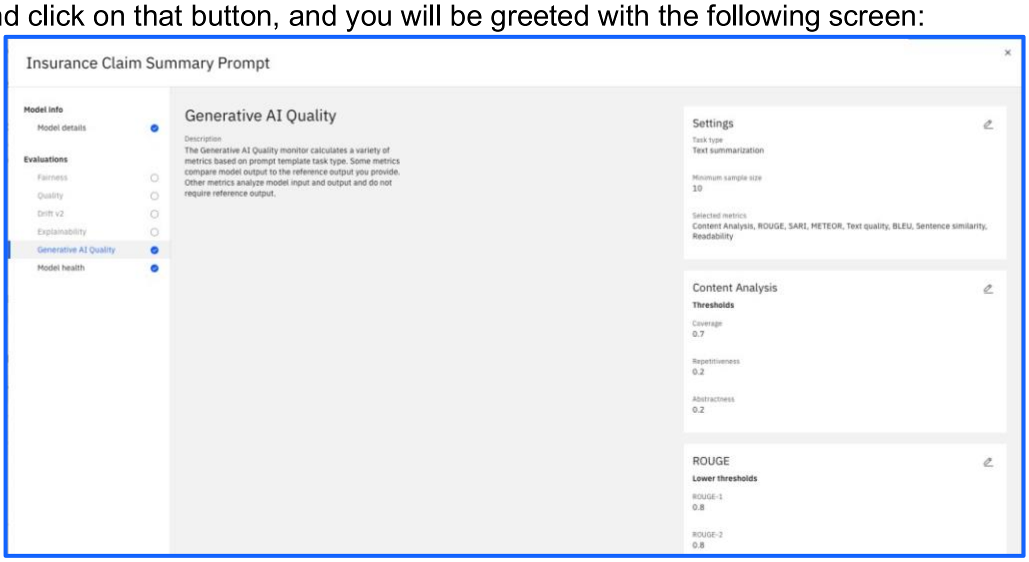
Note: The metrics that you see in the lab may differ slightly to what you see above.

You can see that each metric has an upper and lower bounds with the ideal range indicated by the bolder portion at the left side of each metric's measurement. The upside-down triangles  represent the measured value for each metrics.

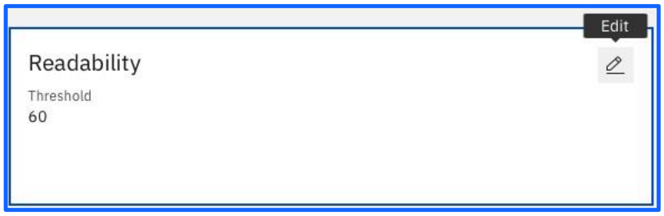
Now, I'll show you how to modify the thresholds for each metric so that your prompt can have looser requirements to pass testing. In a real-world use case, this should only be done if necessary and makes sense for your use case! Head back over to the evaluation screen, where you will see a blue settings button to the right of **Generative AI Quality – Text Summarisation** where we can adjust our thresholds. 



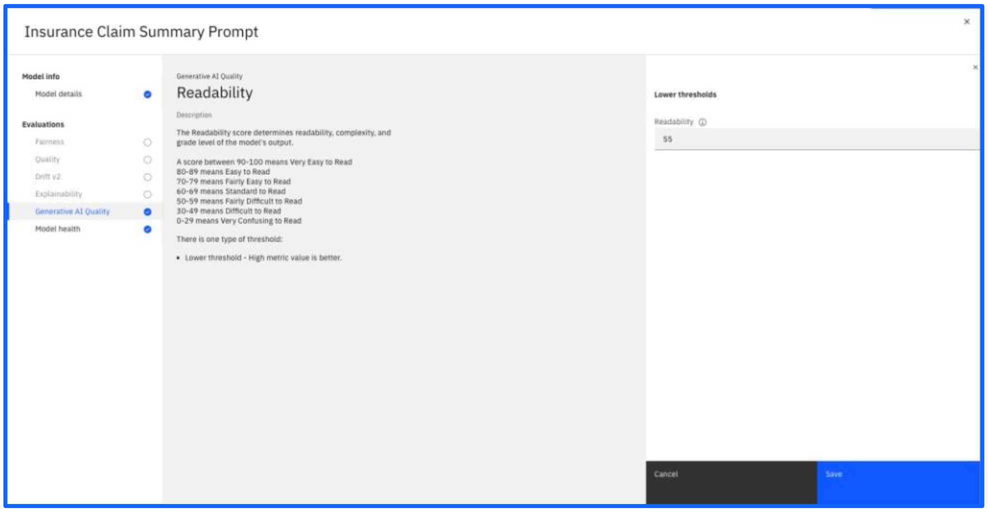
Go ahead and click on that button, and you will be greeted with the following screen:



Here we are going to adjust the **Readability** metric. This metric is higher if your model's generated responses to your prompt are easy to interpret, and lower if they are confusing to read. Scroll down the page and click the edit button to modify it.



Lastly, go ahead and change the readability slightly, say to 55, which puts the requirement into the "Fairly difficult to read" range. After, that, you can go ahead and click save to finalise your changes. Just a reminder here that these metrics should not be altered too much in a real-world use case!

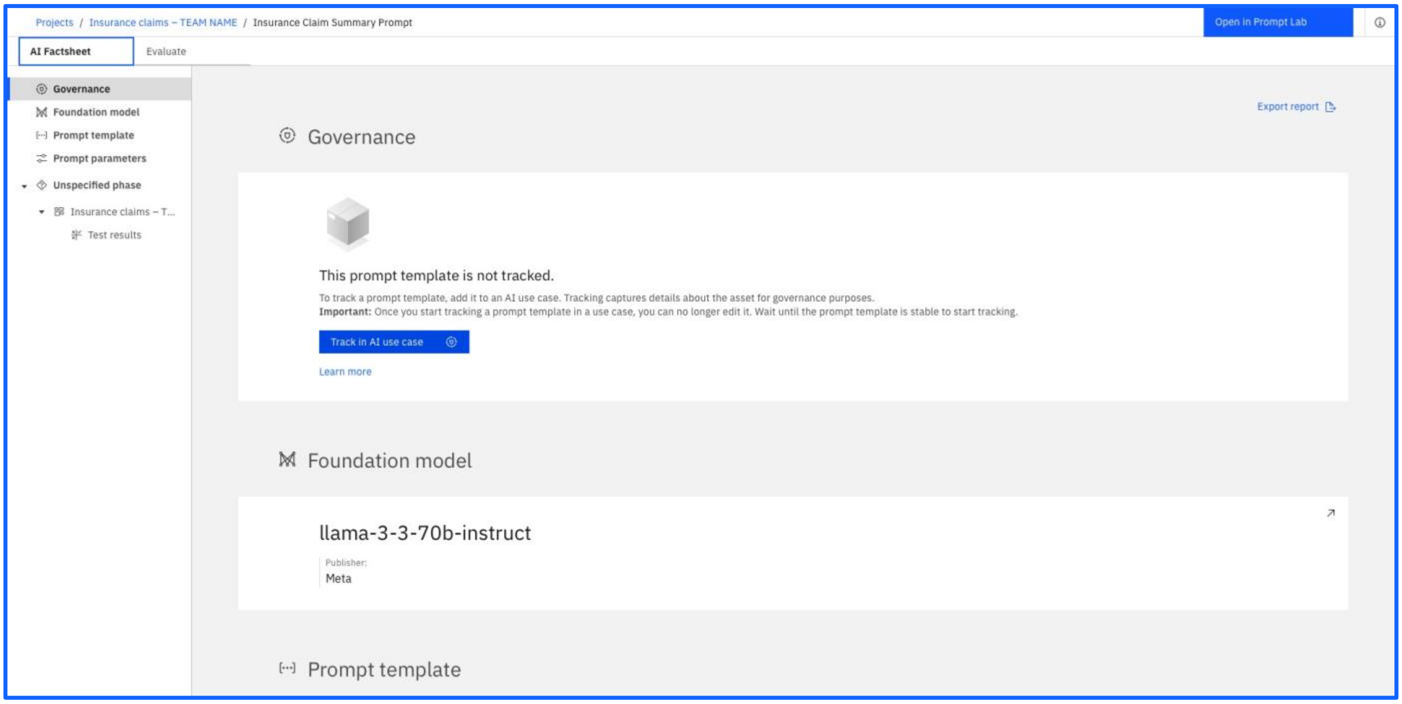


Govern your prompt template

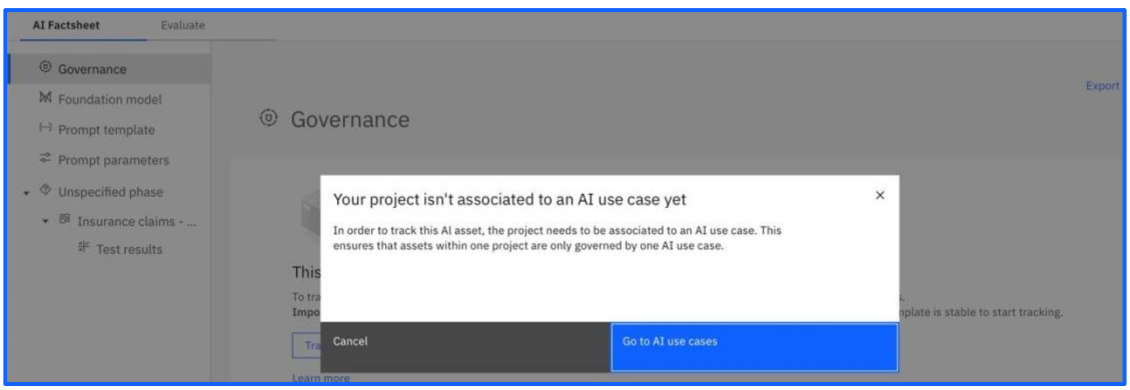
The next section of this lab it to govern your prompt template.

In order to do this, you will need to first associate the project that you created earlier (most likely the Sandbox environment, unless you created one) to the relevant AI Use Case.

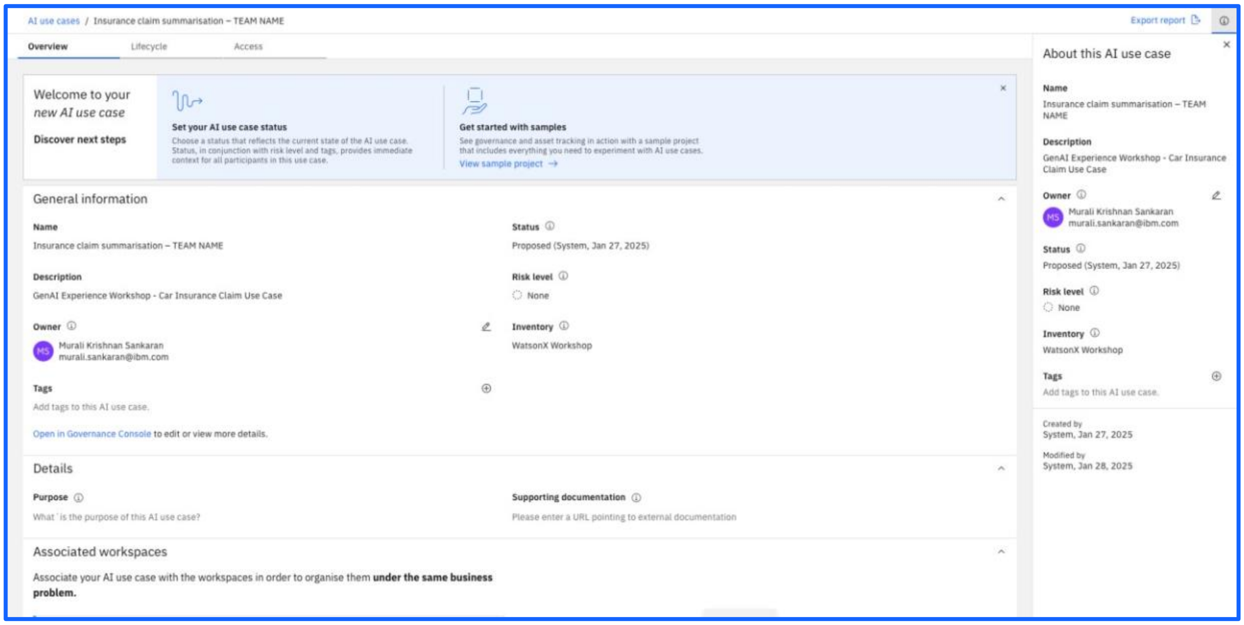
To do so, click into the Track in AI use case button within the ***Governance*** tab:



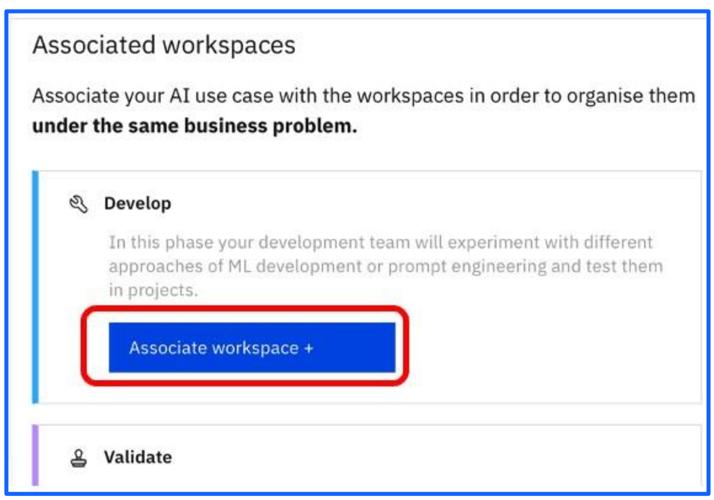
Then, select Go to AI use cases to open up a list of AI use cases.



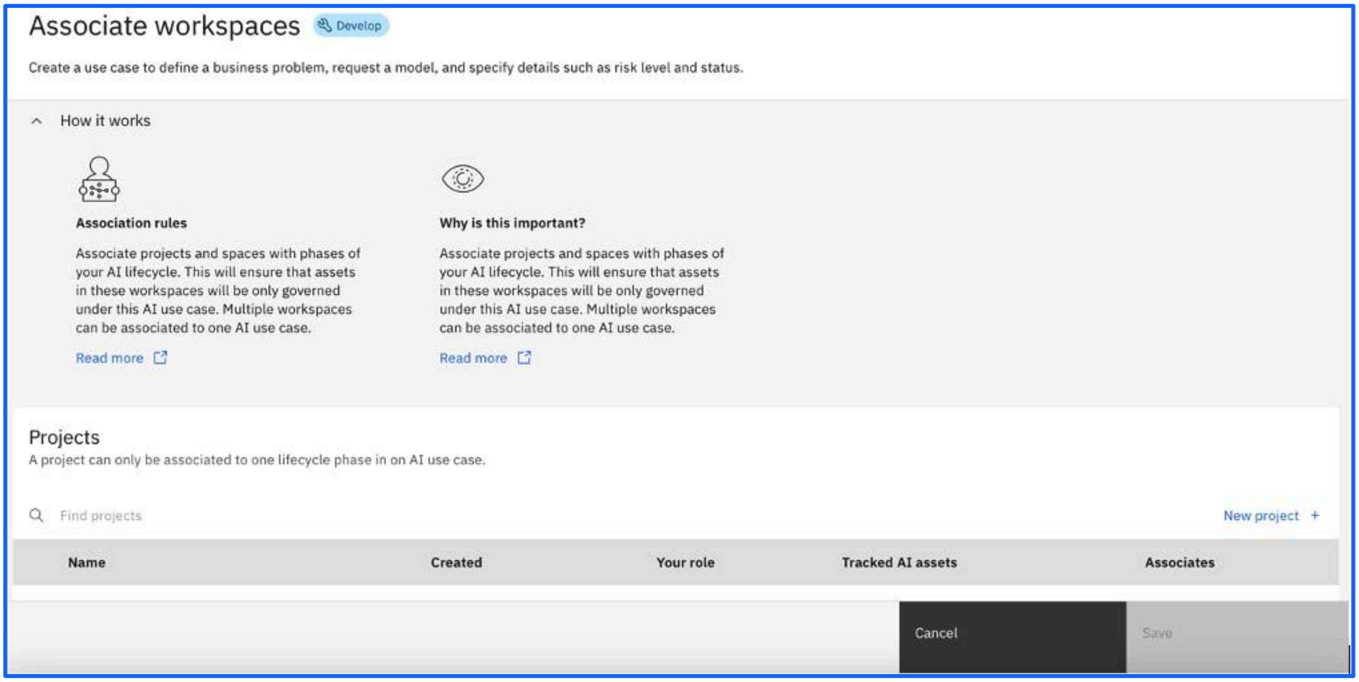
Select the Use Case that you created earlier, and you will be met with this screen:



Scroll down to the **Associated workspaces** section. In this lab, the relevant phase is the '**Develop'**; the prompt requires engineering to develop and further refine it. Click the button shown below to Associate workspace + .



You will see the following screen and will need to select the relevant ***Project*** (most likely your Sandbox environment) by ticking the box on the right.

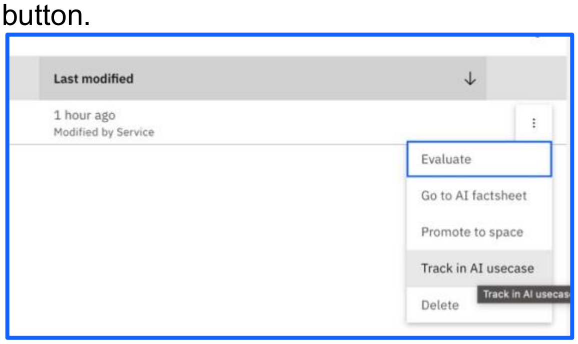


Tracking Prompt Template in a Use Case

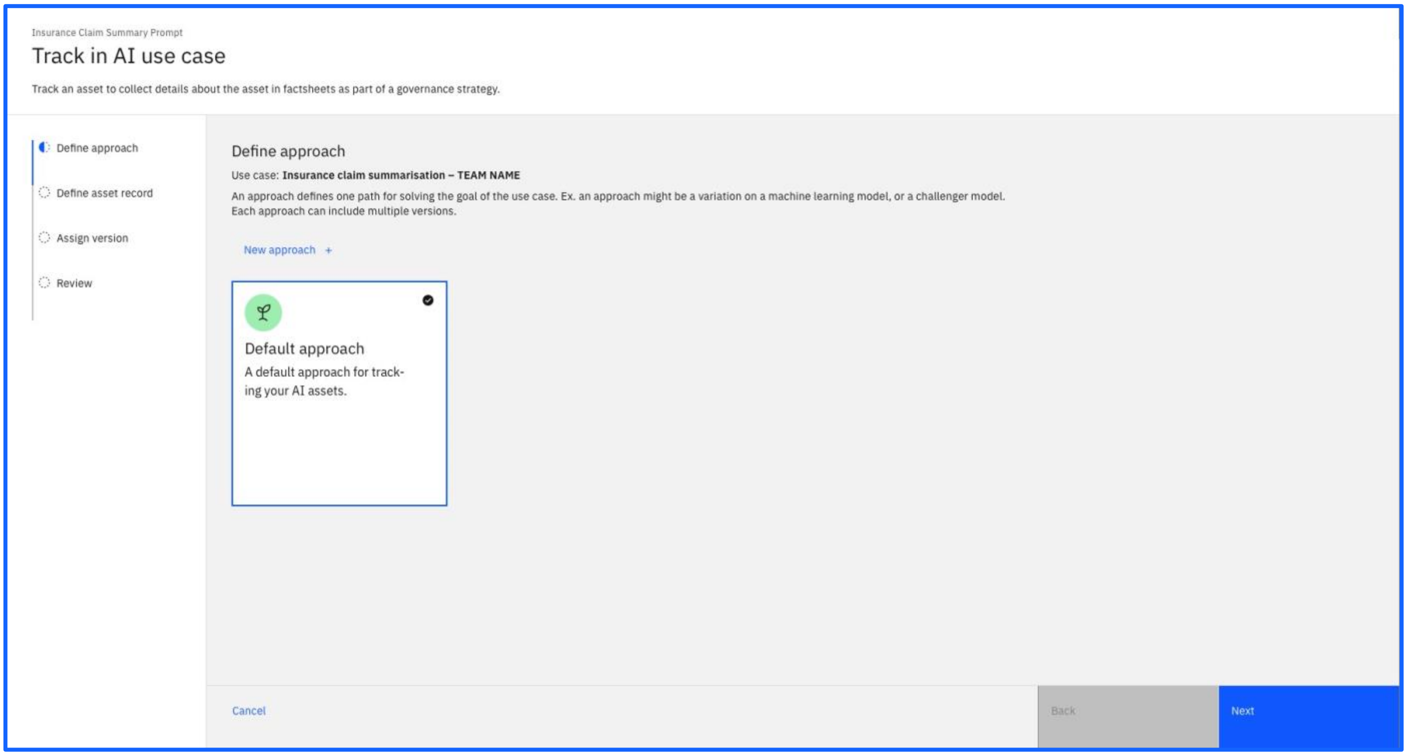
Now that you have associated your project to your AI use case, you can track your prompt template in your AI use case. The idea behind tracking prompt template in a Use Case is to save AI Engineers time and give transparency into the work they're doing. You can automatically save prompt templates and use versioning to easily identify the different prompts and where they are in the lifecycle.

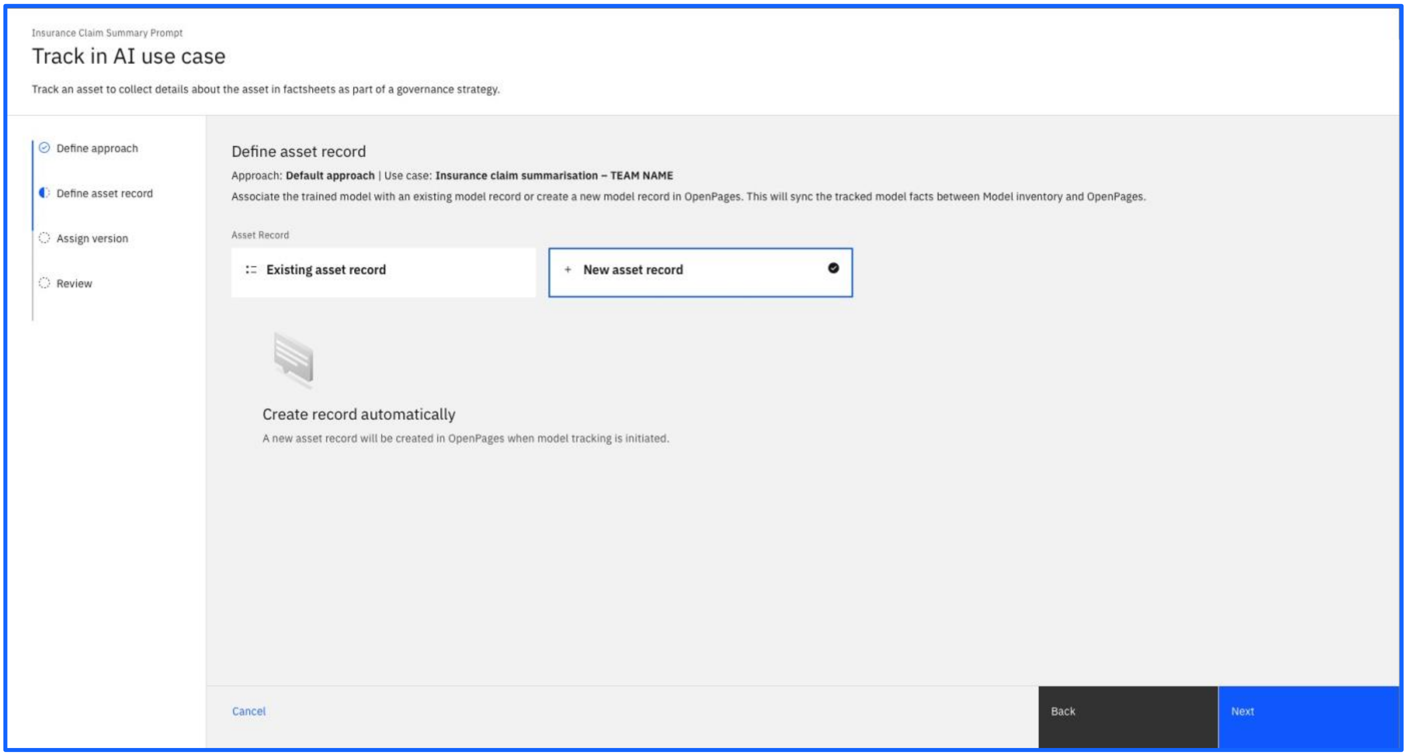
Select the navigation menu in the top right again, and under **Projects**, click into **View all projects** to navigate to your project. Click into this and find the prompt you made earlier. If you struggle to find your way to your project, your facilitator will be able to help.

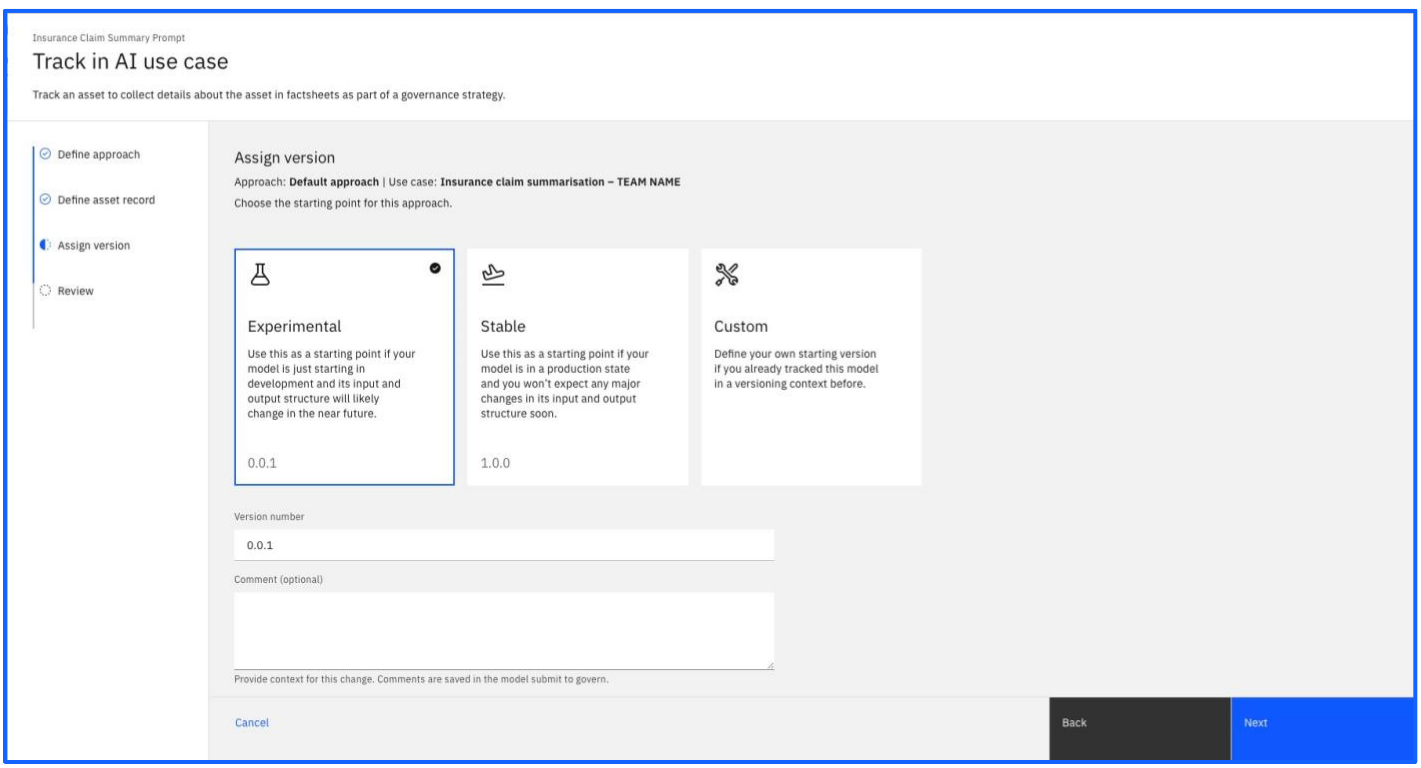
If you click the three dots on the right, you will be able to see that the prompt needs to be tracked – click on the Track in AI usecase button.



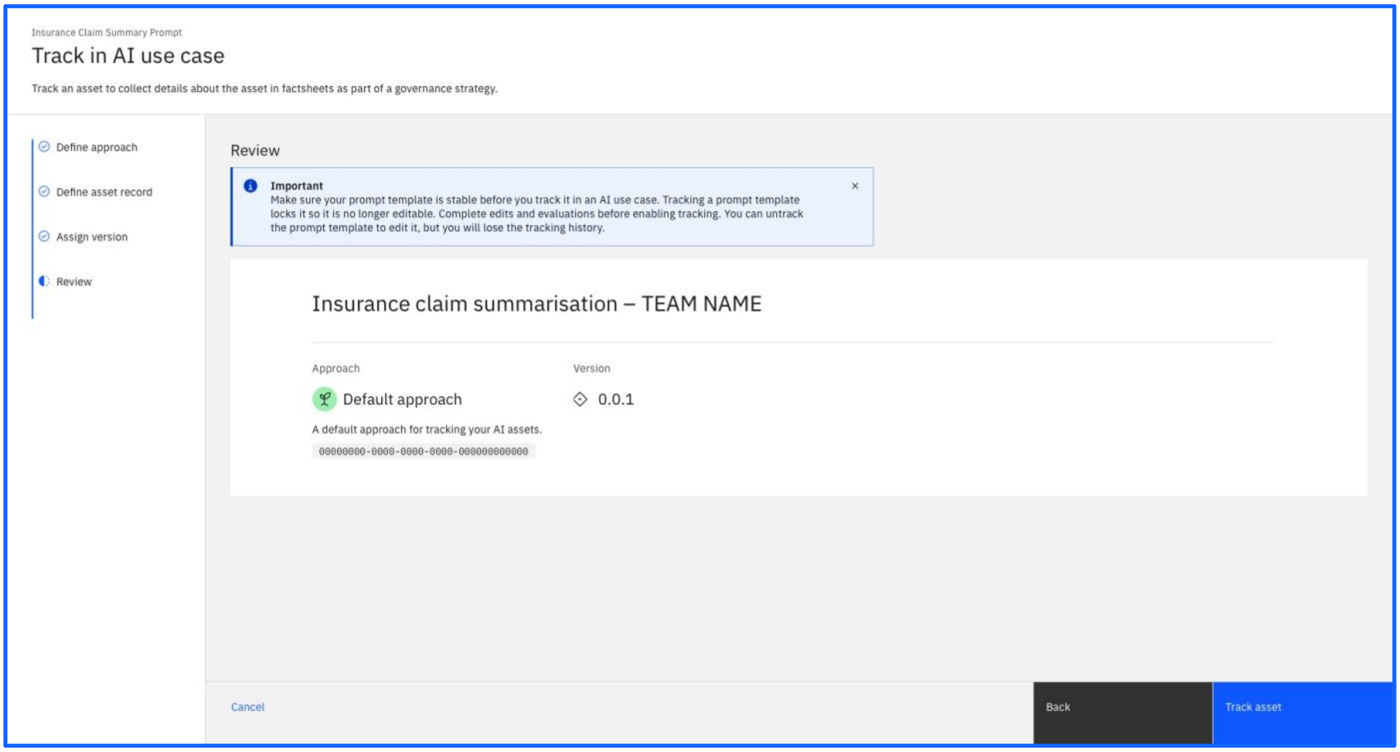
Select the **Default approach**, select to automatically create a **New asset record** and make it **Experimental**.

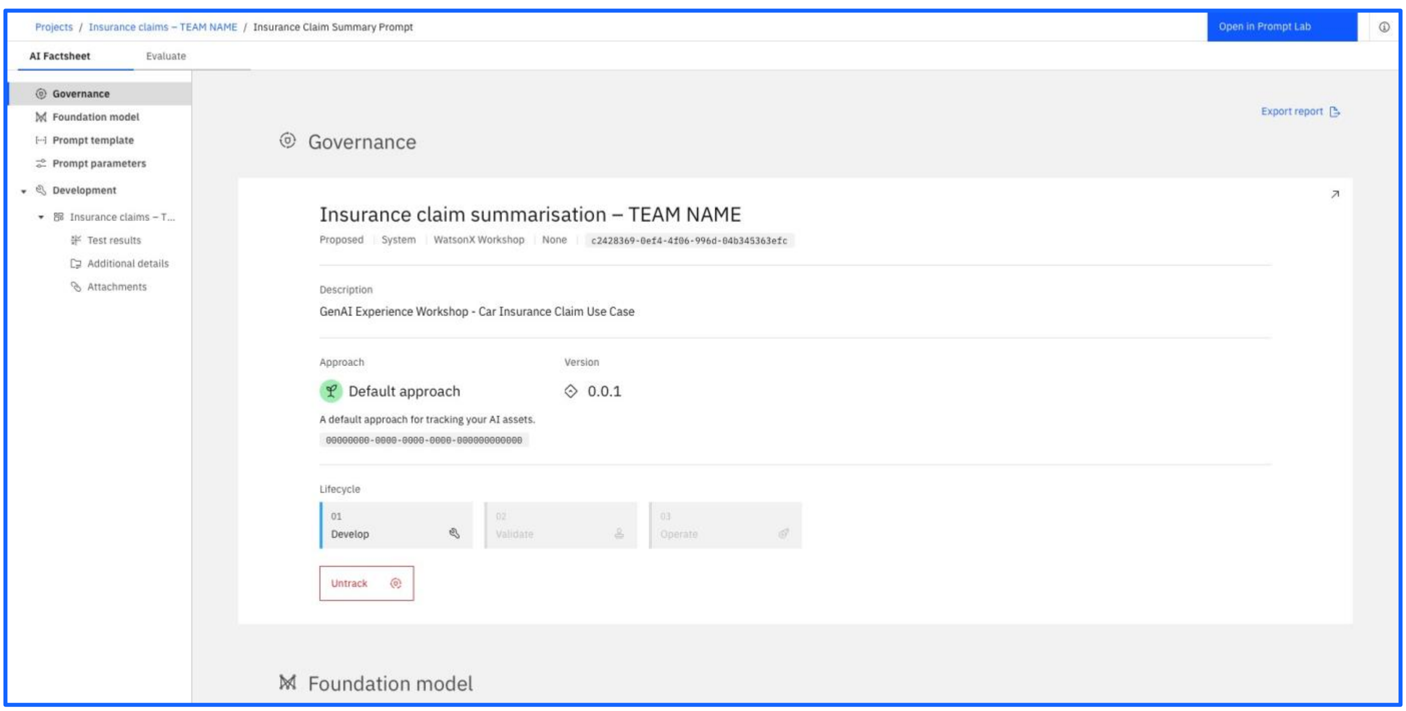

Insurance Claim Summary Prompt
Track in AI use case
Track an asset to collect details about the asset in factsheets as part of a governance strategy.
Define approach
Define asset record
Assign version
Review
Define approach
Use case: Insurance claim summarisation – TEAM NAME
An approach defines one path for solving the goal of the use case. Ex. an approach might be a variation on a machine learning model, or a challenger model.
Each approach can include multiple versions.
New approach +
Default approach
A default approach for tracking your AI assets.
Cancel
Back
Next


Insurance Claim Summary Prompt
Track in AI use case
Track an asset to collect details about the asset in factsheets as part of a governance strategy.
Define approach
Define asset record
Assign version
Review
Define asset record
Approach: Default approach | Use case: Insurance claim summarisation – TEAM NAME
Associate the trained model with an existing model record or create a new model record in OpenPages. This will sync the tracked model facts between Model inventory and OpenPages.
Asset Record
Existing asset record
New asset record
Create record automatically
A new asset record will be created in OpenPages when model tracking is initiated.
Cancel
Back
Next

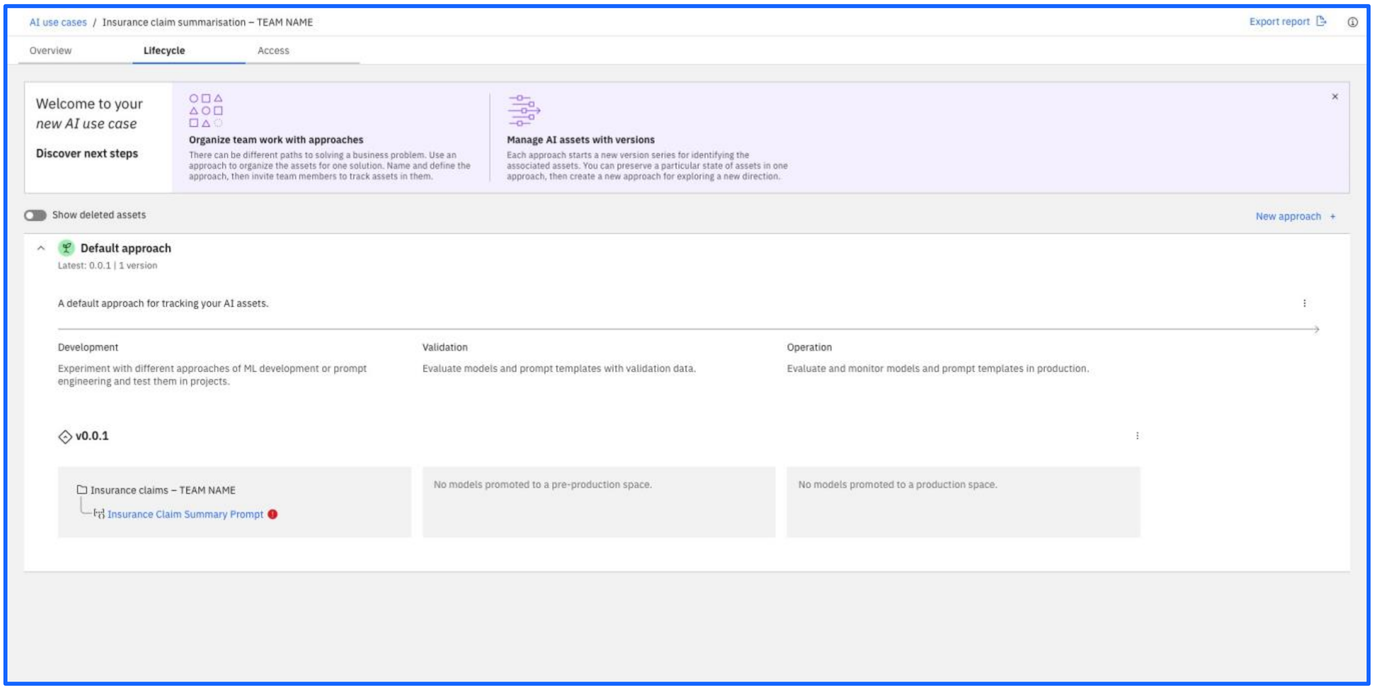

Insurance Claim Summary Prompt
Track in AI use case
Track an asset to collect details about the asset in factsheets as part of a governance strategy.
Define approach
Define asset record
Assign version
Review
Assign version
Approach: Default approach | Use case: Insurance claim summarisation – TEAM NAME
Choose the starting point for this approach.
Experimental
Use this as a starting point if your model is just starting in development and its input and output structure will likely change in the near future.
0.0.1
Stable
Use this as a starting point if your model is in a production state and you won't expect any major changes in its input and output structure soon.
1.0.0
Custom
Define your own starting version if you already tracked this model in a versioning context before.
Version number
0.0.1
Comment (optional)
Provide context for this change. Comments are saved in the model submit to govern.
Cancel
Back
Next

Review your choices and start Tracking the Asset





Using the hamburger menu, navigate to your **AI Use Case,** switch to the **Lifecycle** tab see your prompt template lifecycle being tracked. If it doesn't appear immediately refresh your screen.



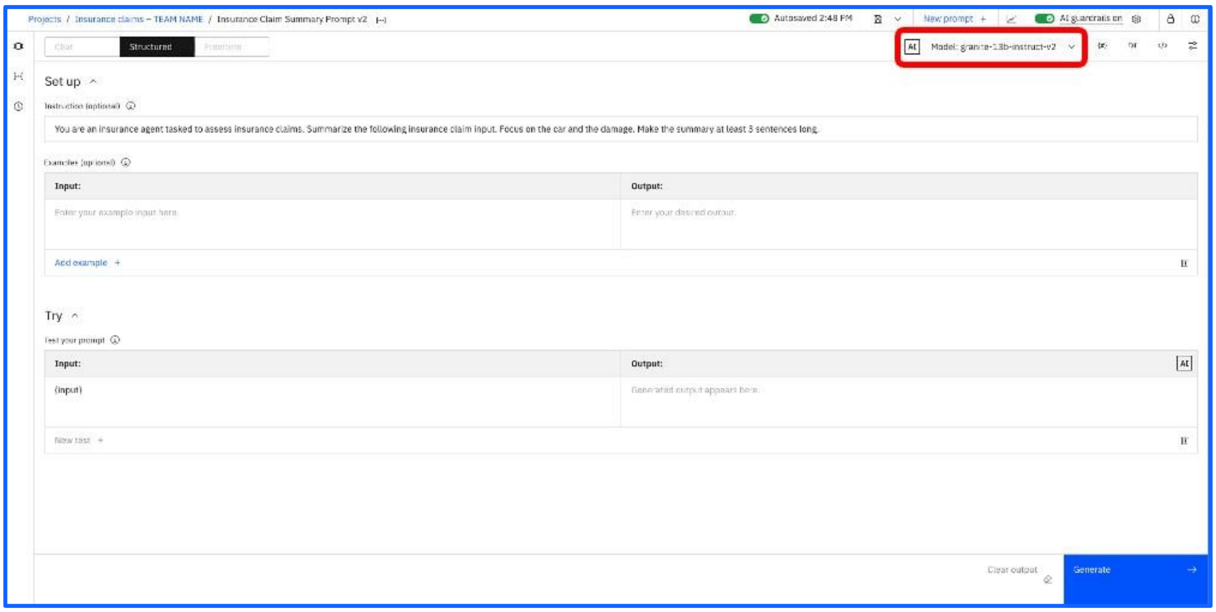
This is where you can view any changes to the prompt template as we continue with the lab.

Update your prompt template

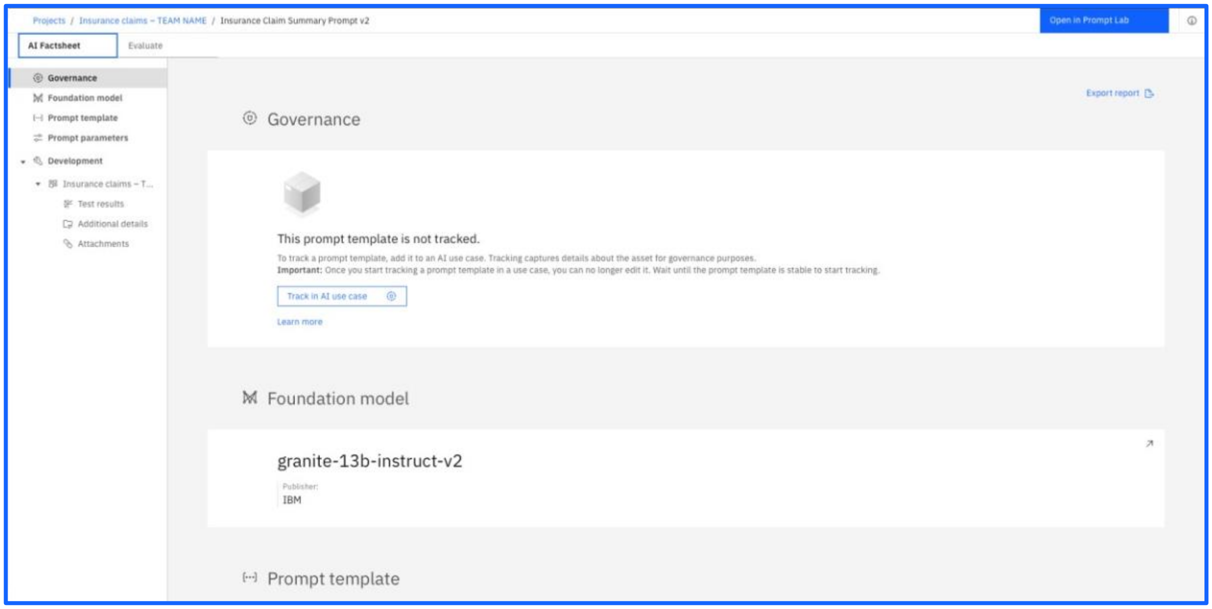
We will now make additional changes to the prompt template, which we’ll do by creating a new version of the original prompt.

For that, use the main menu to navigate to your project. We’ll create a new prompt that is a copy of the first one. Do that by opening your template from the **Assets** tab and use **Save as** to give it a new name, such as “Insurance Claim Summary Prompt v2”. That opens it for editing again.

Change the model to “**granite-13b-instruct-v2**” to introduce a change in this new version of the prompt template!

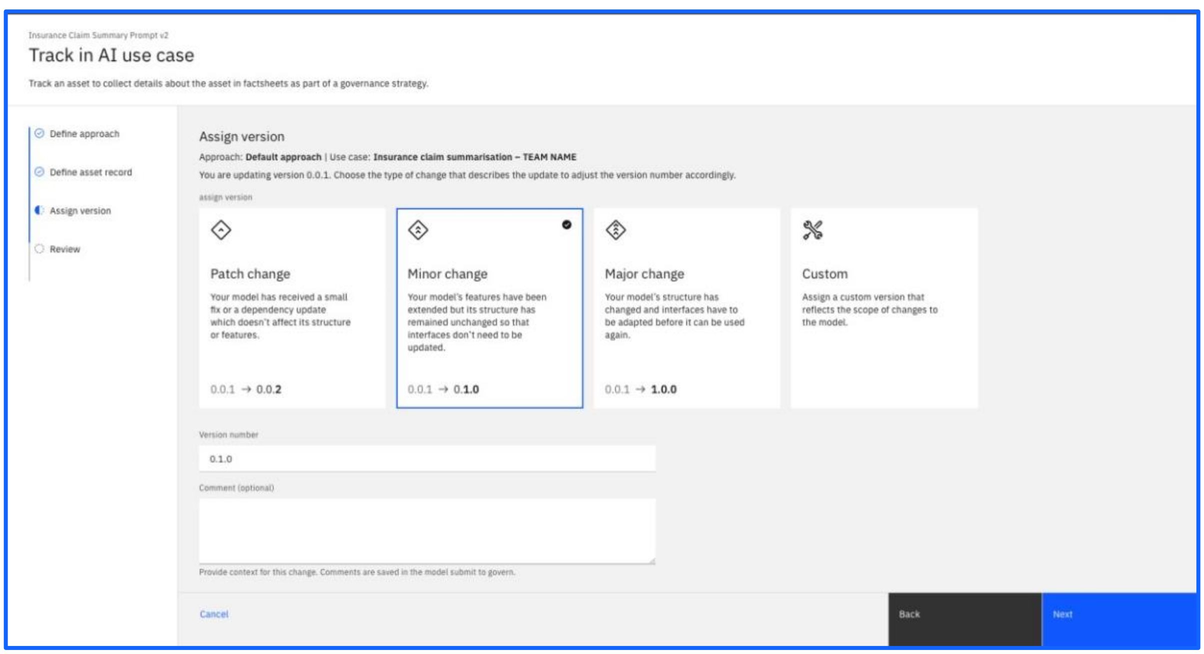


Evaluate your prompt template by following the same steps as before ([Evaluating your prompt]()).

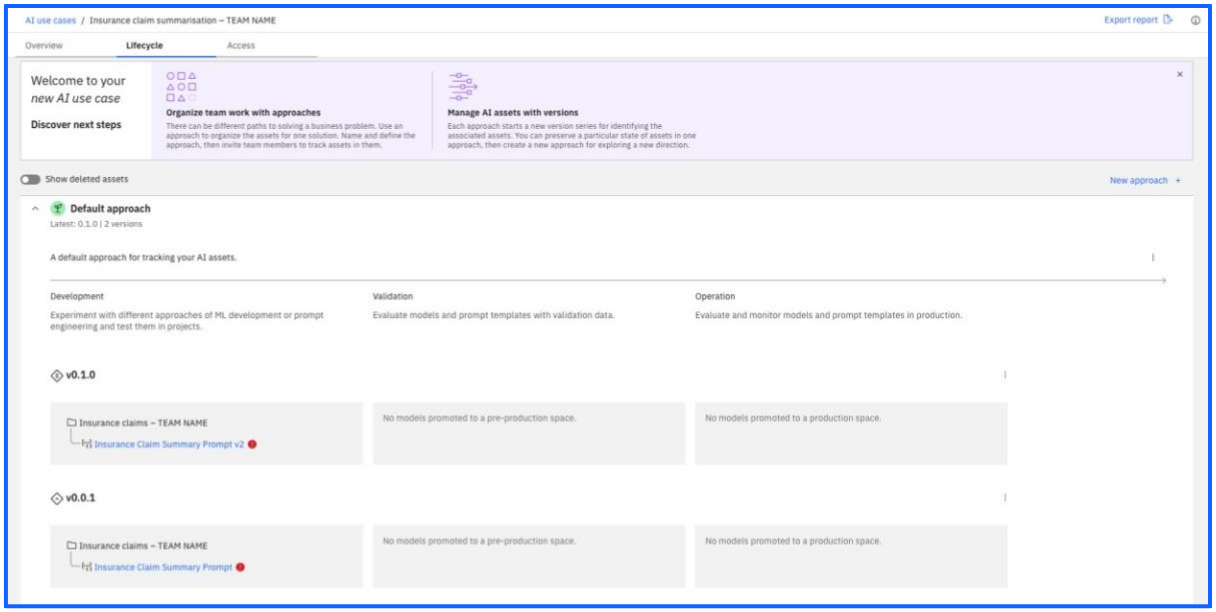


Once you have completed the evaluation of this new prompt template, you can track it in your AI use case. Go to your project by selecting the project name in the path at the top.

Track it in your AI use case by selecting **Minor Change (version 0.1.0)**.



Check that it appears in the **Lifecycle** tab of your AI use case**.**

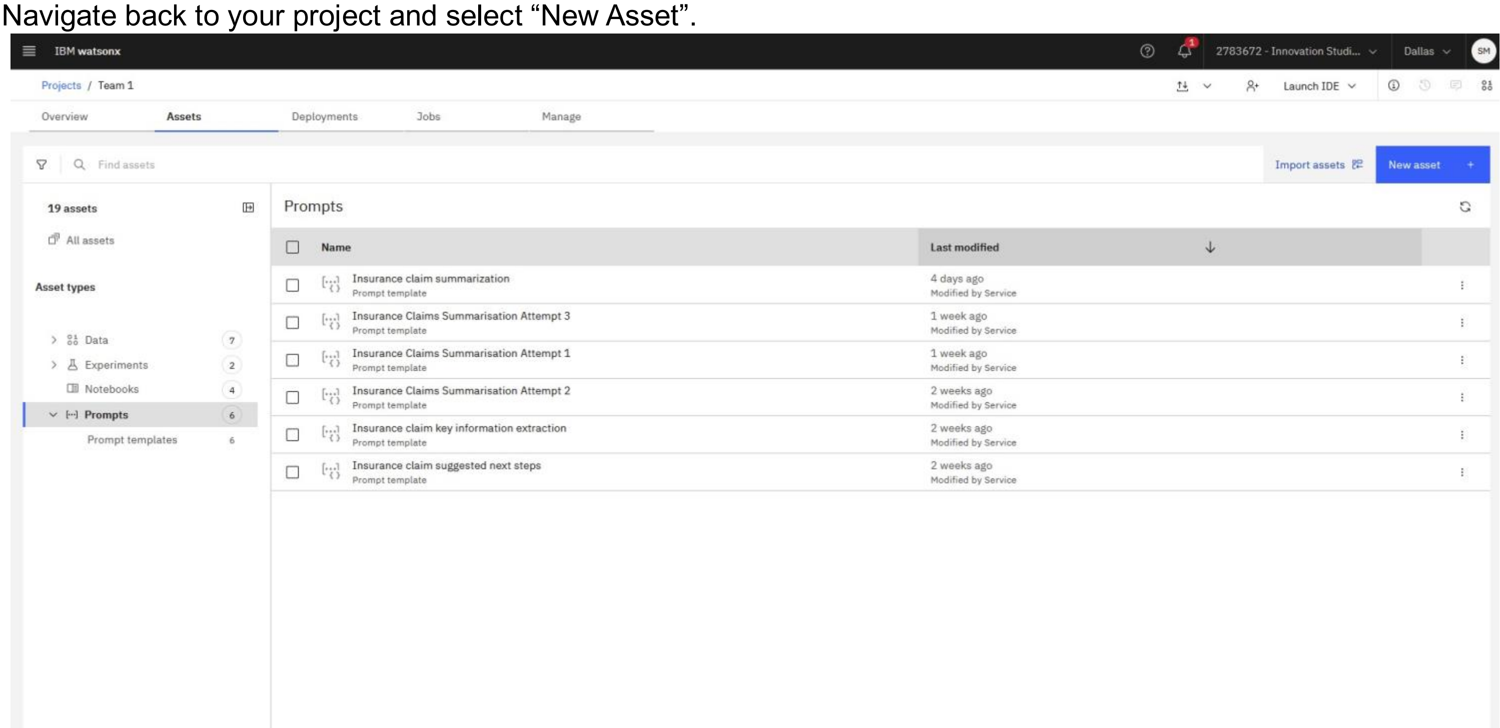


You will see that the two prompts are now tracked separately within the same AI use case, making it easy to track different prompts, models or parameters for a single use case/application.

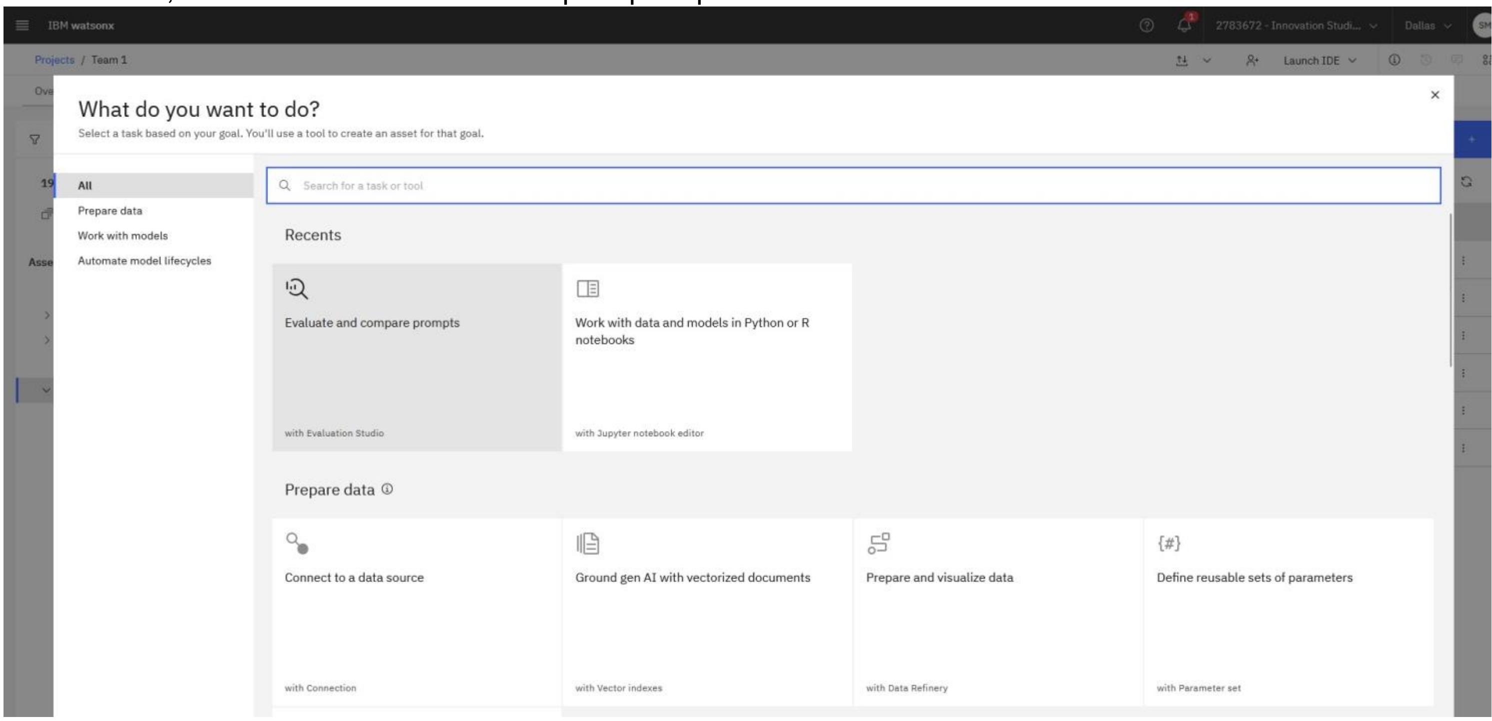
Evaluation Studio

The purpose of the Evaluation Studio is to look at different prompts and identify which is the strongest performing. For the lab, you will compare the two prompt templates you've created. Note if you didn't get this far, then there are existing prompts you can use. The following lab explains how simple it is to perform an evaluation using the Evaluation Studio.

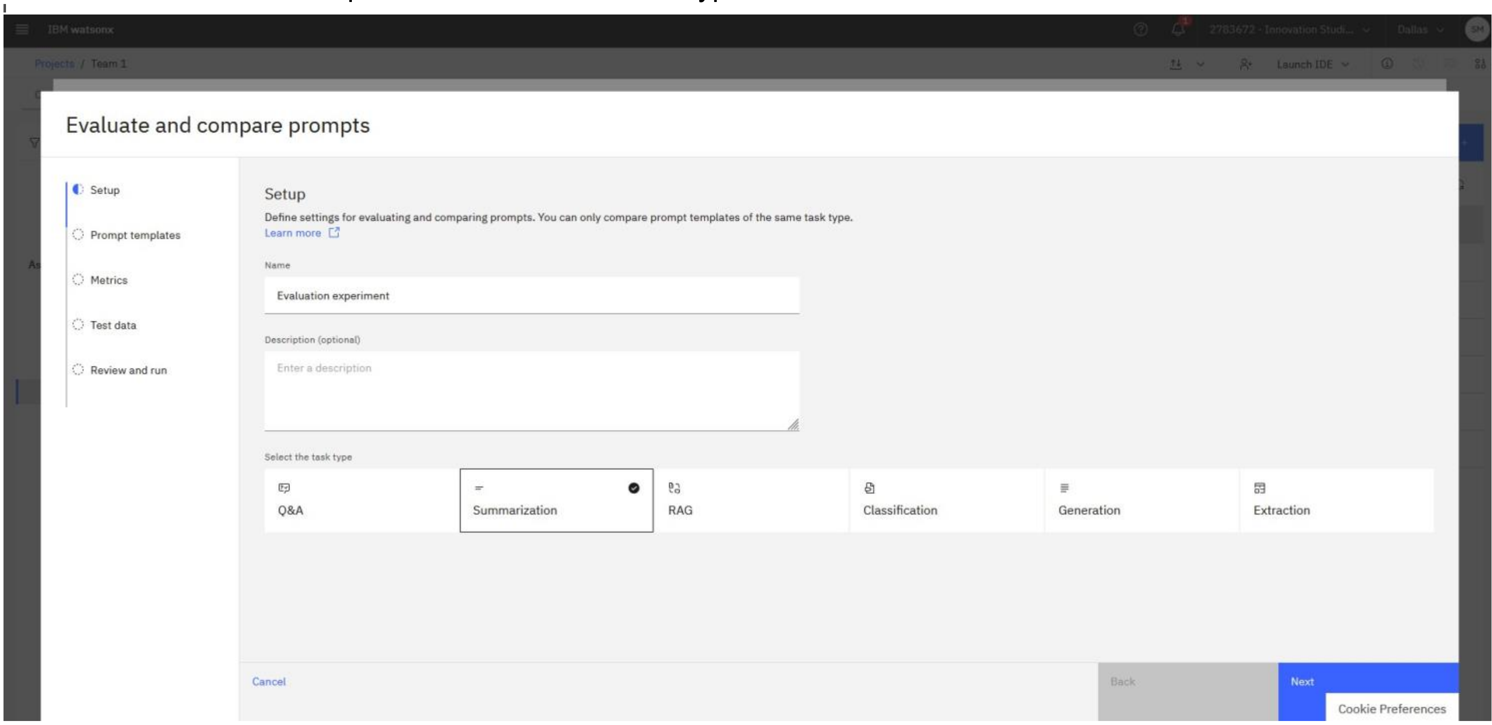
Navigate back to your project and select "New Asset".



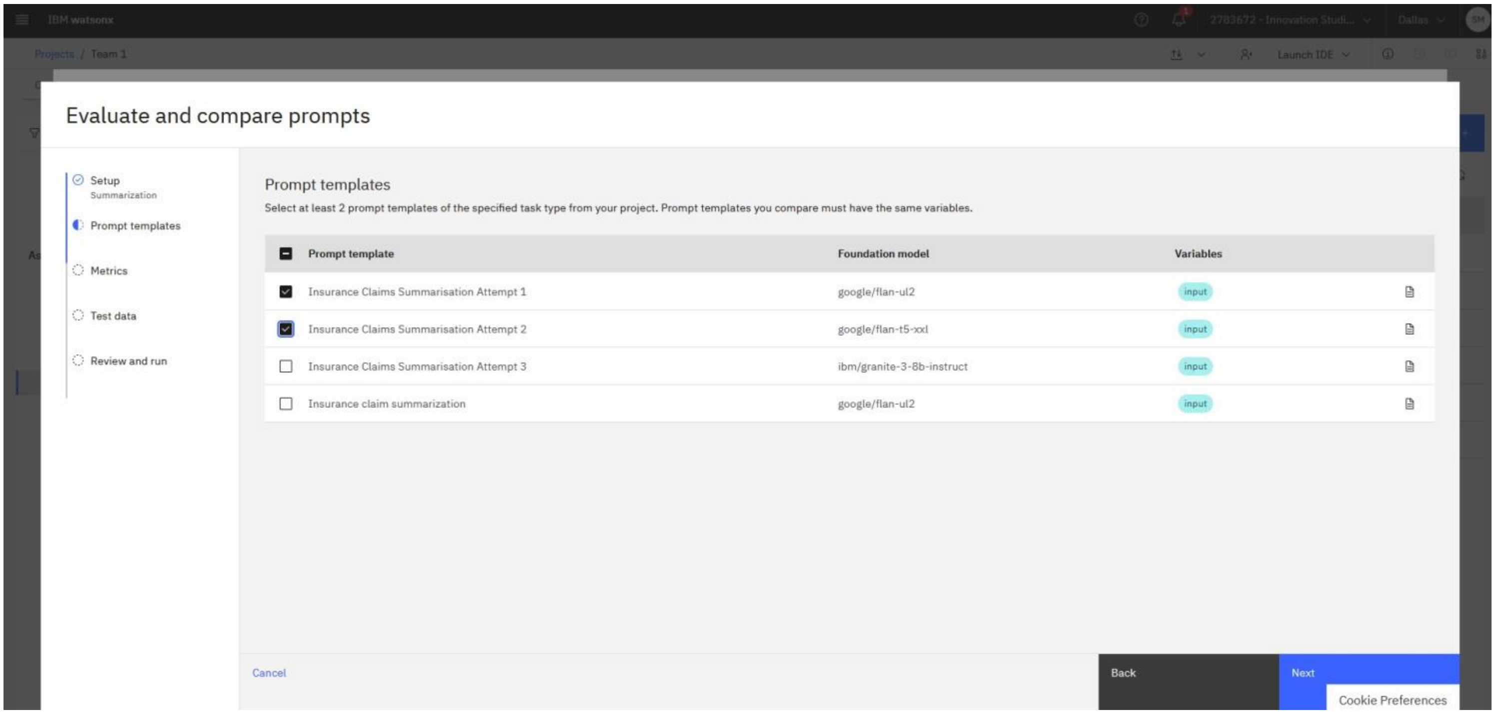
In the new UI, search for Evaluate and Compare prompts.



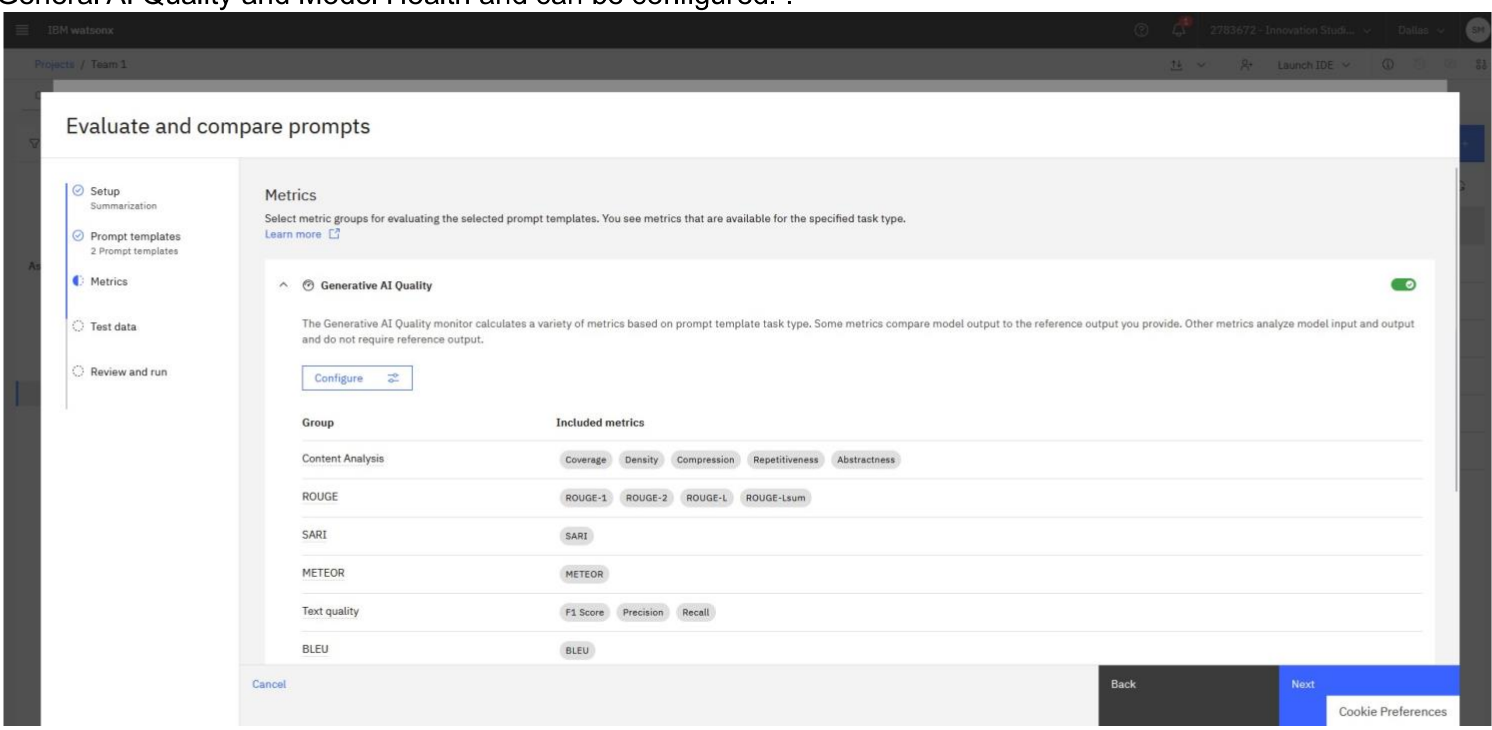
Leave the name and description as it. Select the task type as Summarisation



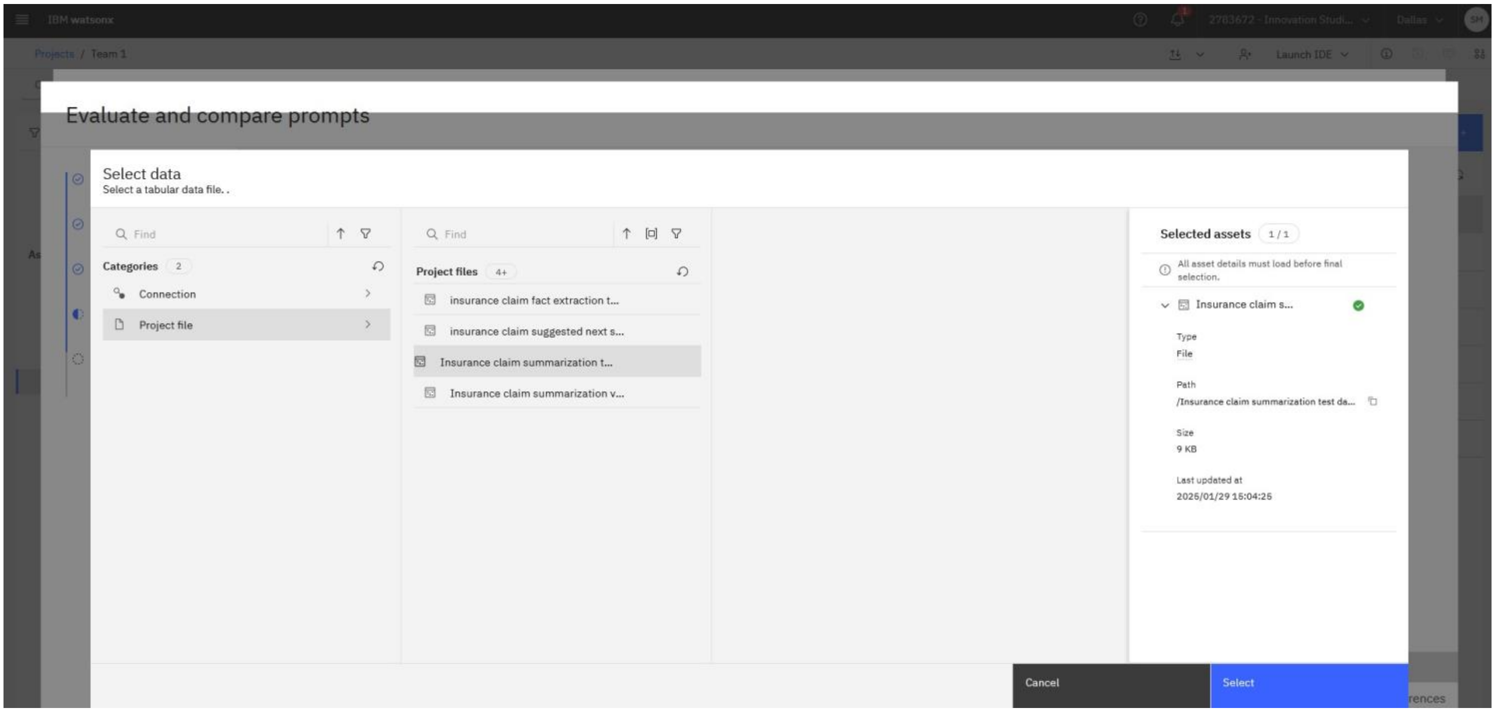
Select the two different prompts you created earlier. In this example they are labelled Insurance Claims Summarisation Attempt 1 and 2. Click Next.



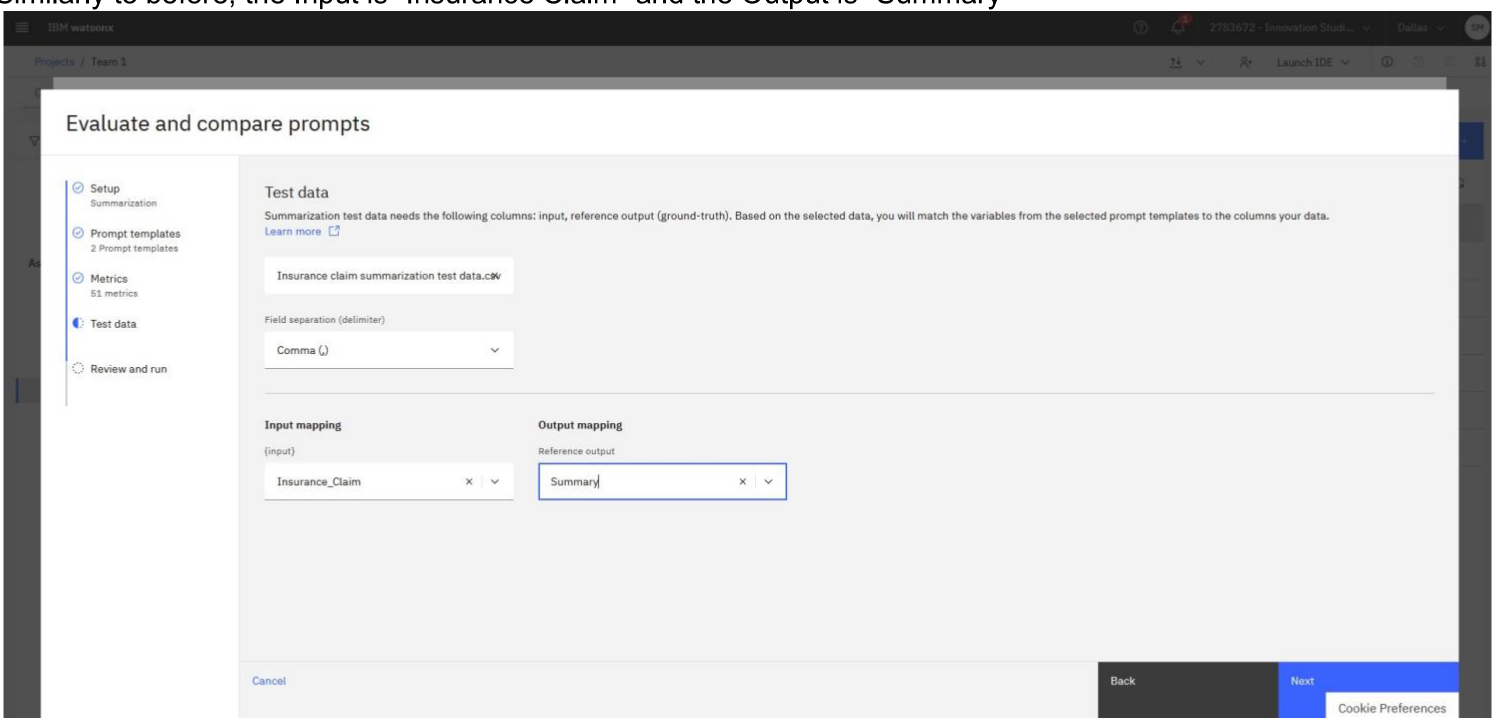
Review the different metrics which you are evaluating against. Again these fall under the categories General AI Quality and Model Health and can be configured. .



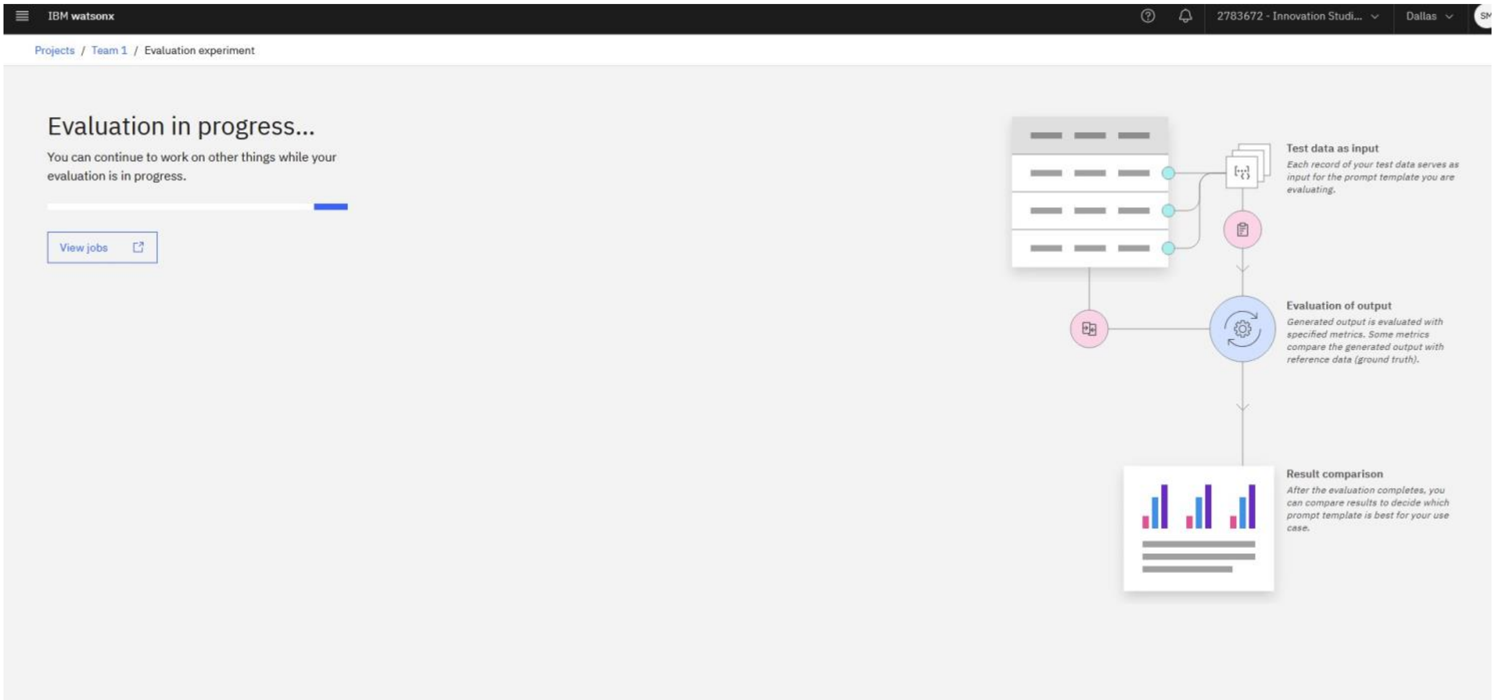
On the Test Data screen, please select the Insurance Claims Summarisation test data csv from the project. This is under the project file



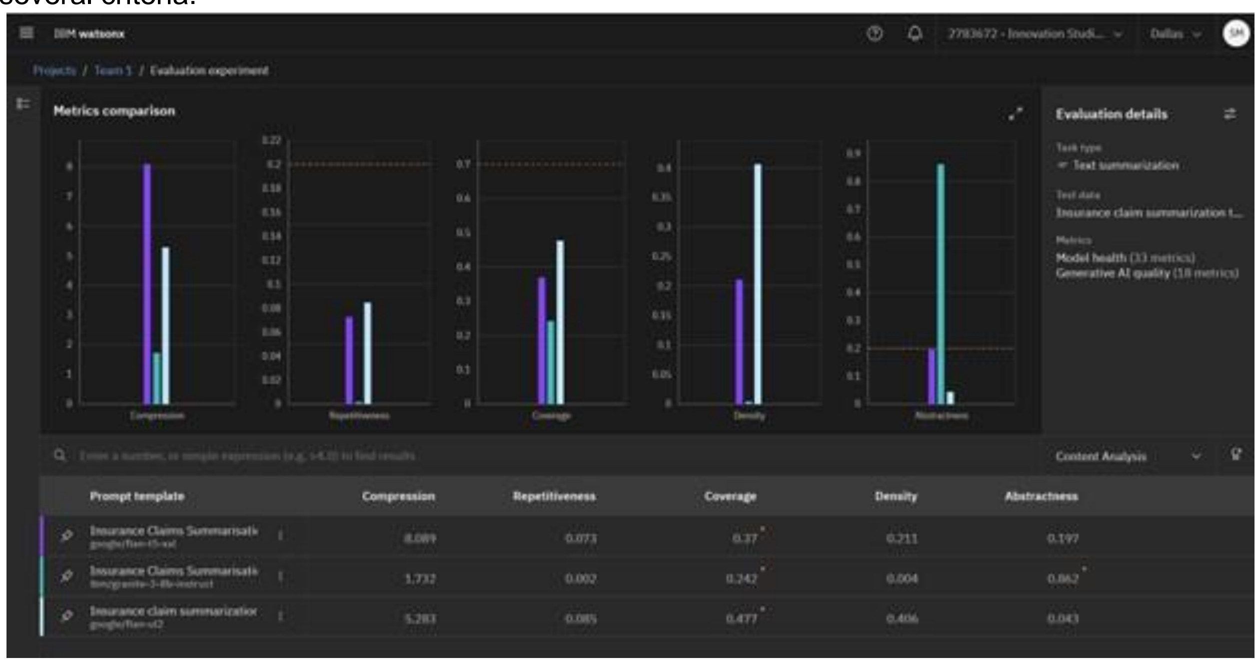
Similarly to before, the Input is “Insurance Claim” and the Output is “Summary”



Review the evaluation before running it, this will take a few minutes.



Spend some time going through the results, identifying how each attempt performs differently across several criteria.



Once done, navigate back to the project nad you will see these results exist as an evaluation result.

Phase 4: Deployment

If we consider that you've iterated the prompts and found the right combination of model, parameters, model variables until you're happy. You'd now look to promote that asset into production. When you put an asset in production, there are a different set of Risks associated with it for example Drift. This next part of lab will show you how to promote the asset into production and the features available plus turning on the evaluation for drift.

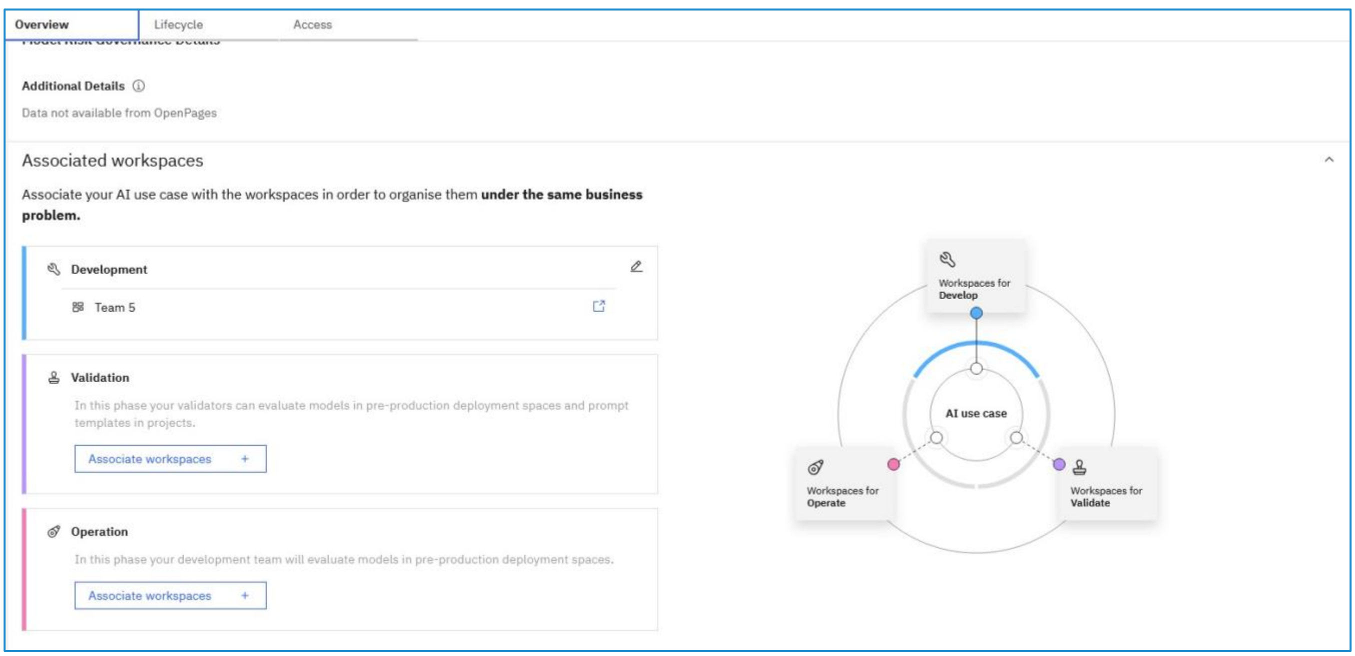
Deployment Space

A prompt template is only useful if it is accessible by an application; this prompt template needs to be made available in a consumable way. What's more, the deployment process needs to be completed in an orderly manner. For example, you cannot allow everyone to put prompt templates/models into production. For this purpose, the watsonx.ai environment includes deployment spaces.

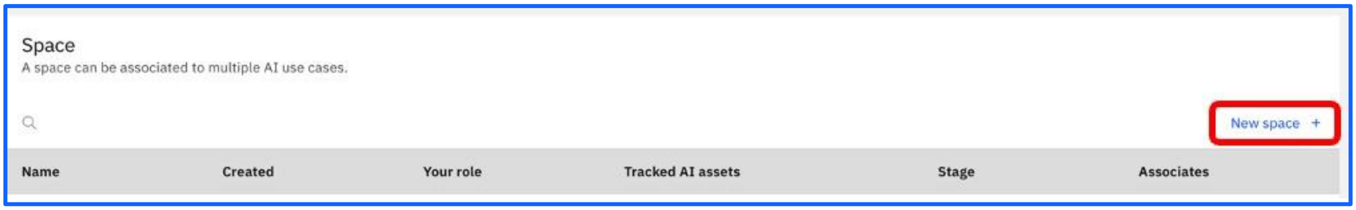
Anyone can create their own personal deployment space and promote prompt templates/ models to it. However, in an enterprise, designated administrators should be the only ones able to promote a model to a production deployment space.

Go back to your Use Case and scroll down to Workspaces, click to add a Workspace in Validation:

Go back to your Use Case and scroll down to Workspaces, click to add a Workspace in Operate:

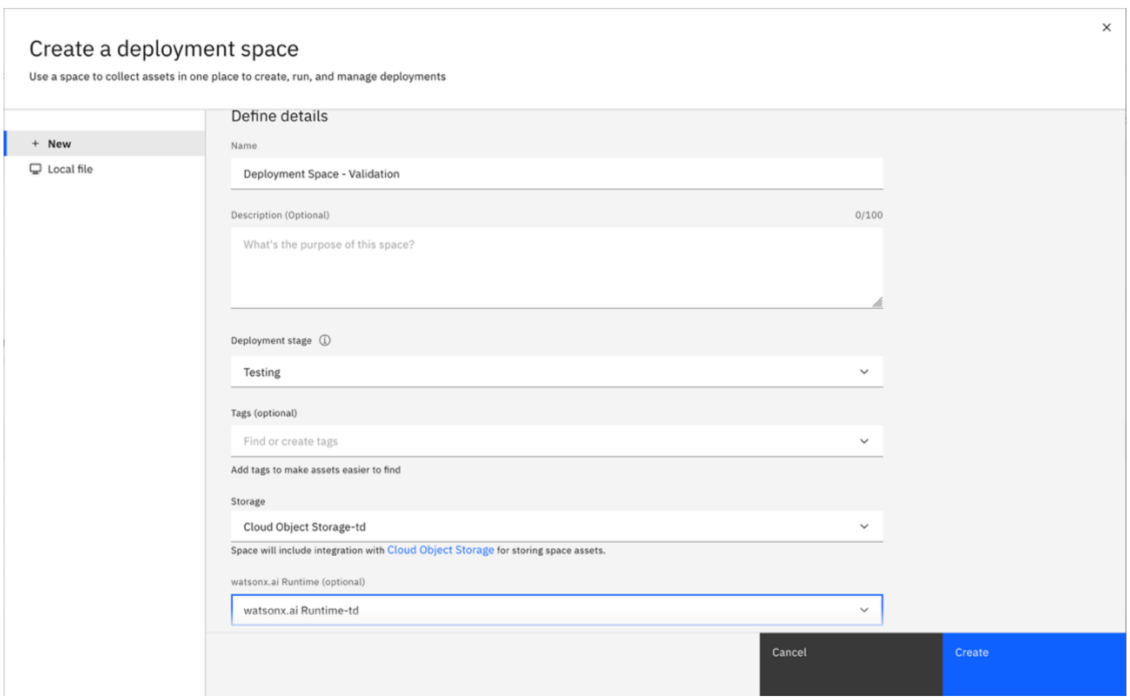


Click on the button shown below to create a New space.



You will see the window shown on the next page.
Fill in the details as specified below:

- Name the new deployment space something appropriate
- Select the storage service and machine learning service from the options that are available under the respective dropdown menus on the right side
- Select the Deployment stage as ‘Validation’
- Select the Storage as “Cloud Object Storage-td”
- Associate the “watsonx.ai Runtime-td service”
- Click **Create**

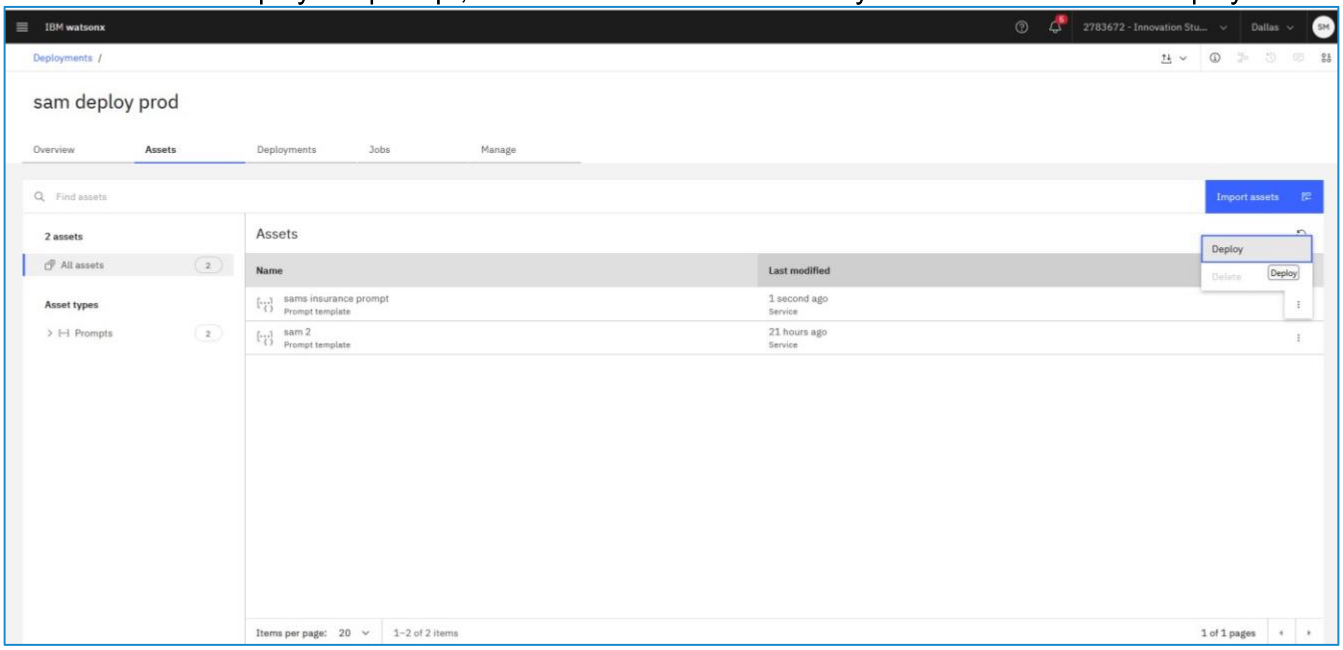


Once you have created this deployment space, press Save to associate it to the workspace **Validation.**

-

Next, find your second prompt version (the one created during the Develop phase) within **Projects** and promote it to this new workspace. Select the Deployment space you created earlier.

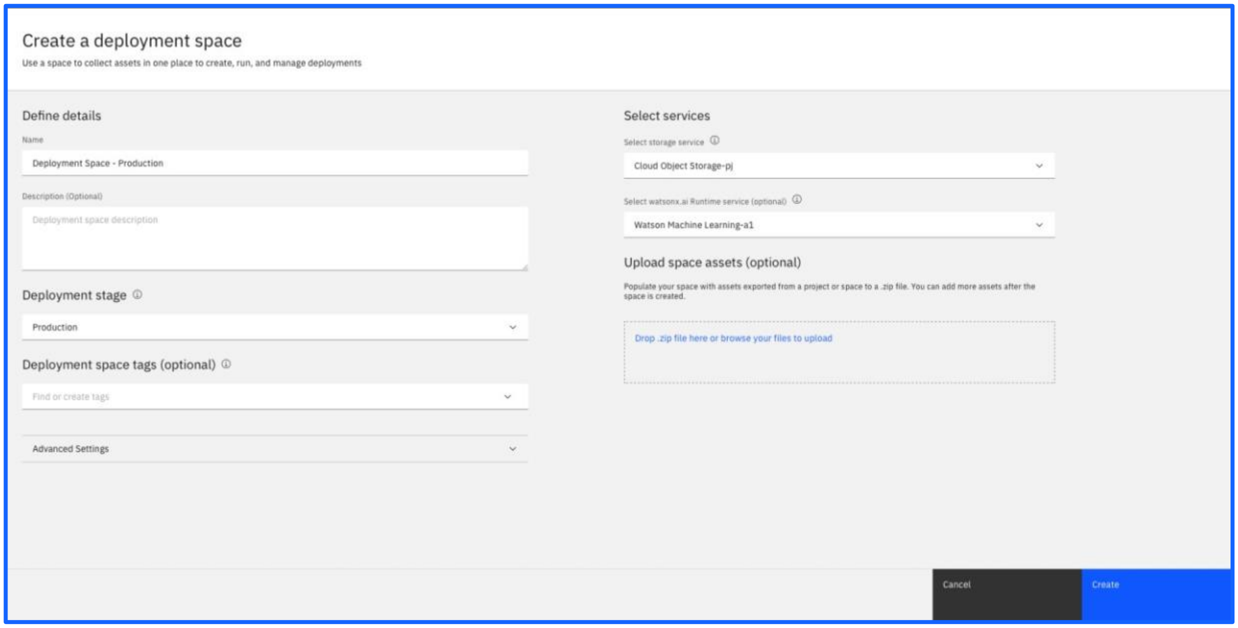
We then need to deploy the prompt, click on the 3 buttons next to your asset and choose deploy:



Now repeat the same steps in **Deployment Space** for a Workspace in **Operate**:
Fill in the details as specified below:

- Name the new deployment space something appropriate
- Select the storage service and machine learning service from the options that are available under the respective dropdown menus on the right side
- Select the Deployment stage as 'Production'

- Press Create.
-



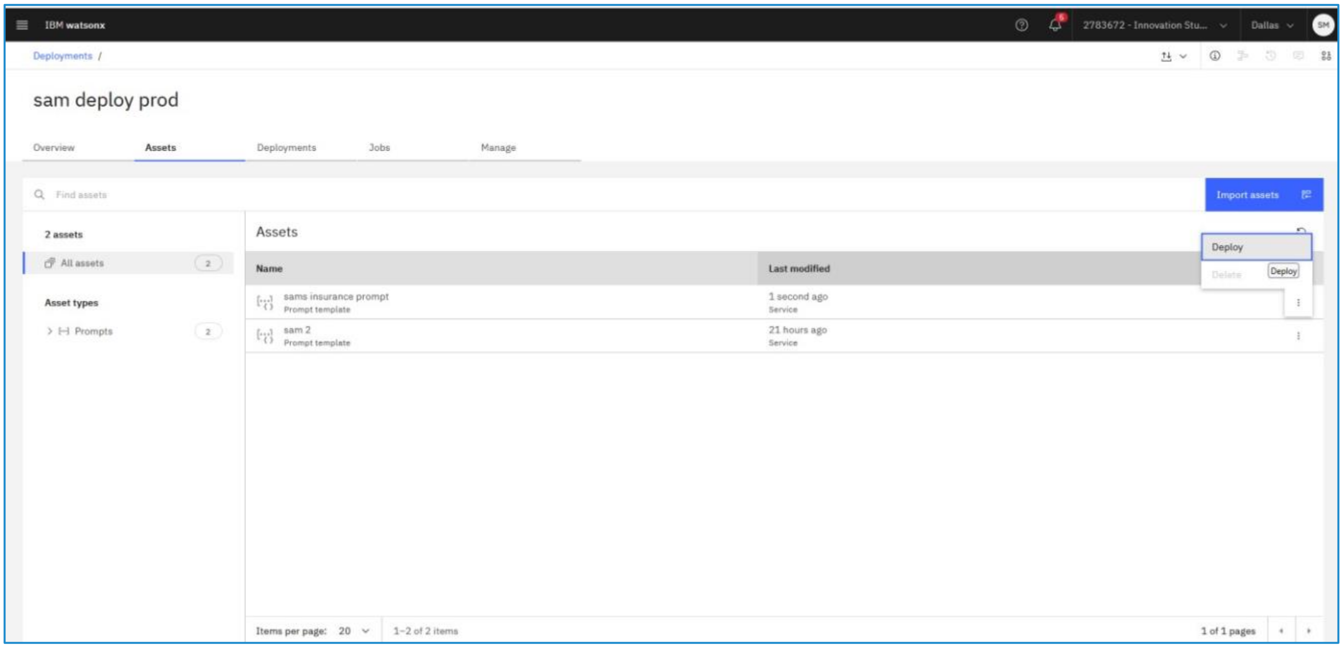
-

Once you have created this deployment space, press Save to associate it to the workspace **Operate.**

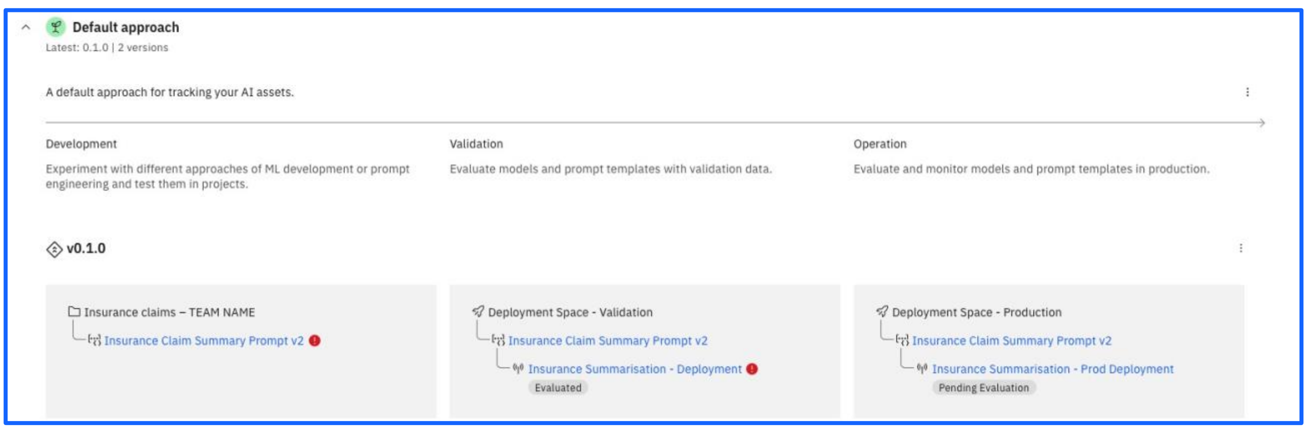
-

Next, find your second prompt version (the one created during the Develop phase) within **Projects** and promote it to this new workspace. Select the Deployment space you created earlier.

We then need to deploy the prompt, click on the 3 buttons next to your asset and choose deploy:



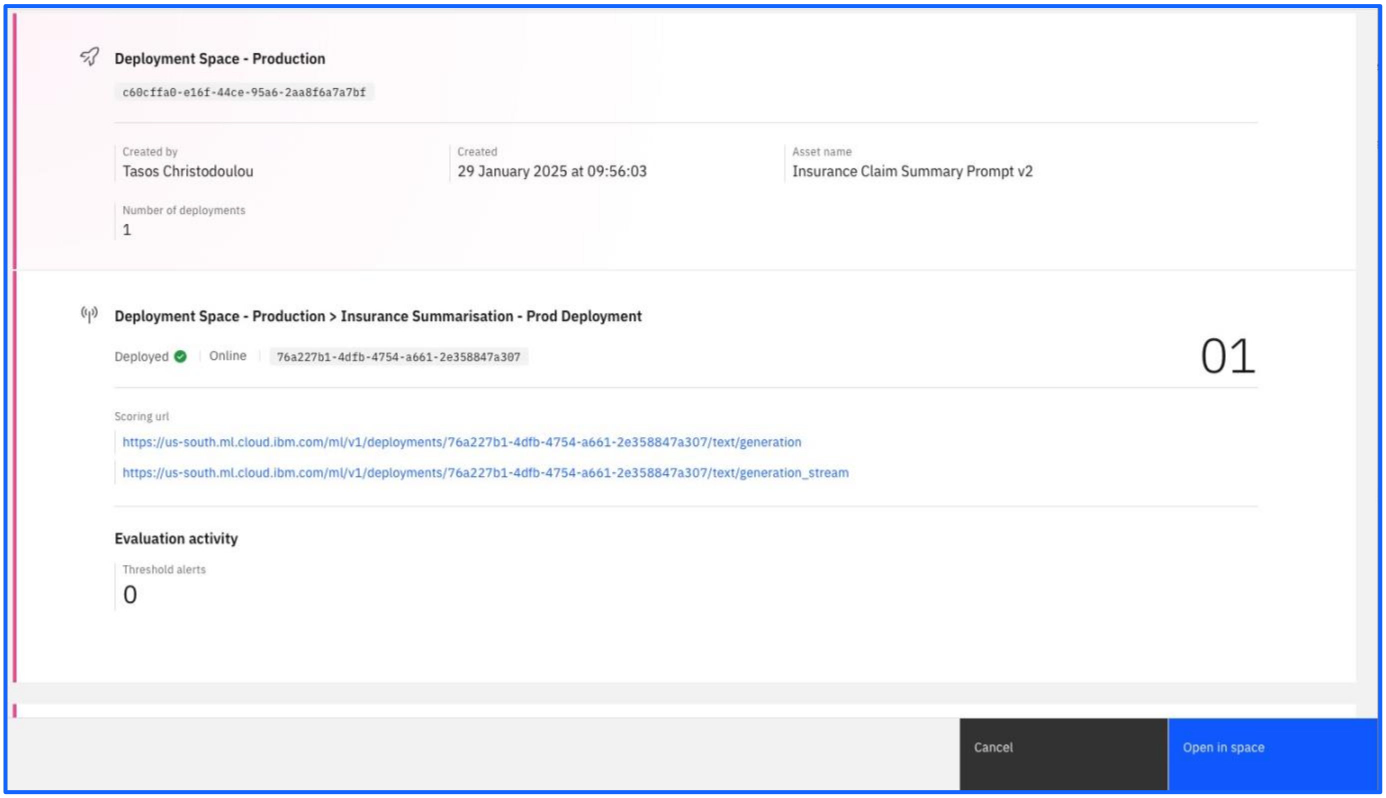
To confirm the steps worked, you should see the prompt template in the *Production deployment space* on the far-right panel, with the *Pending Evaluation* label



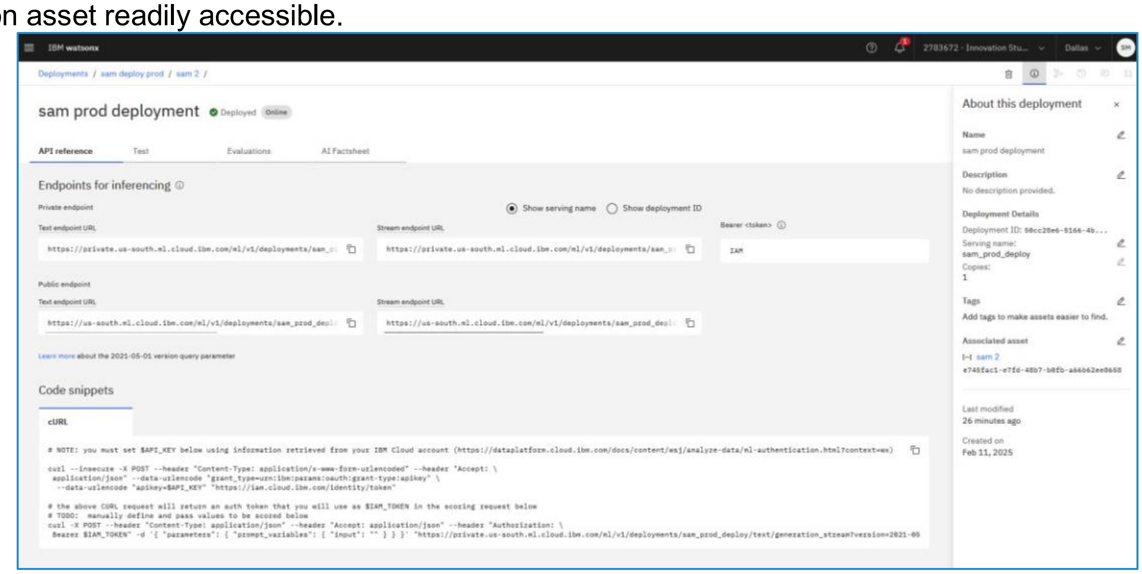
Well done, you have successfully deployed your first asset into production.

Managing Assets in production

Click on the recently created production deployment space (the one that is shown to be pending evaluation), and click Open in Space.

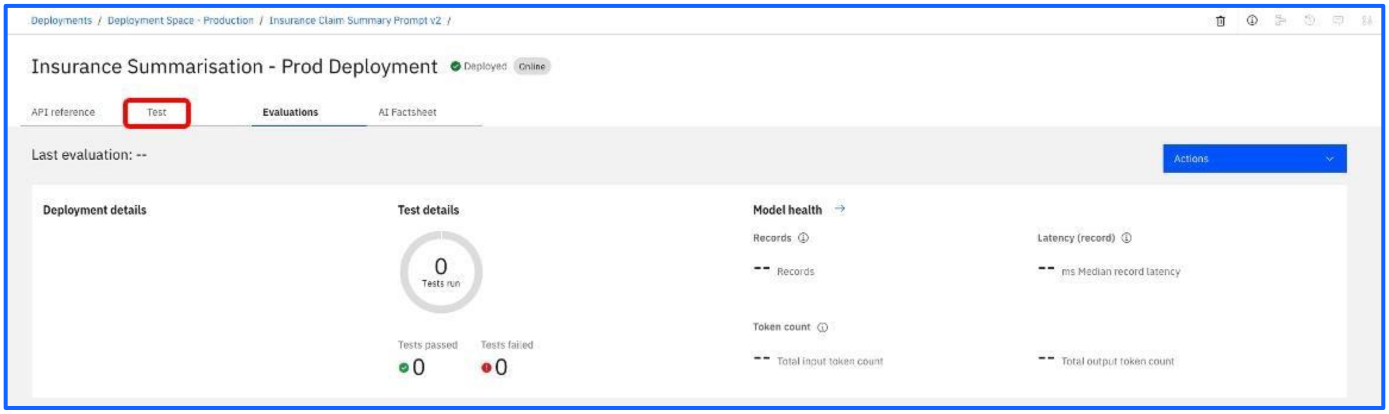


The first thing to talk through is the API reference. As discussed, there is no point in developing a model if it isn't easy to access. With the deployment space, you can see that there are public and private API endpoints to allow an application to call the model. Making it even simpler, there are code snippets available in multiple languages to allow this to be baked into the application code. Thus, making our production asset readily accessible.

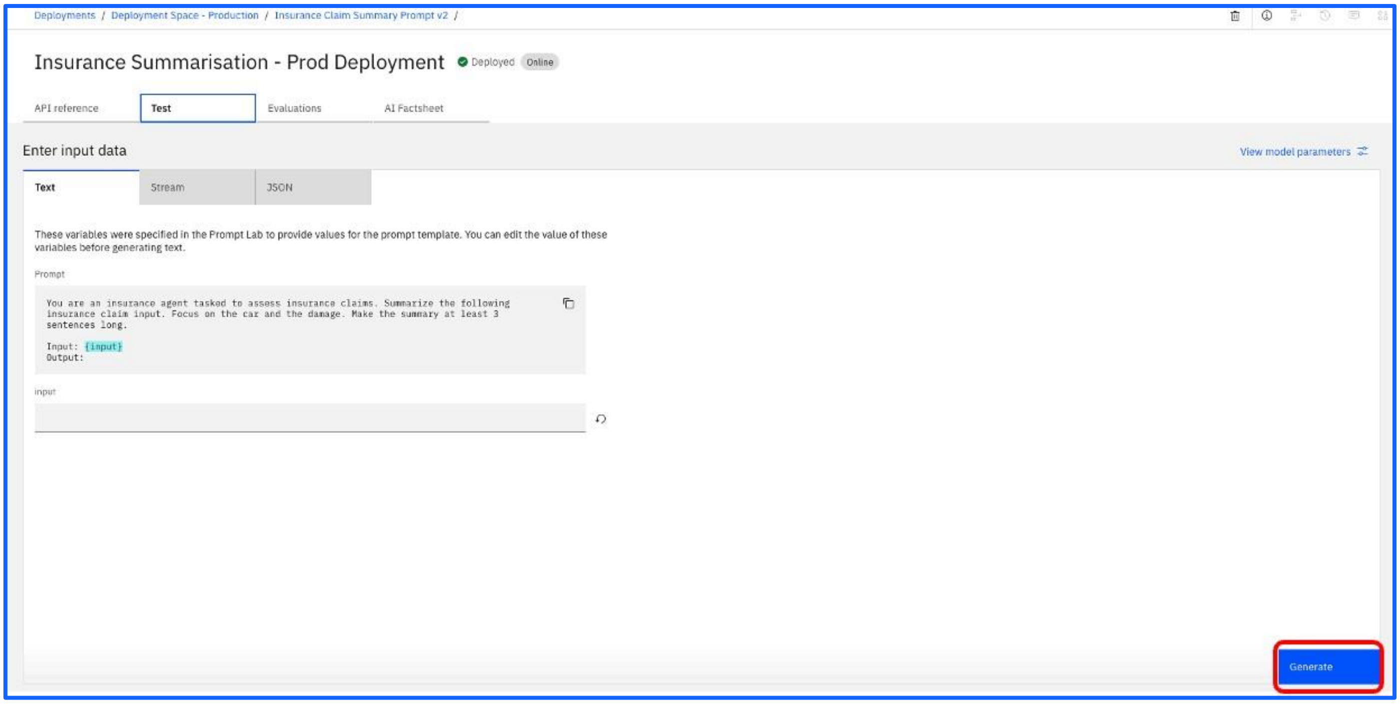


Now, we can test it to make sure it is functioning as expected.

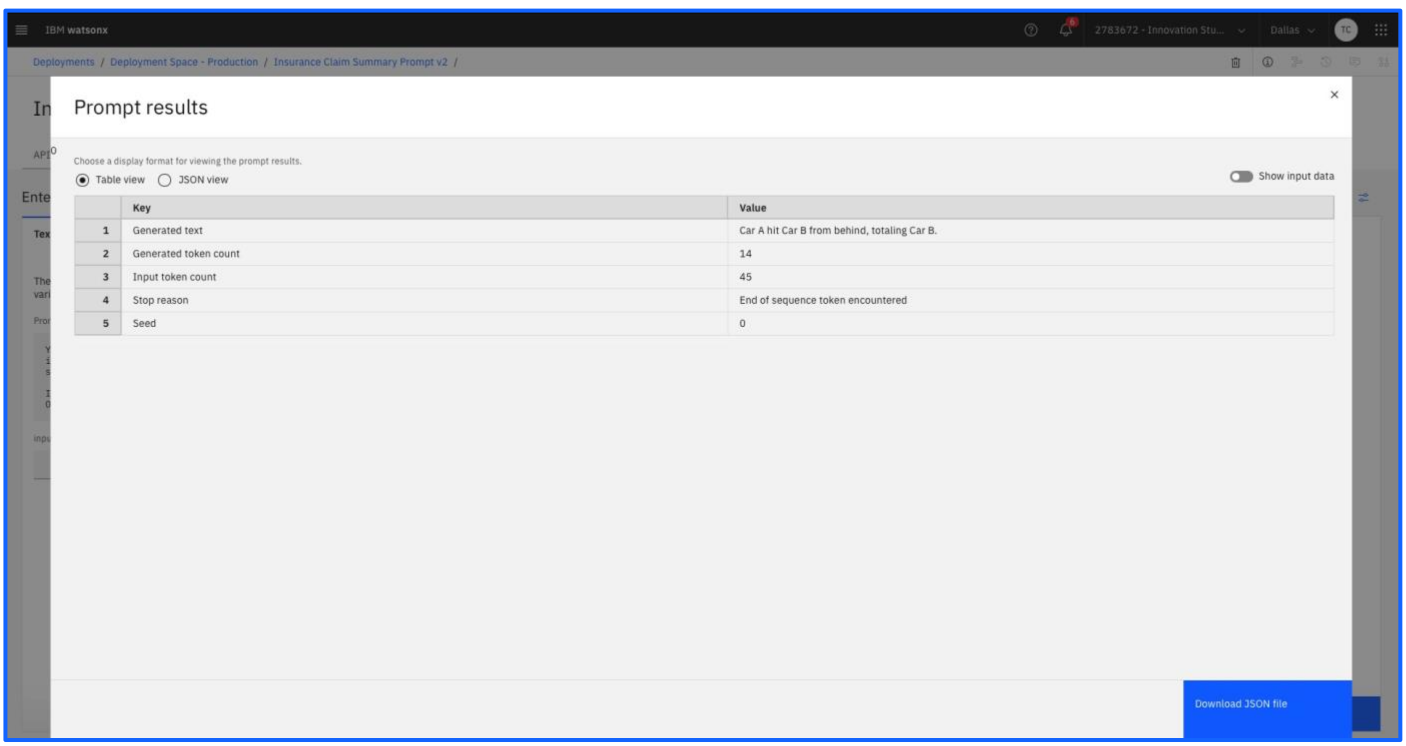
Navigate to the **Test** tab within this **Deployment Space** to ensure the function is working as expected.


Deployments / Deployment Space - Production / Insurance Claim Summary Prompt v2 /
Insurance Summarisation - Prod Deployment
Deployed
Online
API reference
Test
Evaluations
AI Factsheet
Last evaluation: --
Actions
Deployment details
Test details
0
Tests run
Tests passed
0
Tests failed
0
Model health
Records
-- Records
Latency (record)
-- ms Median record latency
Token count
-- Total input token count
-- Total output token count

Within this tab, click on the Generate button to make sure the prompt works.



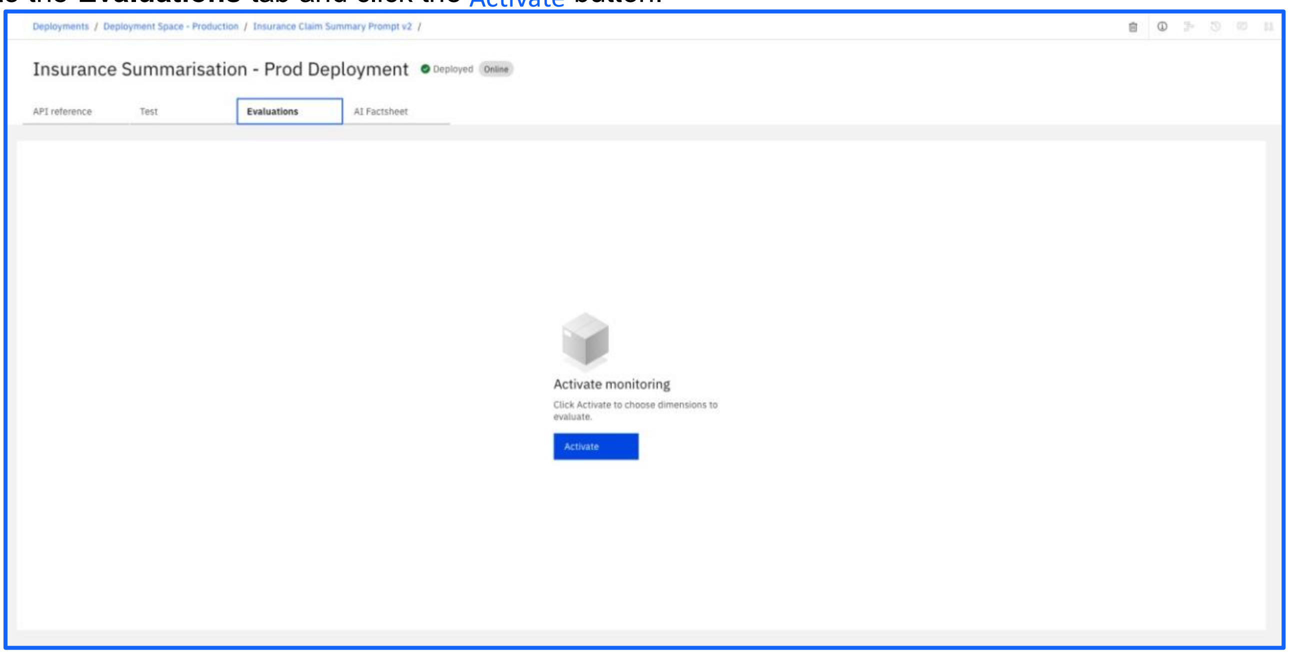
Review the schemas of the payload and the feedback data. The datasets (csv) files need to follow these schemas when we upload them.



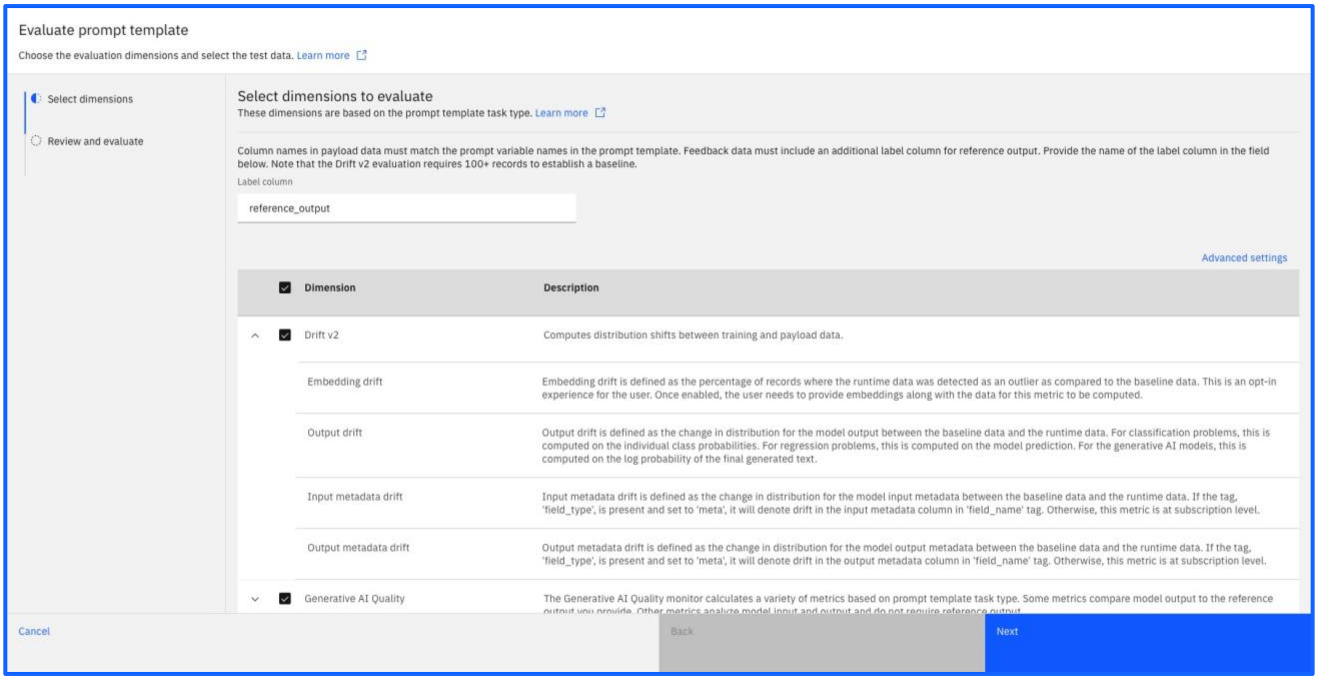
We are now ready to prime the prompt deployment with baseline data.

Key	Value
Generated text	`Car A hit Car B from behind, totalling Car B.`
Generated token count	`14`
Input token count	`45`
Stop reason	`End of sequence token encountered`
Seed	`0`

Go to the **Evaluations** tab and click the Activate button.

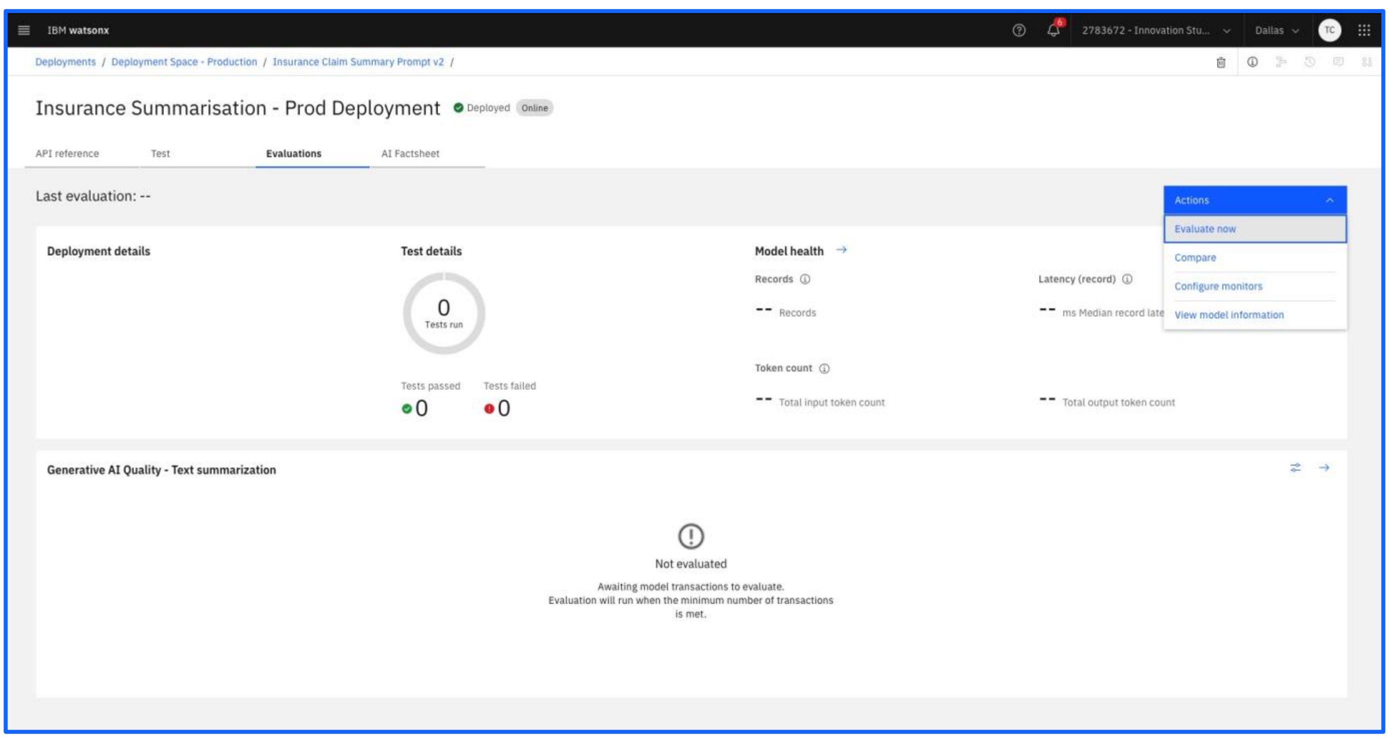


You should notice that now you have activated the monitoring, you will automatically have a new dimension called **Drift v2** as an additional option. As shown in the screenshot below, the description gives details as to how it measures distribution shifts between training and payload data – which we must now upload.



Once you have walked through the steps, click Activate to activate monitoring of the prompt in production. This asset is now being monitored in production on a scheduled basis, metrics captured will be sent to the Governance Console and tracked within the Use Case you created earlier.

Note: The Prompt Results that you see in the lab may differ slightly to what you see above.

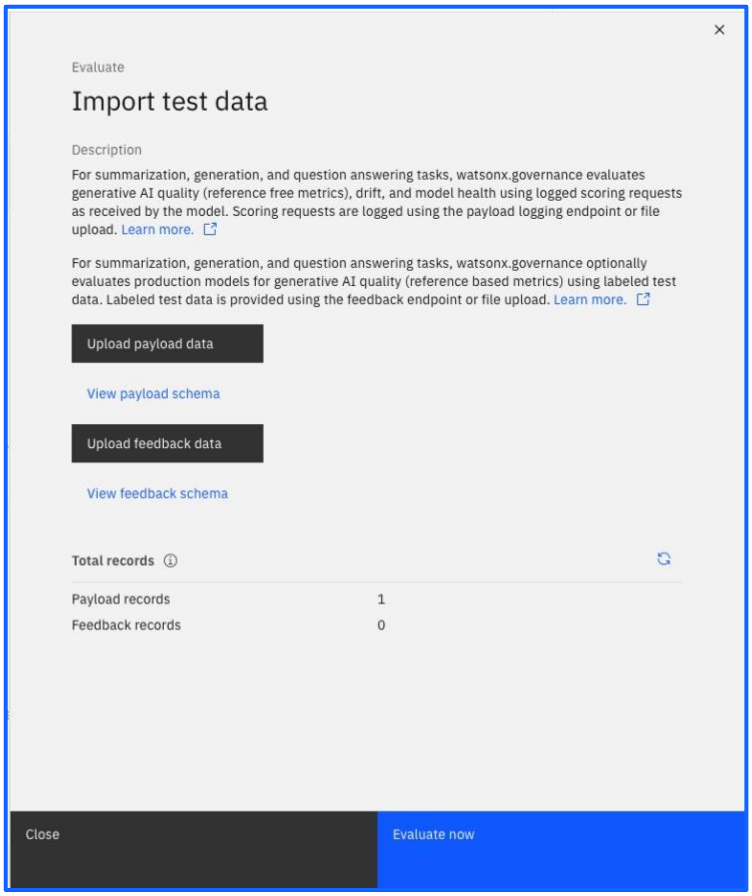


Drift Monitoring (Optional – feel free to progress to the Governance Console section)

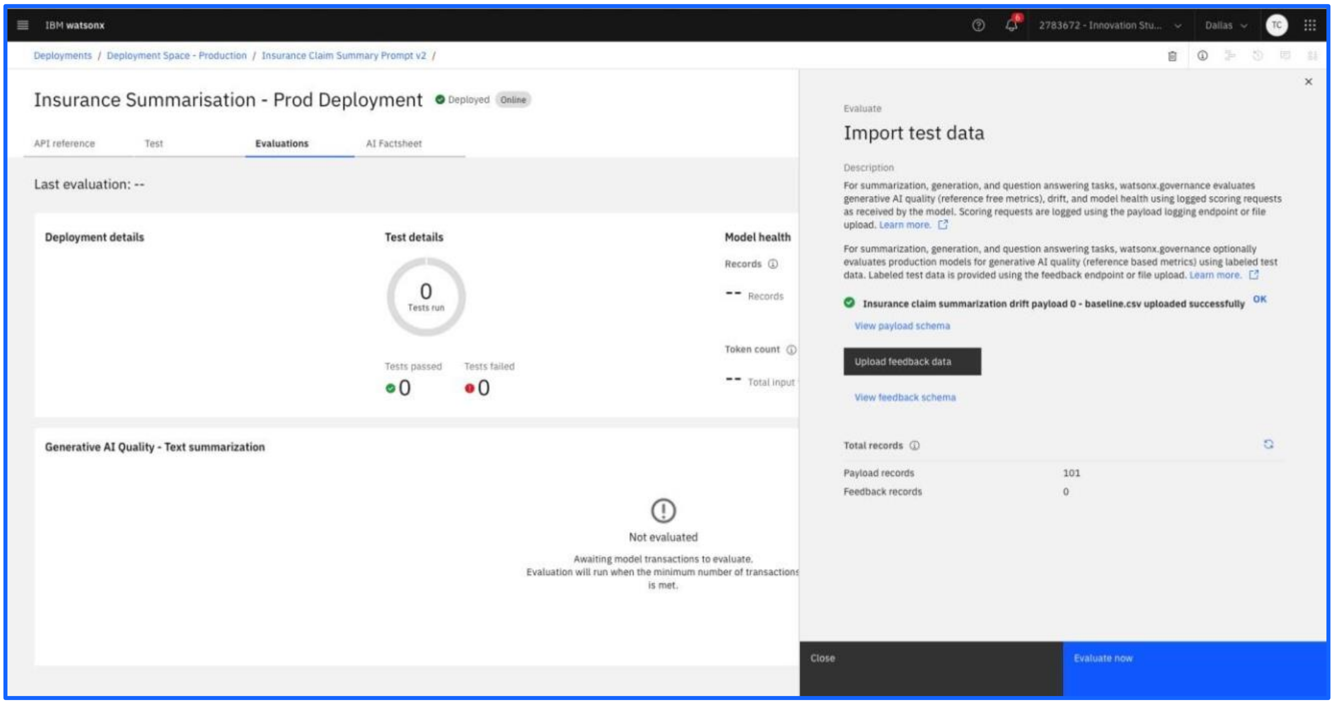
To demonstrate drift we need to establish a baseline. Consider this as the ground truth for the data. That is to say at time of production the data is up to date and accurate. Over time of course, Drift will be introduced as the data becomes outdated and no longer relevant. The calculation of this works off the baseline data.

Select Upload Payload data and then upload the file called:

`Insurance claim summarisation drift payload 0 - baseline.csv`

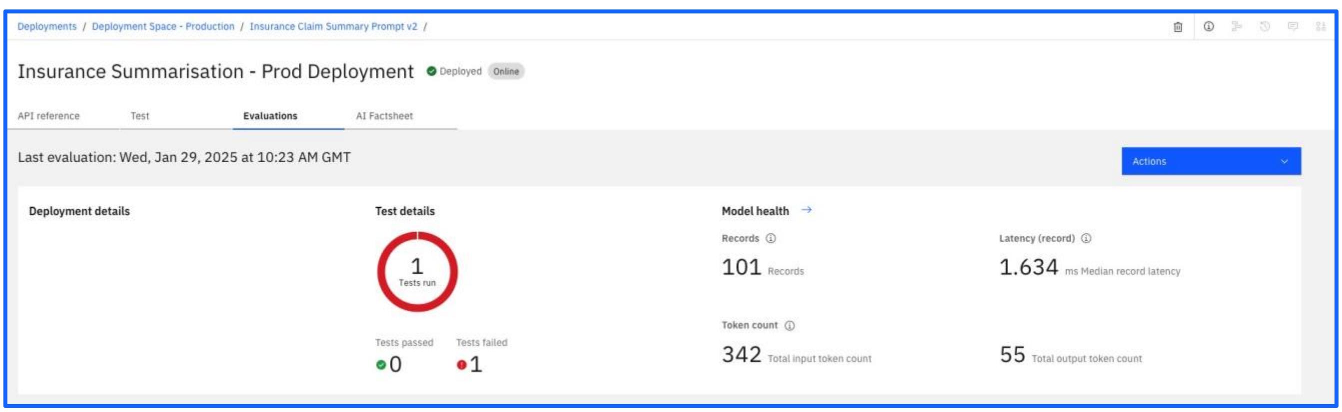


Note: We are not uploading any feedback data at this point.



Once that has been successfully uploaded, Click on the Evaluate now button.

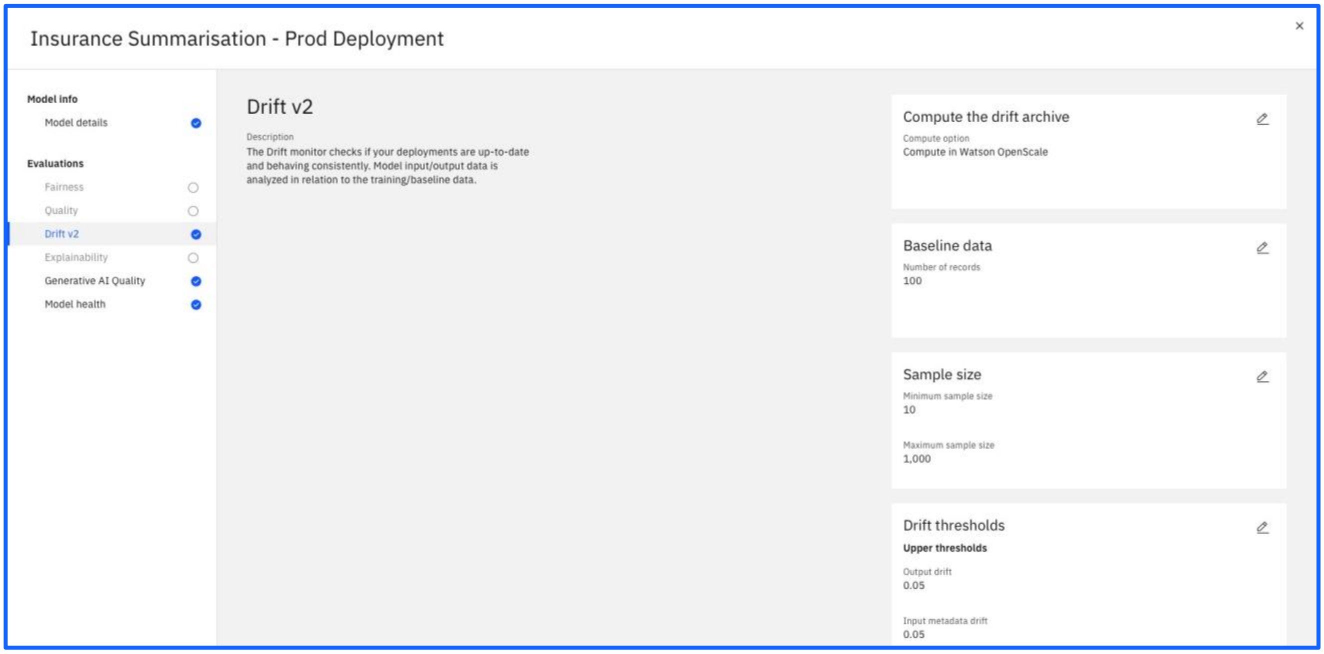
The **Evaluation** will fail due to a violation of the **Coverage** Metric. We could of course amend the threshold of the metric for our **Evaluation** to Pass, however for the purposes of our workshop will accept it as it is.



If you now click on the Configure monitors button in the Actions dropdown menu, you will see **Drift v2** enabled.

As shown in the screenshot below, you will have a baseline of 100 records. You can edit the threshold values here as well which would be important in a real-world scenario, to adjust them to an appropriate level given the use case and prompt.

For now, we'll keep the default values.

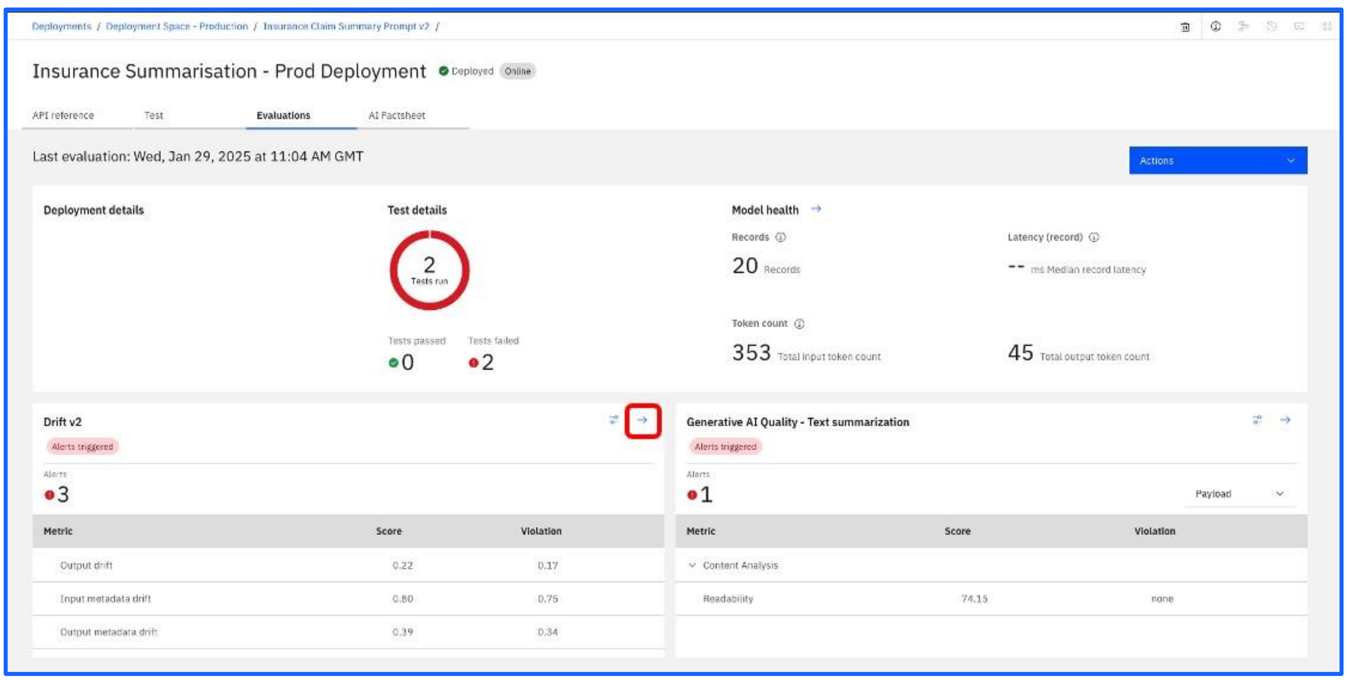

Insurance Summarisation - Prod Deployment
Model info
Model details
Evaluations
Fairness
Quality
Drift v2
Explainability
Generative AI Quality
Model health
Drift v2
Description
The Drift monitor checks if your deployments are up-to-date and behaving consistently. Model input/output data is analyzed in relation to the training/baseline data.
Compute the drift archive
Compute option
Compute in Watson OpenScale
Baseline data
Number of records
100
Sample size
Minimum sample size
10
Maximum sample size
1,000
Drift thresholds
Upper thresholds
Output drift
0.05
Input metadata drift
0.05

The system will now monitor invocations of the prompt and detect any potential drift. We will simulate an example of this by uploading the following datasets.

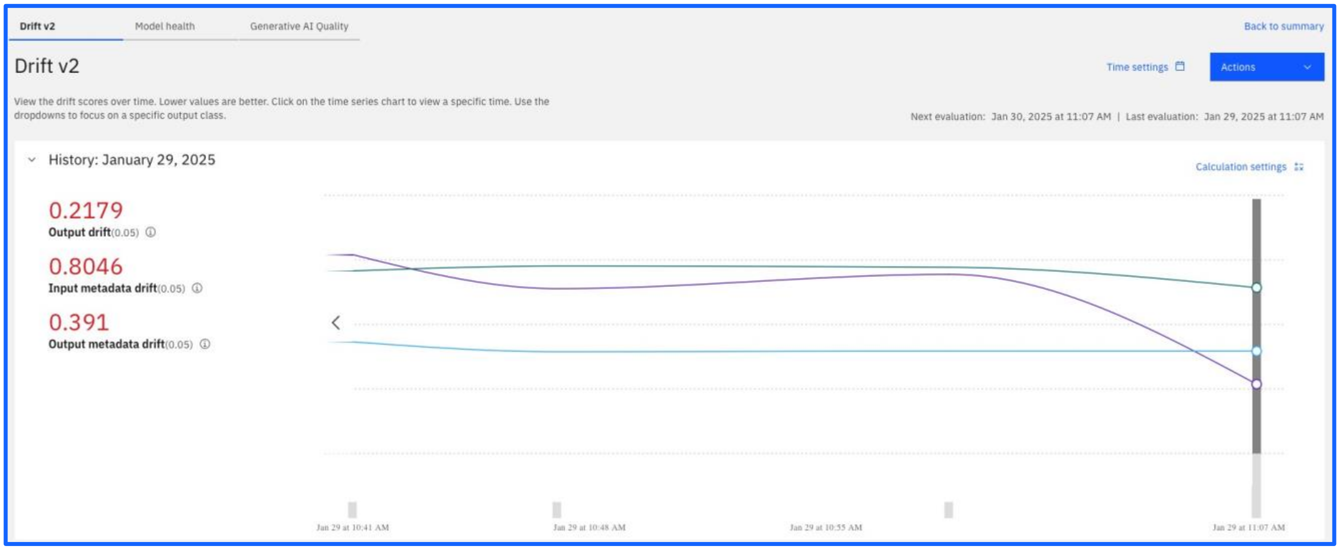
- `Insurance claim summarisation drift payload 1.csv`
- `Insurance claim summarisation drift payload 2.csv`
- `Insurance claim summarisation drift payload 3.csv`
- `Insurance claim summarisation drift payload 4.csv`

Upload all of the csv files as payload data, but be sure to let some time pass between each run, to be sure that the evaluation shows a meaningful timeline (and drift).

Once you have run these evaluations, you will see a sub-tab under **Evaluations** to show a diagram for the drift of input and output. Click on the blue arrow button in the corner of the Drift v2 section to get a more detailed view.



Make sure you have selected the right time settings to see the drift change over time. If you have any issues, ask for assistant from your facilitators.



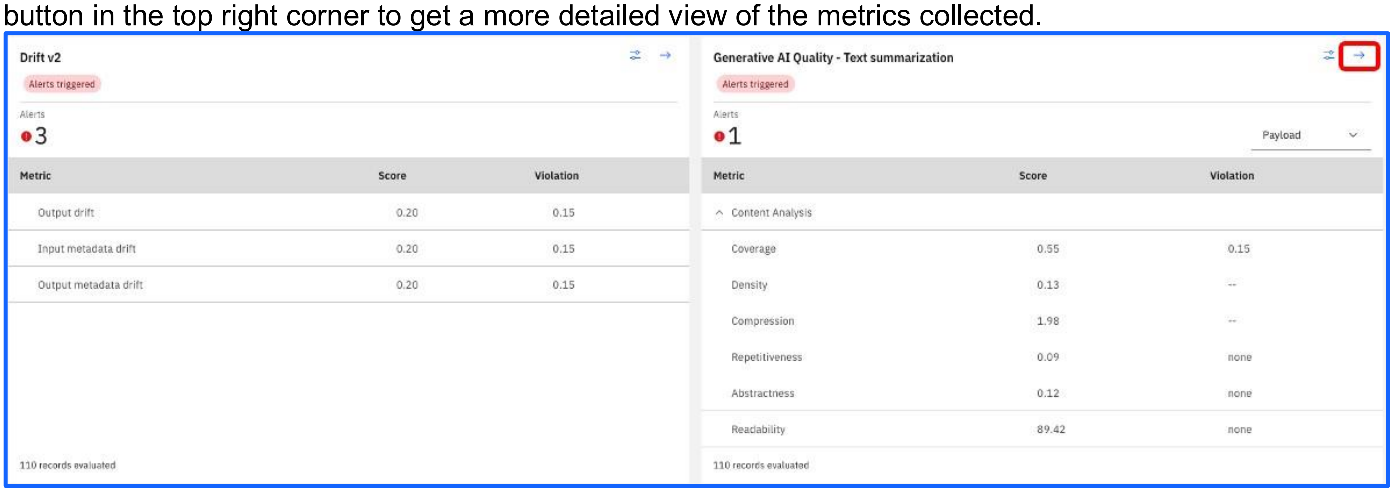
There are only three criteria which go into the drift calculations: *output*, *output metadata drift* and *input metadata drift*.

These metrics do not offer any information regarding the *quality* of the prompt – In order to measure that, we need to upload *feedback* data.

Navigate back to the ***Evaluations*** tab and once again click on the Evaluate now button within the Actions dropdown menu. Upload the following file:

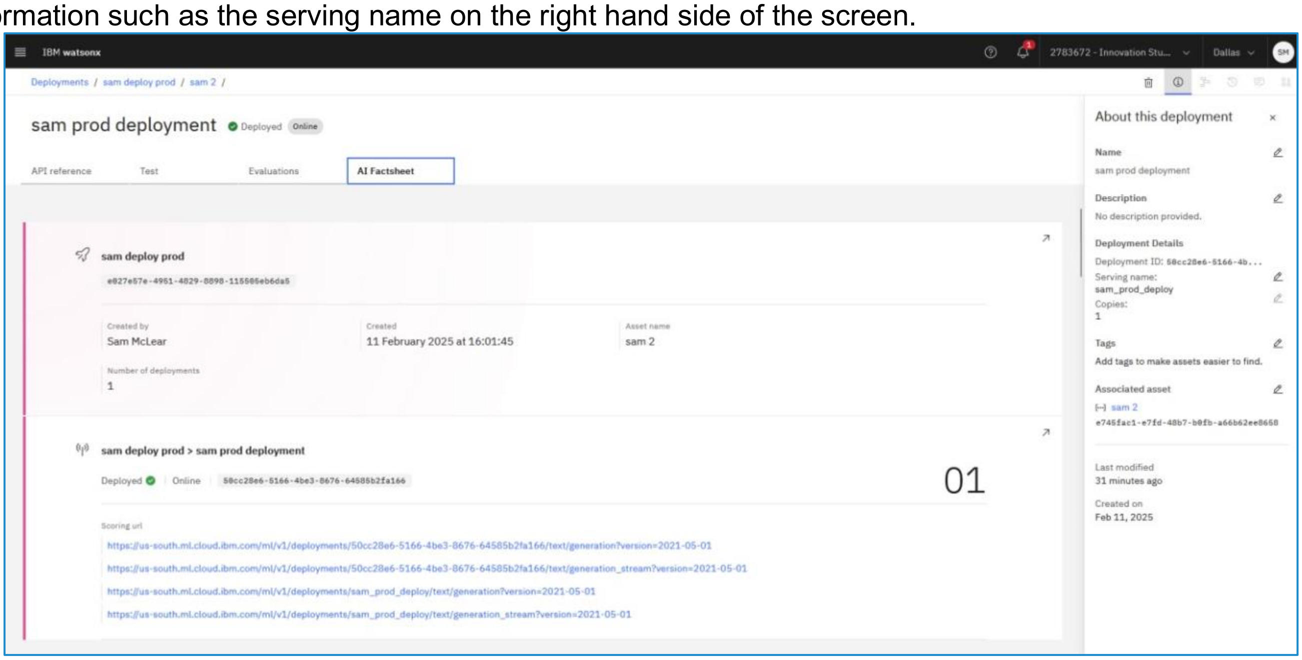
- `Insurance claim summarisation feedback.csv`

When this run is finished, you will see a panel with **Generative AI Quality - Text summarisation** metrics on the lower right of the screen, next to the Drift v2 section. If the option allows, click on the ® blue arrow button in the top right corner to get a more detailed view of the metrics collected.

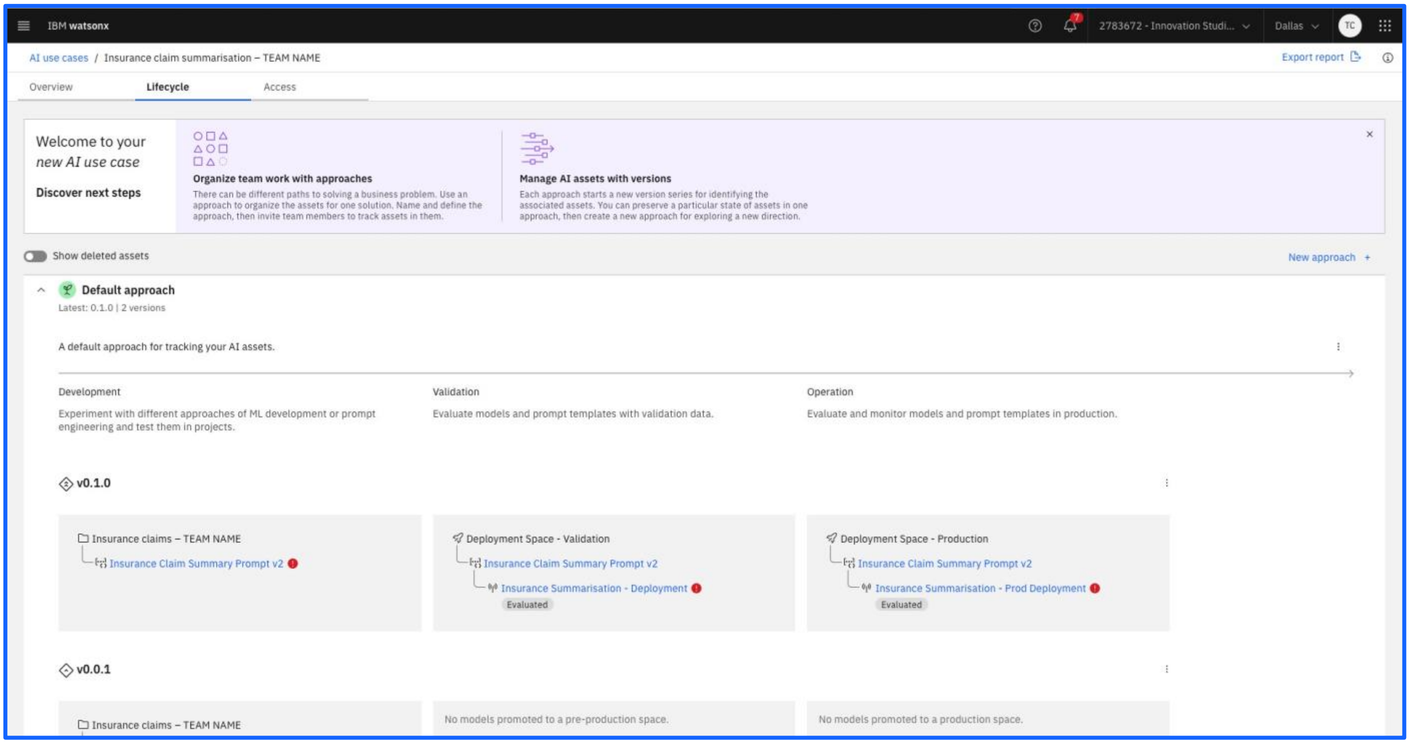


Finally, we are ready to advance the status of the use case, to identify it as a model in **Operation**.

The final tab, AI Factsheet, captures all of the facts about the AI Asset. It is also possible to view information such as the serving name on the right hand side of the screen.



The prompt is deployed in production and is being monitored for both drift and quality. As seen back in the lifecycle screen.

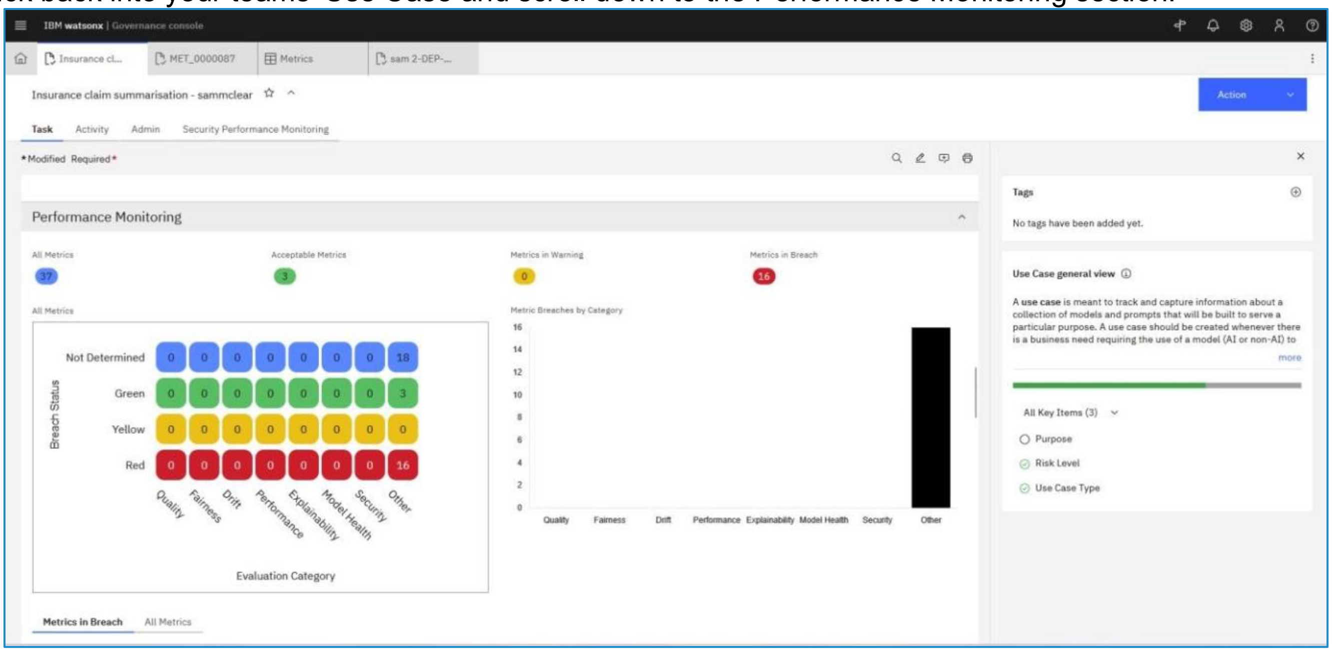

IBM watsonx
2783672 - Innovation Studi...
Dallas
AI use cases / Insurance claim summarisation – TEAM NAME
Export report
Overview
Lifecycle
Access
Welcome to your new AI use case
Discover next steps
Organize team work with approaches
There can be different paths to solving a business problem. Use an approach to organize the assets for one solution. Name and define the approach, then invite team members to track assets in them.
Manage AI assets with versions
Each approach starts a new version series for identifying the associated assets. You can preserve a particular state of assets in one approach, then create a new approach for exploring a new direction.
Show deleted assets
New approach +
Default approach
Latest: 0.1.0 | 2 versions
A default approach for tracking your AI assets.
Development
Experiment with different approaches of ML development or prompt engineering and test them in projects.
Validation
Evaluate models and prompt templates with validation data.
Operation
Evaluate and monitor models and prompt templates in production.
v0.1.0
Insurance claims – TEAM NAME
Insurance Claim Summary Prompt v2
Deployment Space - Validation
Insurance Claim Summary Prompt v2
Insurance Summarisation - Deployment
Evaluated
Deployment Space - Production
Insurance Claim Summary Prompt v2
Insurance Summarisation - Prod Deployment
Evaluated
v0.0.1
Insurance claims – TEAM NAME
No models promoted to a pre-production space.
No models promoted to a production space.

Governance Console

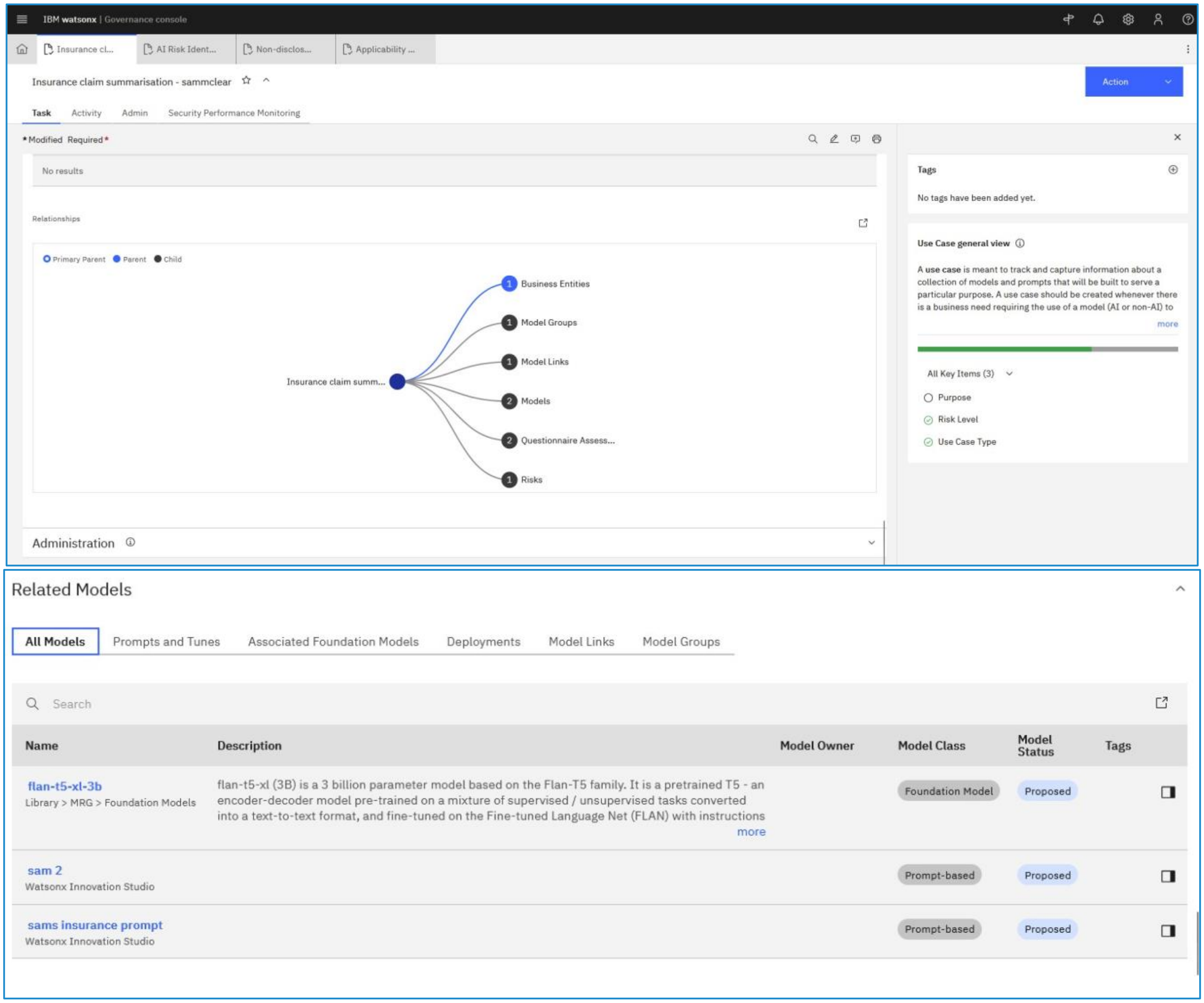
For the last part of the lab, you will see how the work carried out throughout this lab was automatically captured in the Use Case on the Governance Console.

Navigate back to the Governance Console either from the Use Case view or this link https://ibm.biz/BdGmZc

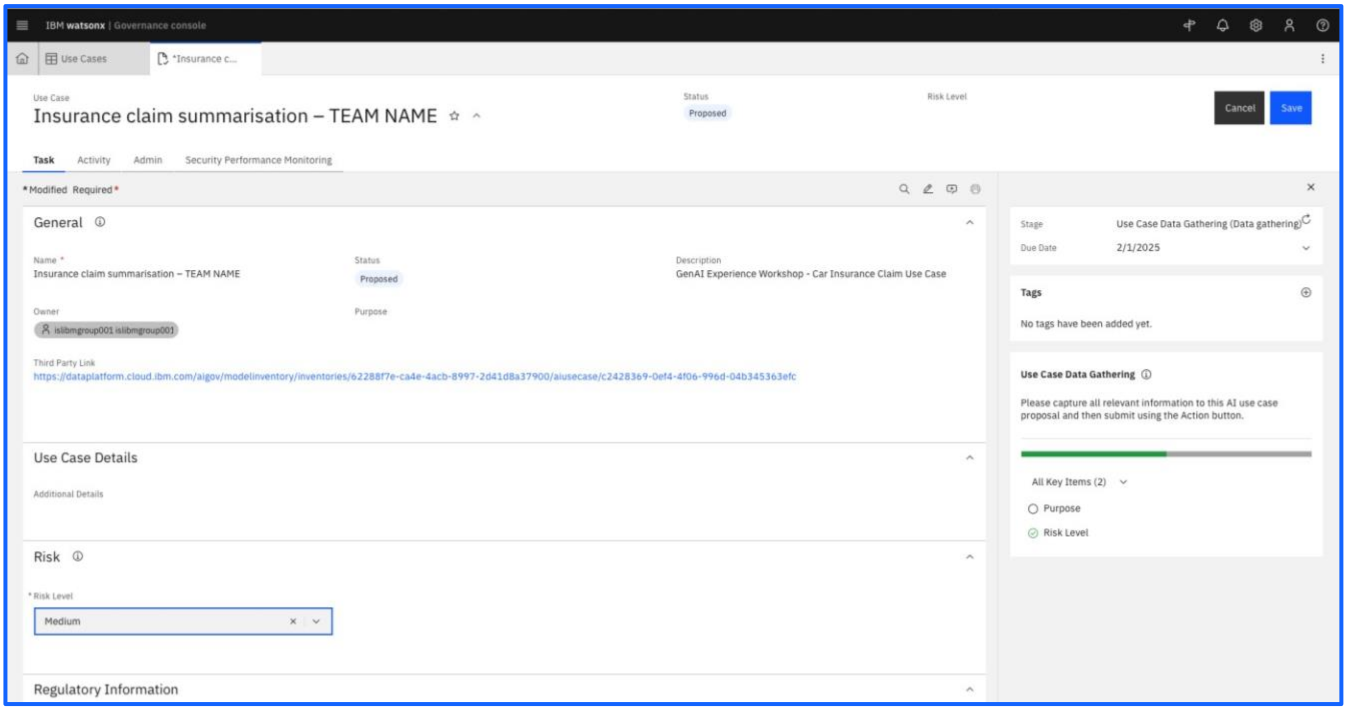
Click back into your teams' Use Case and scroll down to the Performance Monitoring section:



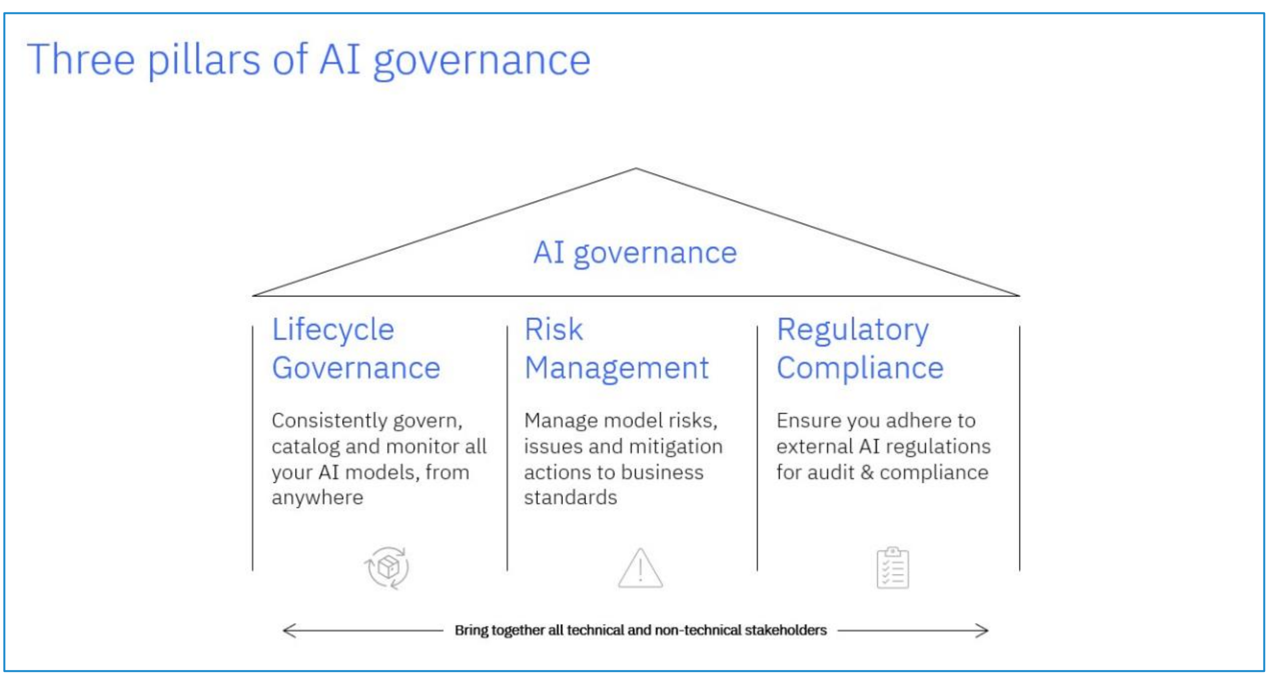
Scroll down to the Relationships screen. You can see the different objects that are now tied to the Use Case. To drill down further into the models themselves and their performance, click on the name and additional objects will appear.



Given the evaluation results, we may want to retrospective actions such as Issue Management or change the Risk level to Medium.



Congratulations! You have completed the Workshop. You have built, deployed and managed an AI Use Case tackling the 3 pillars of AI Governance.

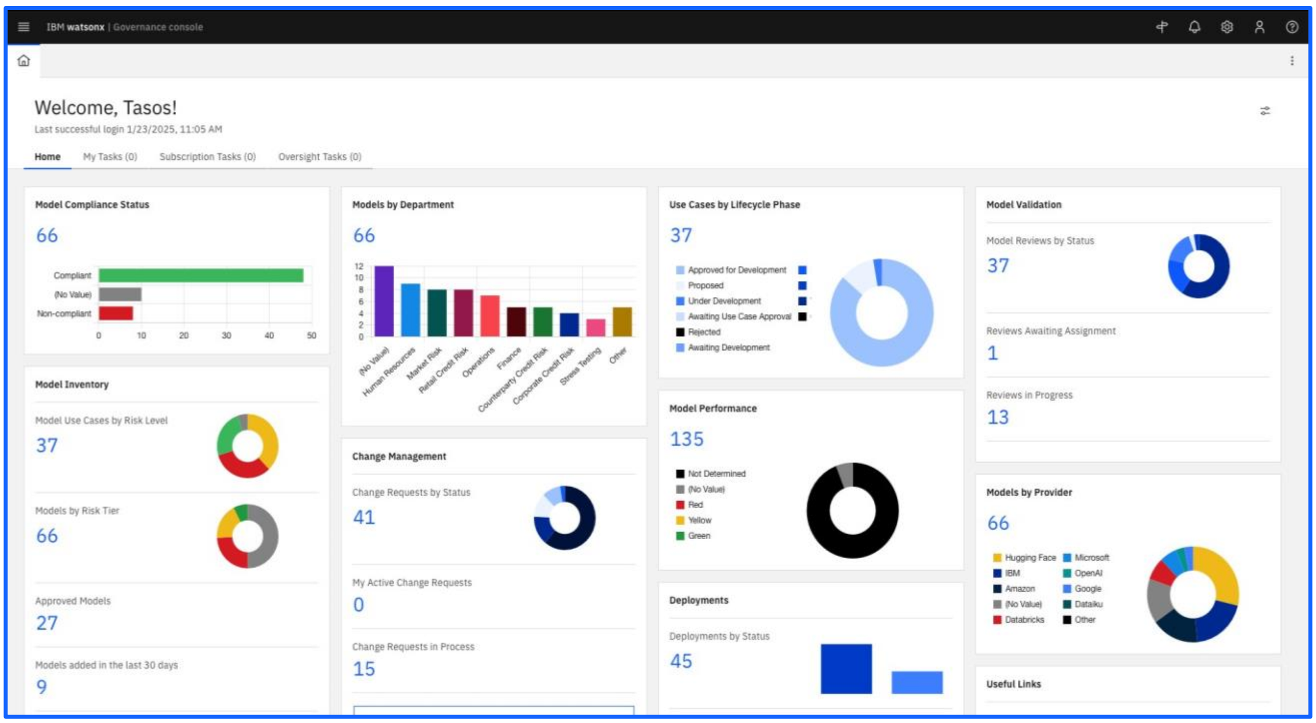


Optional steps: Navigating the UI of the Governance Console

Get Started with your watsonx.governance Workshop

1. Open **Chrome** or **FireFox** and navigate to the following Link: https://f032c83f-74ea-45cb-a1bc-641daaa13009.us-east-1.aws.openpages.ibm.com
2. If not logged in already, **Logon** with your IBM Cloud credentials following Credentials.

watsonx.governance User Interface



After login, you will be displayed the Homepage tab, and in it we will see four tabs.

- **HOME** tab. The dashboard displays panels with content that is personalized for each user, intended to be relevant information for the user. *You may move and arrange them, as per your preferences. Later we will see how to create a new panel.*

The next three Task related tabs, are displayed tasks that I am notified during the workflow lifecycle of the solution components *(Click on each tab to see the content):*

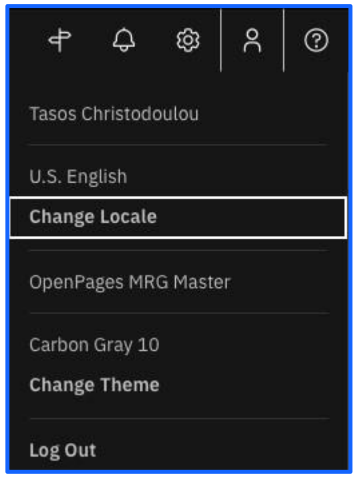
- **My Tasks:** activities that I have been assigned. It is action that I must perform.
- **Subscription Tasks:** activities that I want to know that happen, however I do not have to do anything.

- **Oversight Tasks:** I have full visibility on the whole lifecycle of an element (i.e., a Model Use case). I do not have any task assigned, just visibility.

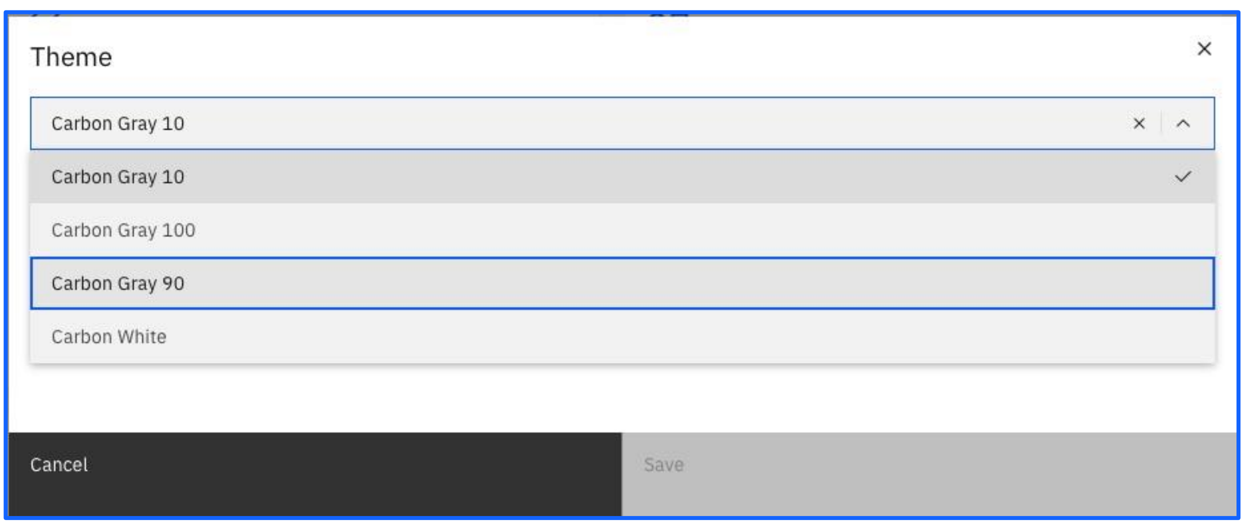
Changing Colour Theme

1. Themes are used to customize the colours that are displayed in OpenPages (watsonx.governance).

2. Users can apply a theme by using the  > Change Theme task.



3. Using the Drop-down list, change your Theme to Carbon Gray 90.

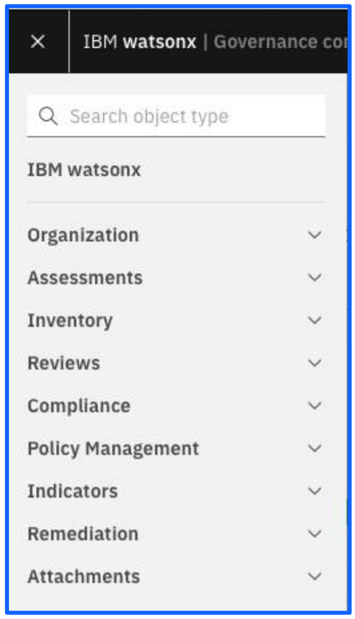


If you do not like the contrast, you may choose any other.

Themes can be configured by the client to allow a corporate colouring of the solution.

The Primary Menu

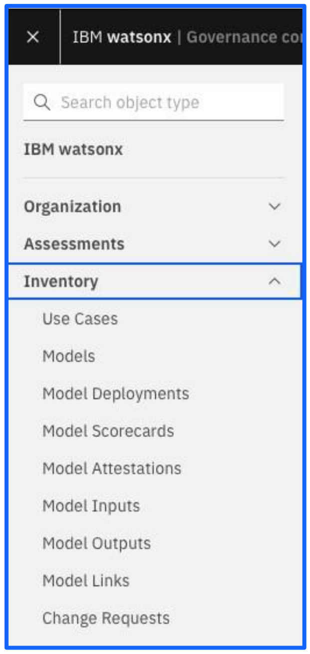
1. On the top left corner, click on the Primary Menu (also known as the “Hamburger menu”)



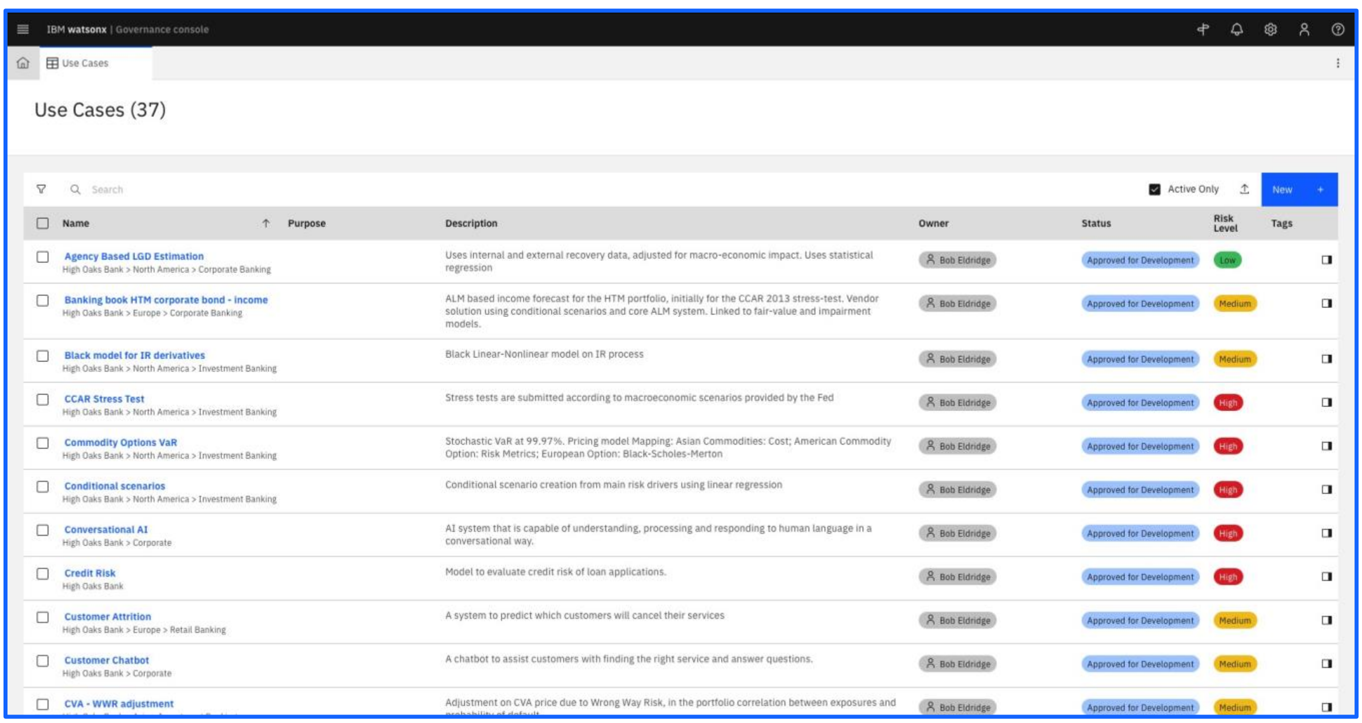
2. The Primary menu lists the categories of object types that you can access in IBM OpenPages® (watsonx.governance).

   Expand and collapse the categories to see the object types. Seeing more or less objects depends on your profile.

3. When you click an object type, for example, **Use Cases** under the section Inventory, a list of objects opens.



4. Click on , expand **Inventory**, click on **Use Cases.**



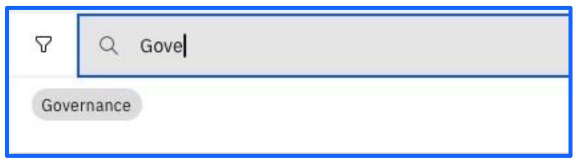
The list of Use Cases Display. This type of view is called Grid View. When you select an object type from the Primary menu, the **Home**, or a **Task View**, a Grid View is displayed.

There are several actions that might be taken from a grid view.

Search Use Cases

1. Whilst in the Use Cases Tab, place the cursor on the Search box.

2. Start Typing **"Governance"**. You are presented with a Tag Suggestion

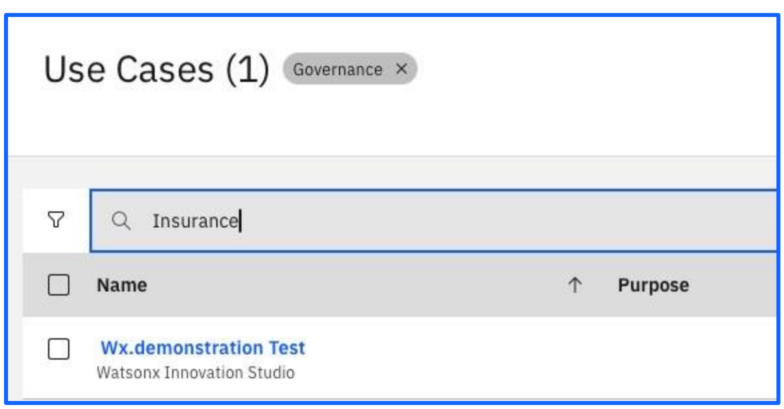


3. Select the Tag "Governance".

Notice the filter indicator displayed.

4. Place the cursor on the Search box once again.

5. Type "**Insurance**". This is a Text Filter.



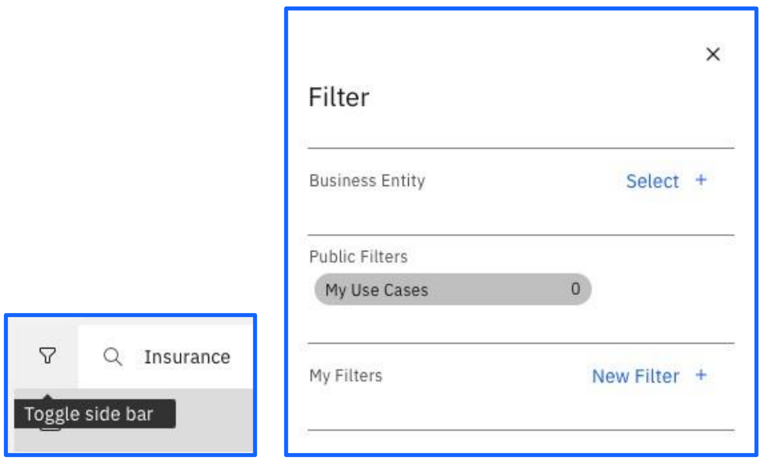
Notice that both filters have been applied.

6. Remove the filter display **Governance**.

Notice that now we have only a text search based on the word “insurance”.

These filters are **Ad hoc** and cannot be reused, as you must define them each time. Ad hoc filters are private filters that are not saved and only used in one session. Use an ad hoc filter if you do not frequently access the search criteria.

7. Click on the funnel symbol on the left to open the filter side bar.



Notice that we have a section named “Public Filters” and “My Filters” (private filters)

An administrator defines public filters. Users can customize private filters for their own use.

Use and customize private filters to find objects based on criteria on field values on objects and, optionally, parent objects. Users can customize private filters for their own use.

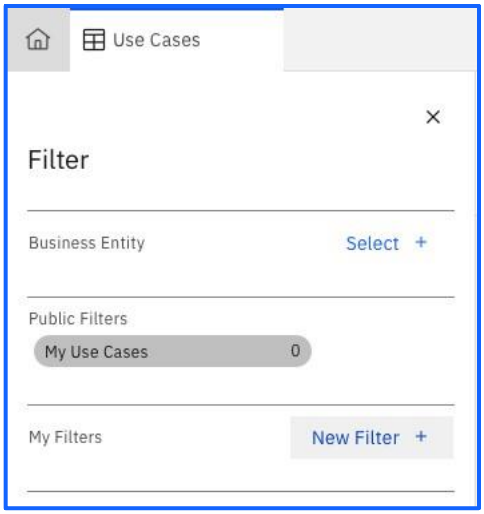
Only one public or private filter can be applied at a time.

You cannot simultaneously use a public filter and a private filter.

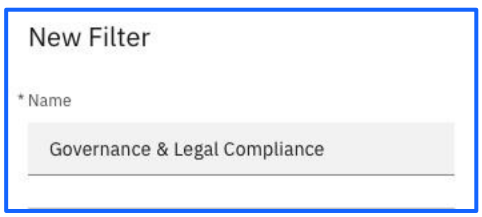
Let us create a filter that we may reuse.

Create Search Filters

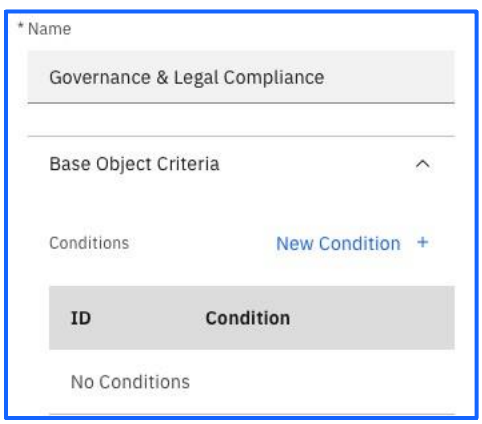
1. Click on **New Filter +**.



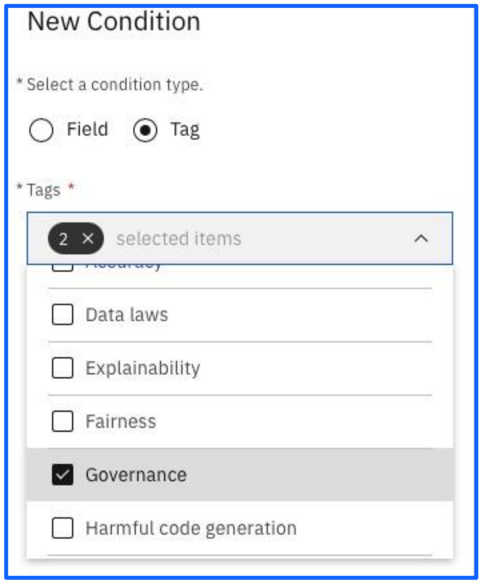
2. Provide the following **Name:** Governance & Legal Compliance



3. Click on **New Condition +**.

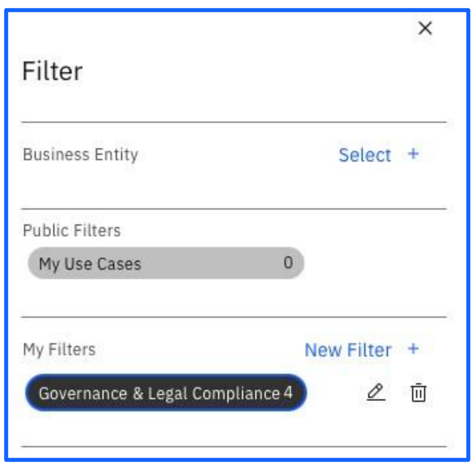


4. Select **Tag** & from the drop-down list select Governance & Legal Compliance.



5. Click **Save**.

   Notice that now appears your filter under "My filters".



6. Your Filter is now displayed by the object type text.

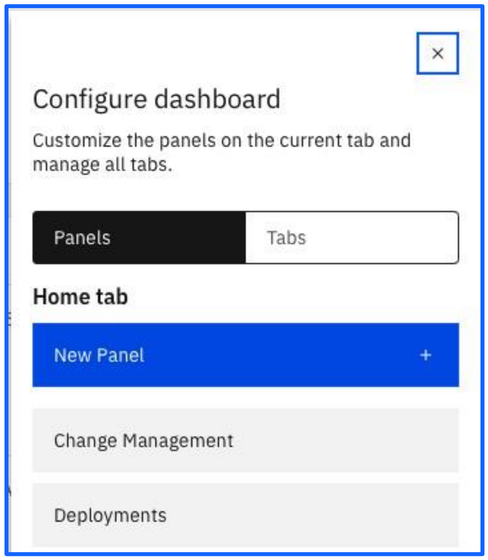
7. Remove the filter and return to the homepage tab.

Configuring your home page: create a new panel

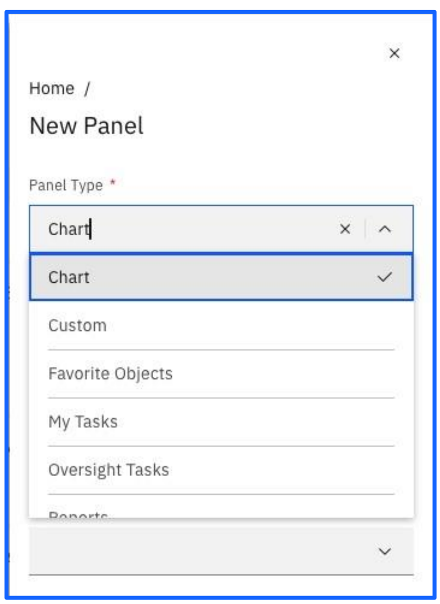
Now we are going to create a new panel, as we want to focus temporarily on Governance & Legal Compliance Use cases. We are going to design a panel that uses our just created private filter.

1. Click on the **Configure** icon on the right, above the tabs.

2. Select **New Panel +** .

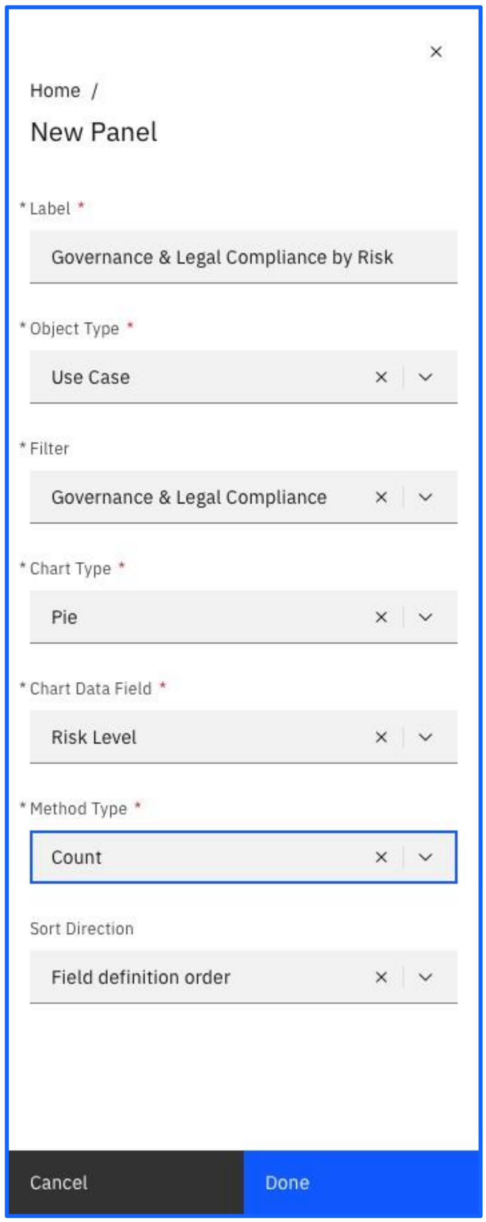


3. From the drop-down list select **Chart**.



4. Accordingly provide the following parameters:

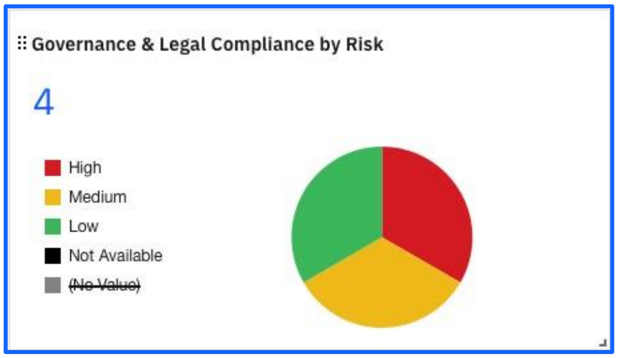
Label: Governance & Legal Compliance by Risk
Object Type: Use Case
Filter: Governance & Legal Compliance
Chart Type: Pie (you may also want to choose any other, except Gantt, to see how it looks like)
Chart Data Field: Risk Level
Method Type: Count



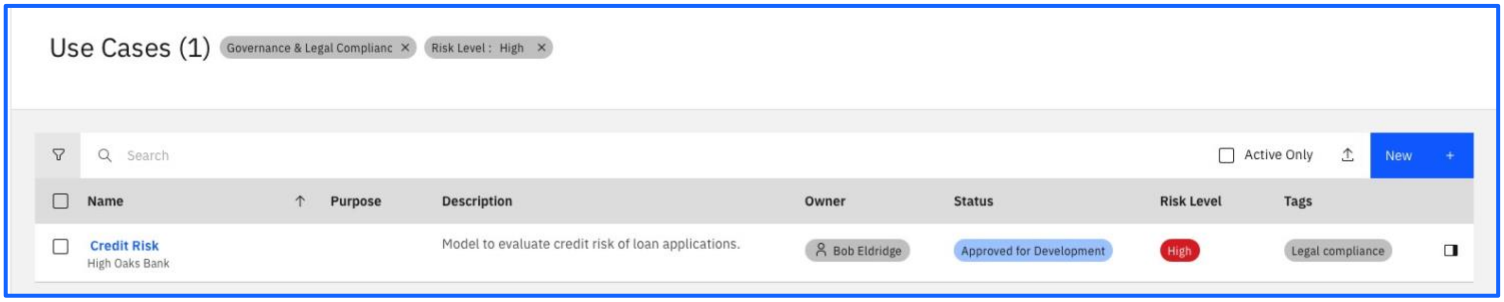
5. Click **Done** & **Save**.

A new panel will appear in your dashboard.

6. Click on the categories in the legend to **hide/unhide** them.

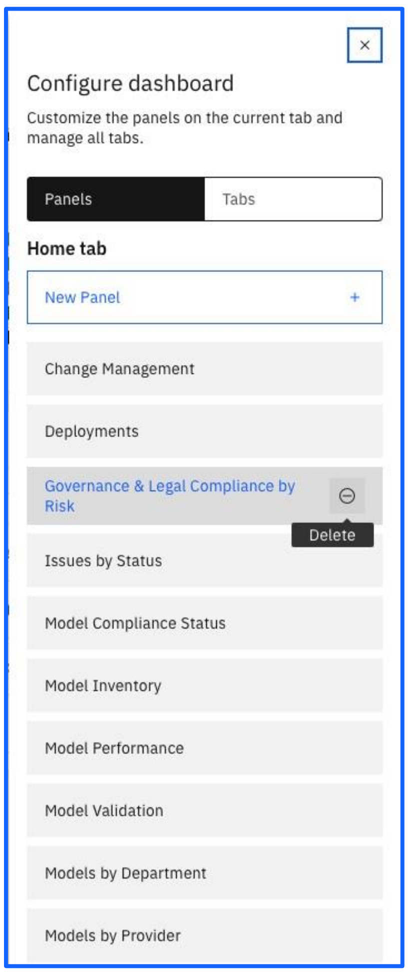


7. Select any section on the chart to drill down on a Risk Level category and you will be taken to a grid view displaying your filter name and the filter based on the risk level you have clicked on.



8. If you wish to remove your panel from the homepage, click on the **Configure** icon on the right, above the tabs.

9. In the displayed list of panels, you would look for the one you would want to remove, hover the mouse over it and you would remove it by clicking on the minus sign.

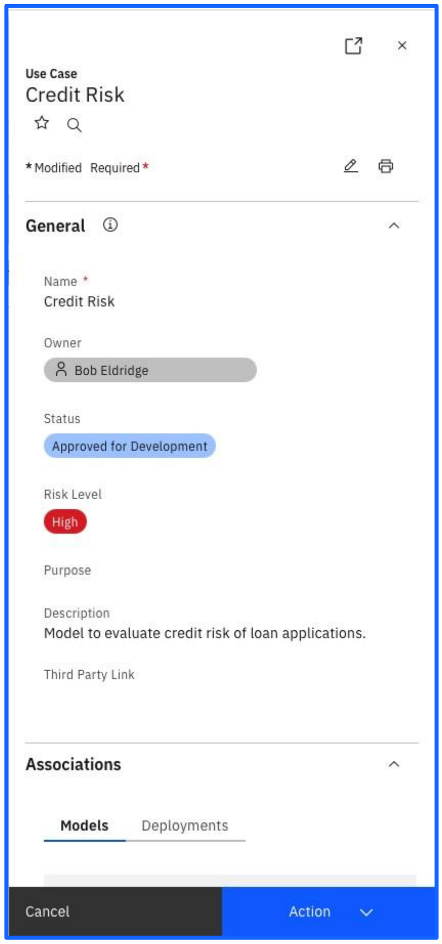


Do NOT remove the newly created panel as we will use it later on.

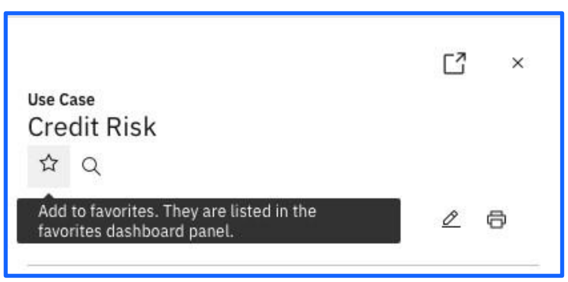
Navigation from an object

1. From the Homepage click on the “High Risk” level area in the chart of the panel you have just created.

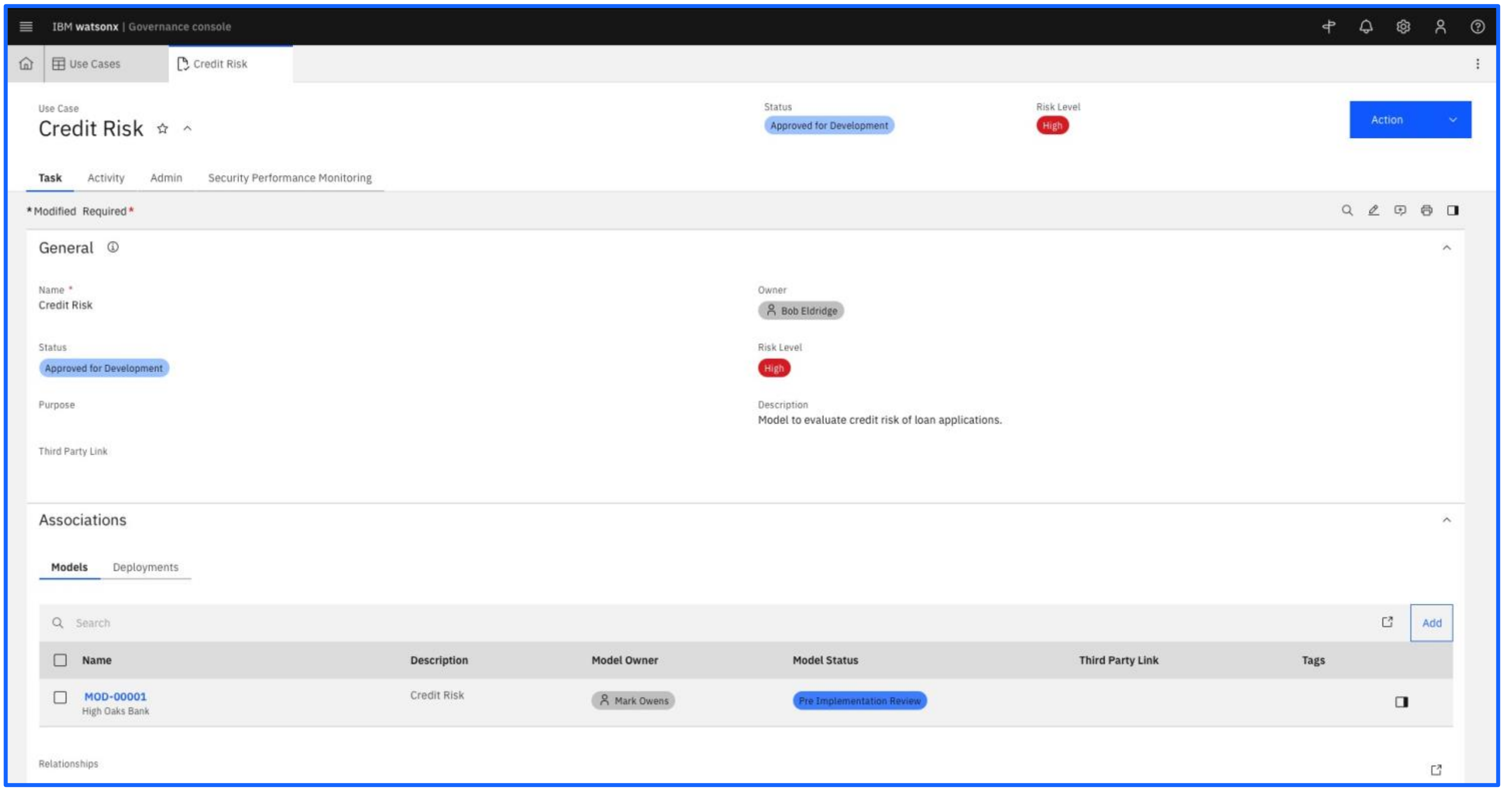
2. Do a quick inspection of the record by clicking on the **Open Side Panel** on the right. 



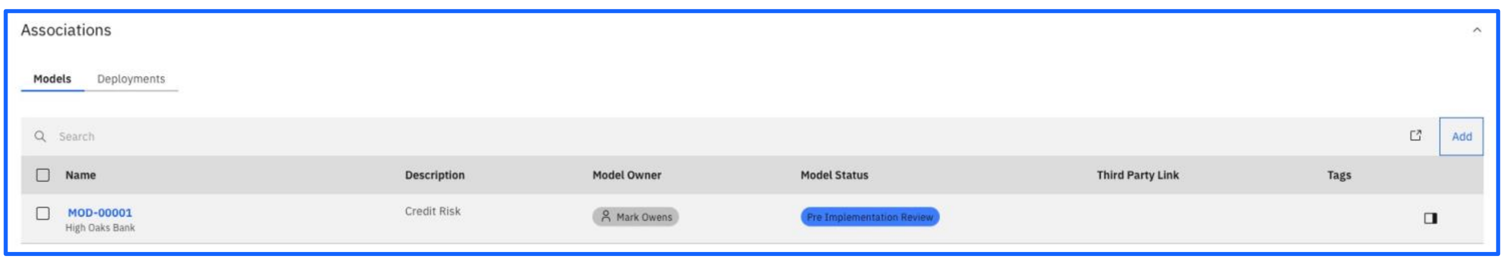
3. Mark this Use Case as Favorite (it will make it appear in your “My Favorites” panel in your homepage),by clicking on the star icon to flag/unflag the record as Favorite.



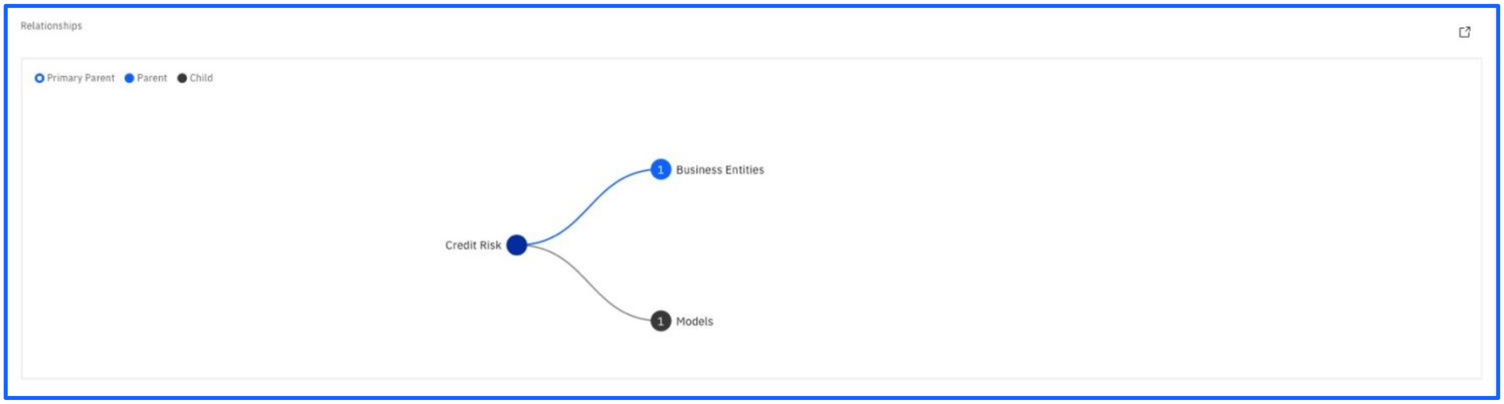
4. Select the Use Case to open it in a new tab, as we want to investigate further.



5. Scroll down to the section **Related Models**. The Tab **All Models** displays the Models related to this Use Case.

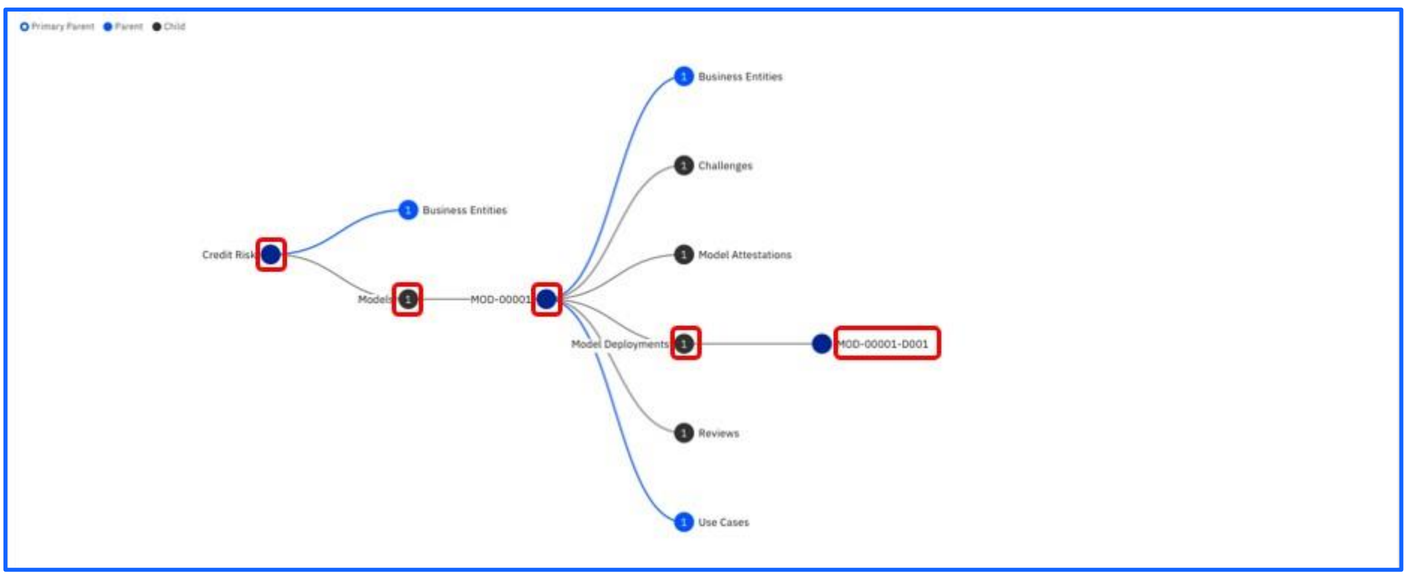


6. Scroll down to **Other Relationships**. The relationships tree view is displayed.



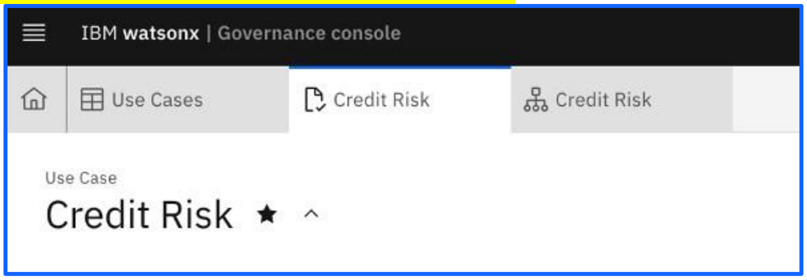
7. Open the tree diagram in a new tab by clicking on the **View fullscreen** icon

8. Following the picture below, click on the circles to expand the content. Click on the individual item to drill down on it. Notice the blue area that explains the relationships between the objects.

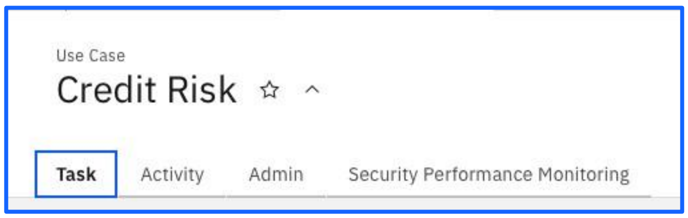


Audit Trail

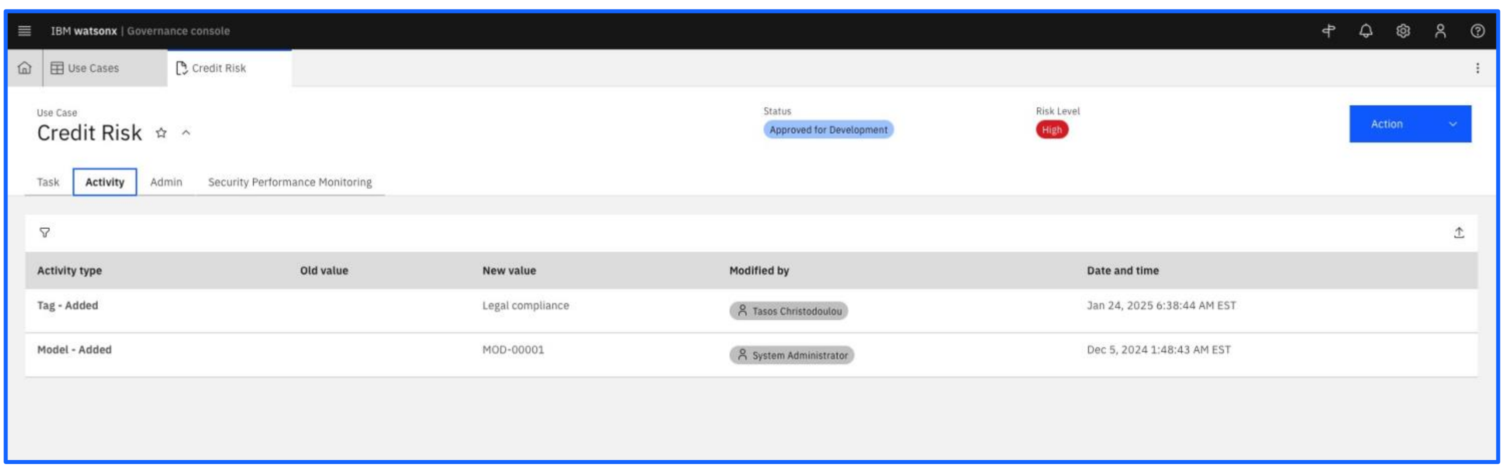
1. Click back into the tab for the Credit Risk Use Case



2. Switch from the **Task** tab to **Activity** tab.



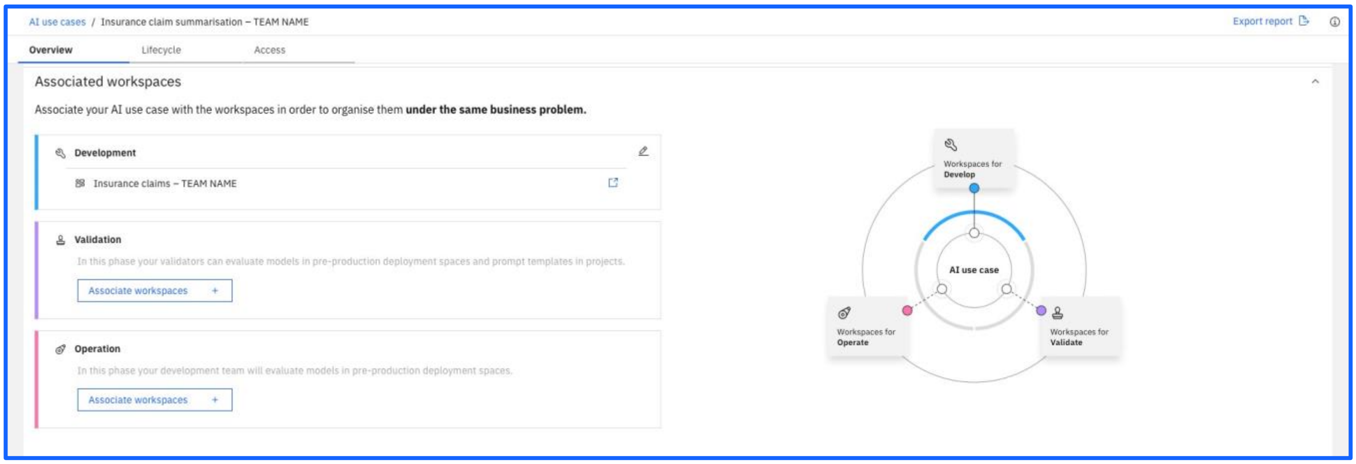
3. You may see all the changes made to this element. The details cannot be edited.



Promoting your prompt to Validation

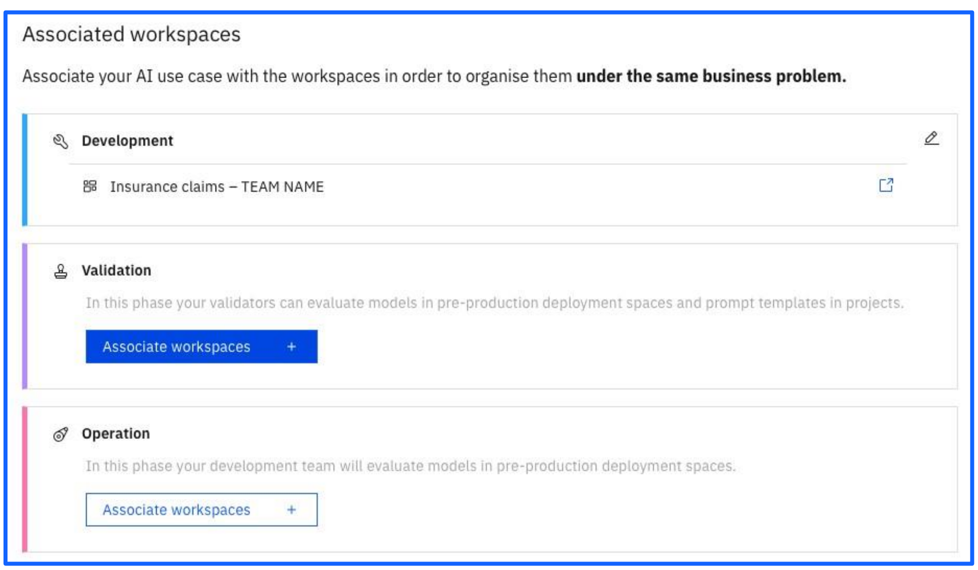
N.B. This stage is skipped to save time. In real life, enterprises would you validation as a testing envrionemtn.

In between development and production their is a validation step. In this next section, we will walk through how to promote and validate your prompt through the stages of the model lifecycle.
To start, navigate back to the **Overview** tab.



Since we now want to promote the second prompt template to the validate phase, we will create a new deployment space and associate it to the **Validation** stage.

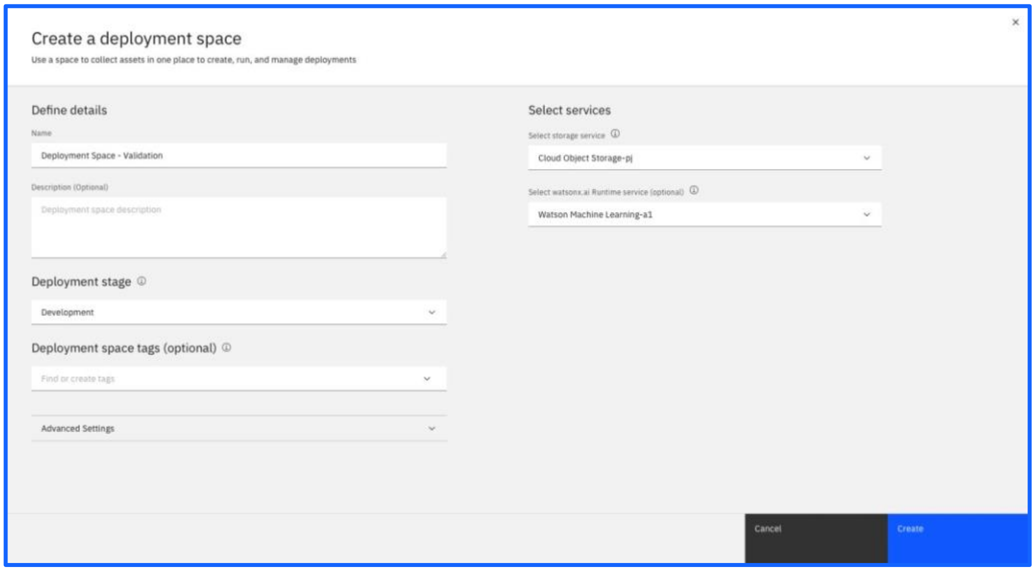
Click on the Associate workspace + button under the Validate space.



We will create a new **Deployment Space** for validation, to promote your prompt into.

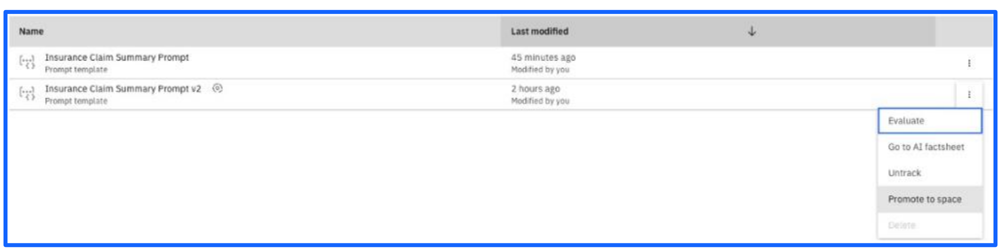
Give the **deployment space** a unique name as appropriate to this **validation stage** (i.e `Deployment Space - Validation`), so we can distinguish from other **deployment space** be will be creating.

Select **Testing** as the deployment stage, as well as the default storage service and machine learning service as you did in the earlier section, before clicking Create**.**

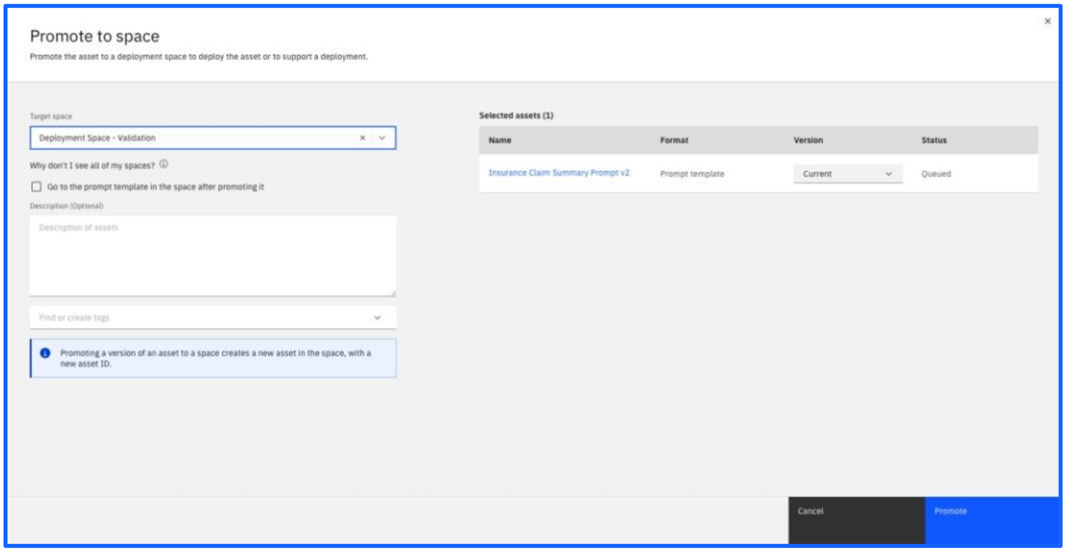


Associate this new Deployment Space under the *Space* section then click Save.

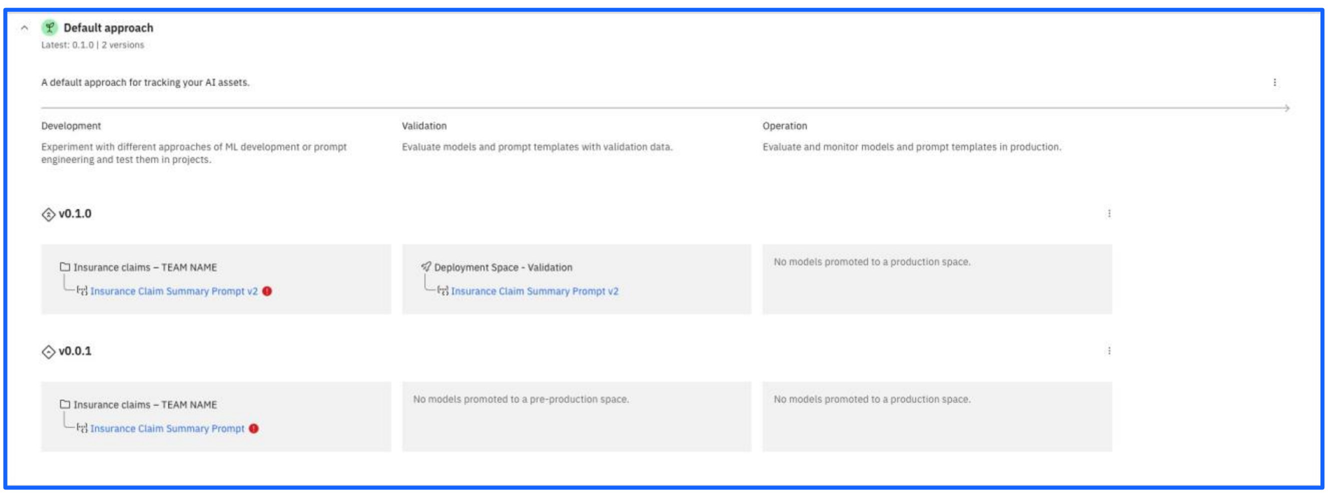
Once this is completed, navigate back to your Project, and press the menu for the most recently created prompt template, and click Promote to space



Select the newly created deployment space under the **Target space** dropdown menu, and once completed click the Promote button.



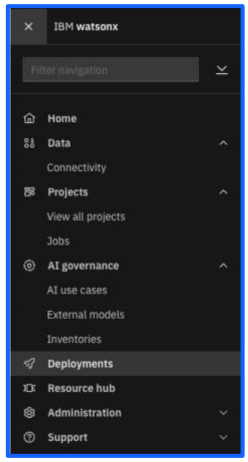
Navigate to the **Lifecycle** tab in your AI Use Case and note that it should appear under the Validate stage as opposed to the Develop stage of the AI use case lifecycle.



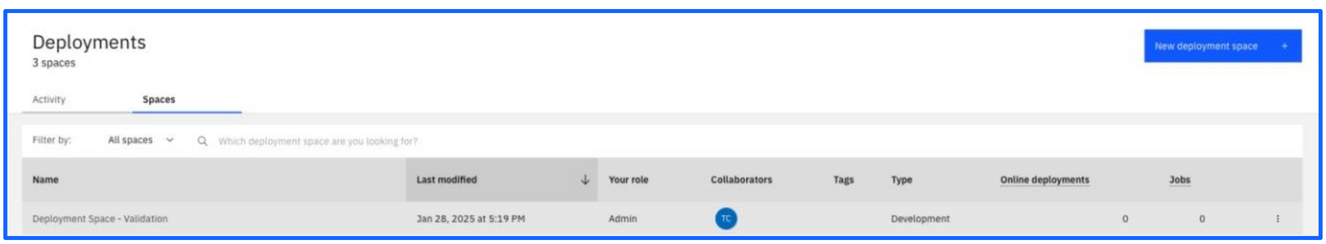
The prompt has not been deployed yet – so this is what we will be doing in the next section.

Deploying the prompt

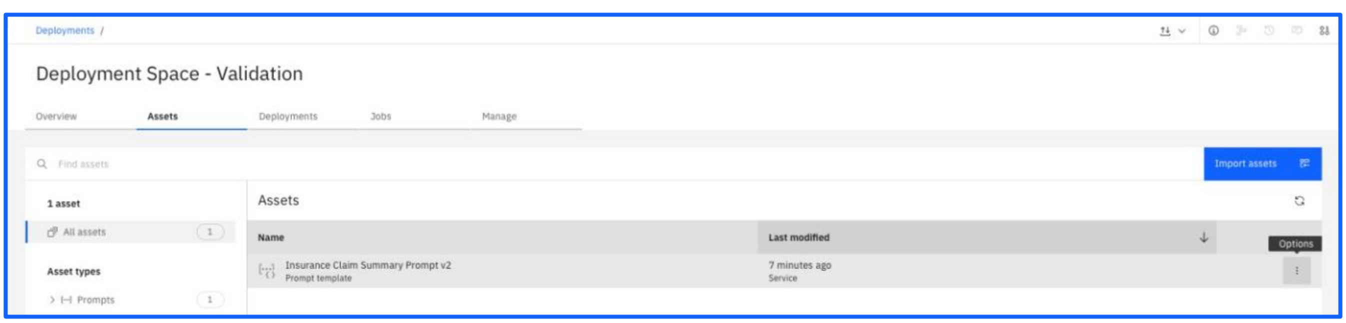
Use the navigation menu to move into the **Deployments** section.



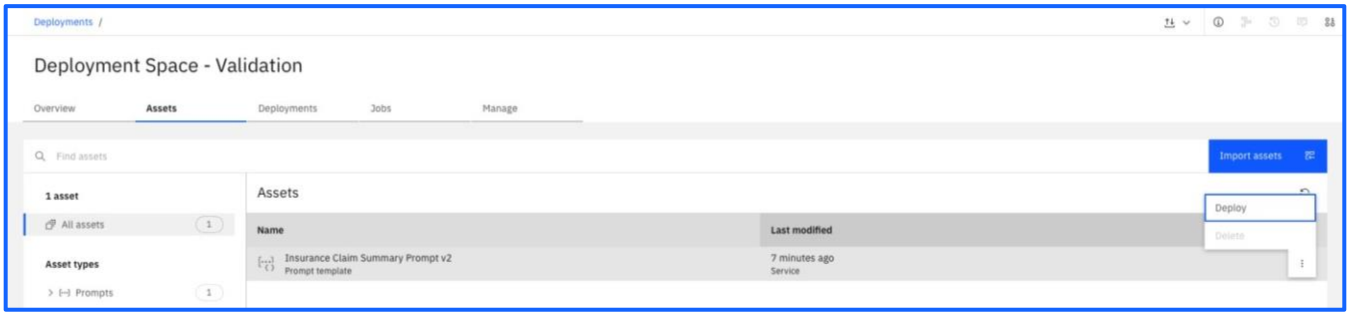
Click into the relevant validation space (i.e. the space you created in the previous section) within the **Spaces** tab.



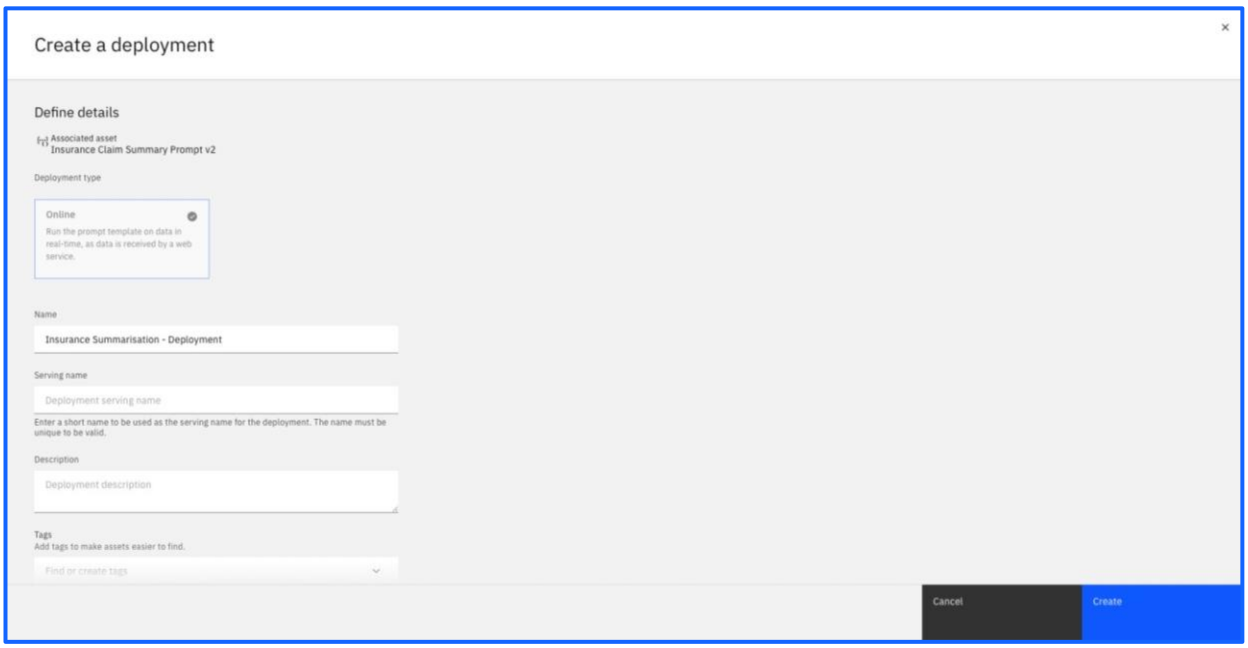
Move into the on the **Assets** tab, and click the three dots to the right of the most recent prompt template as shown:



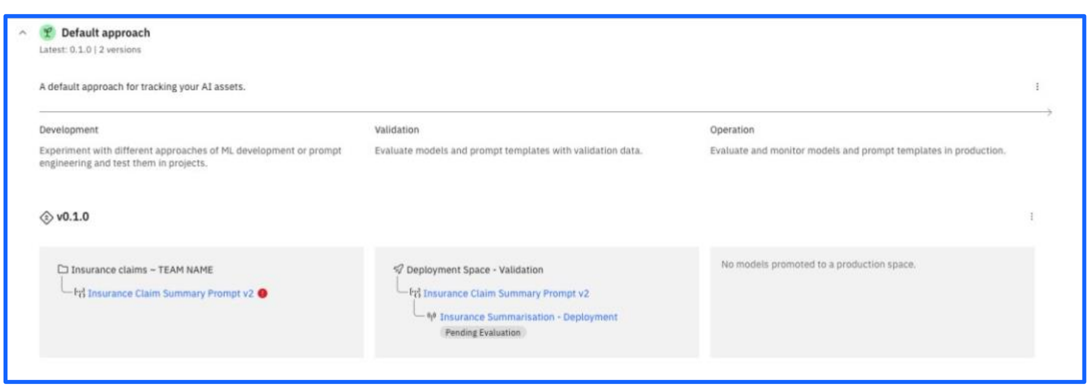
Finally, click the option to Deploy the prompt.



Within the deployment wizard, give an appropriate name (i.e. Insurance Summarisation - Deployment) before clicking Create.



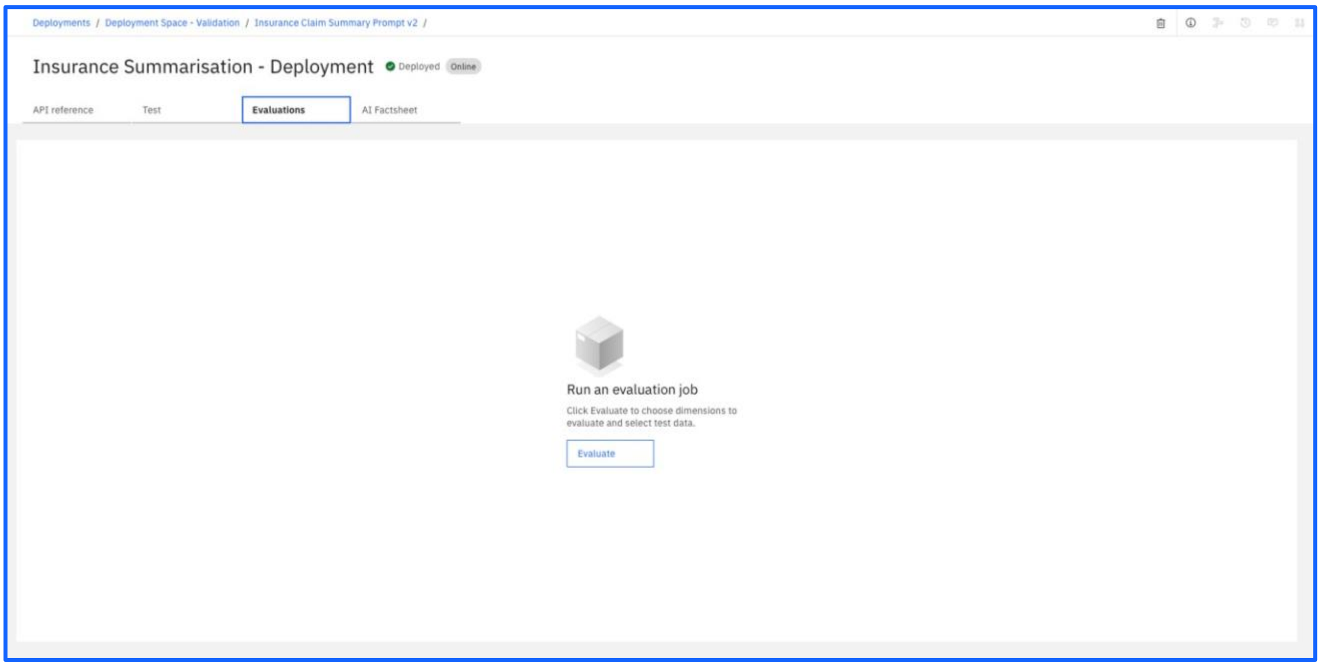
If you navigate back to the AI use case, you will see the prompt as Pending Evaluation.



This is the stage at which an appropriate stakeholder would evaluate and start monitoring the prompt – which we will be doing in the next and final section.

Activate prompt monitoring (Evaluation)

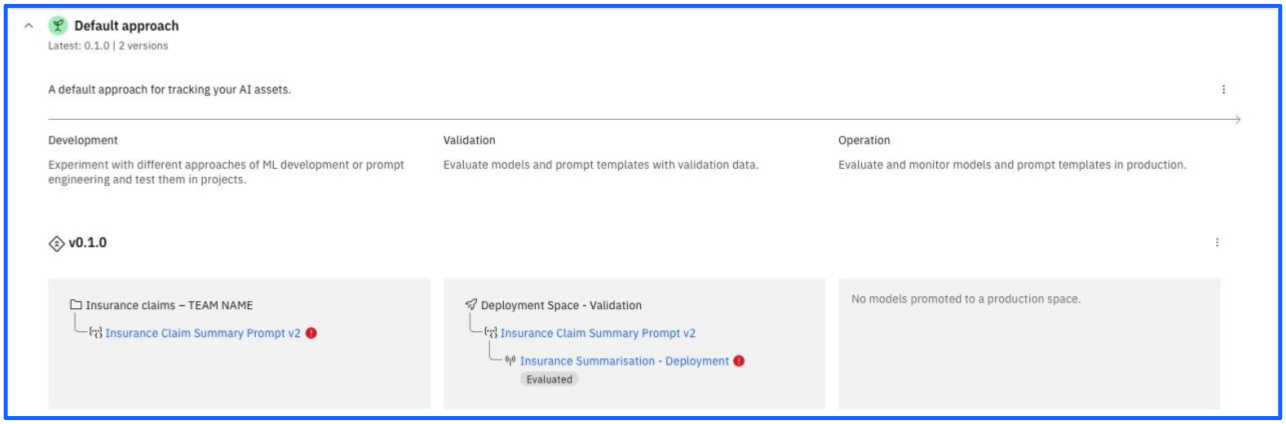
Click on the prompt that is pending evaluation, and move into the **Evaluations** tab to be taken to the screen shown below:



Click on the Evaluate button in the centre of the screen to start another evaluation before checking the results. It may seem redundant to run evaluations with the same prompt and the same data set over and over, but it is important to remember:

- The evaluations would be performed by different people in different roles, e.g. the prompt developer would not be the one evaluating it for use in production.
- In a real-world scenario, the data sets would differ between evaluations.
- There is a strong likelihood that the different models will change the evaluation scores.

After clicking Evaluate, navigate back to the AI use case to confirm that the status has changed from Pending to **Evaluated**


Default approach
Latest: 0.1.0 | 2 versions
A default approach for tracking your AI assets.
Development
Experiment with different approaches of ML development or prompt engineering and test them in projects.
Validation
Evaluate models and prompt templates with validation data.
Operation
Evaluate and monitor models and prompt templates in production.
v0.1.0
Insurance claims – TEAM NAME
Insurance Claim Summary Prompt v2
Deployment Space - Validation
Insurance Claim Summary Prompt v2
Insurance Summarisation - Deployment
Evaluated
No models promoted to a production space.